# Updates

| No. | Date | Cross section | Source | Comments |
|---|---|---|---|---|
| 812 | 19.02.2026 | $^{nat}$Zr(p,p$_0$)$^{nat}$Zr | M.Wang+(2025), Nucl. Instr. Methods. Physics Research B 559 (2025) 165576 | Data from publication<br>P. Dimitriou |
| 811 | 19.02.2026 | $^{16}$O(d,p$_{0,1}$)$^{17}$O | R.Ghasemi+(2025), Nucl. Instr. Methods in Physics research B, Vol.562, p. 165648 | Data from EXFOR<br>P. Dimitriou |
| 810 | 19.02.2026 | $^{12}$C(d,pg$_{1\text{-}0}$)$^{13}$C | A.Jokar+(2024), Nucl. Instr. Methods in Physics research B, Vol.557, p. 165549 | Data received from A. Jokar<br>P. Dimitriou |
| 809 | 19.02.2026 | $^{12}$C($^3$He,p$_{0\text{-}2}$)$^{14}$N<br>$^{12}$C($^3$He,a$_0$)$^{11}$C<br>$^{12}$C($^3$He,$^3$He$_0$)$^{12}$C | F.Maragkos+(2024), Nucl. Instr. Methods in Physics research B, Vol.556, p. 165517 | Data from EXFOR<br>P. Dimitriou |
| 808 | 05.02.2026 | $^2$H($^3$He,p$_0$)$^4$He | V.Foteinou+(2025), Nucl. Mat. Eenrgy, Vol.42, p. 101833 | Data received from V. Foteinou<br>P. Dimitriou |
| 807 | 05.02.2026 | $^7$Li($^3$He,p$_{0,2}$)$^9$Be<br>$^7$Li($^3$He,d$_0$)$^8$Be | C.Lopez Perez+(2025), Nucl. Instr. Methods in Physics research B, Vol.558, p. 165564 | Data received from C. Lopez Perez<br>P. Dimitriou |
| 806 | 05.02.2026 | $^{10}$B($^3$He,p$_{1,3}$)$^{12}$C<br>$^{11}$B($^3$He,p$_{0,1,2,3}$)$^{13}$C<br>$^{11}$B($^3$He,d$_0$)$^{12}$C | T.Dunatov+(2025), Rad. Phys. Chem., Vol.226, p. 112298 | Data received from T. Dunatov<br>P. Dimitriou |
| 805 | 05.02.2026 | $^{10}$B(d,a$_{0,1}$)$^8$Be<br>$^{10}$B(d,p$_2$)$^9$Be<br>$^{11}$B(d,a$_{0,2}$)$^9$Be | S.Taskaev+(2024), Nucl. Instr. Methods in Physics research B, Vol.557, p. 165527 | Data received from S. Taskaev<br>P. Dimitriou |
| 804 | 05.02.2026 | $^6$Li(d,a)$^4$He<br>$^6$B(d,p)$^7$Li<br>$^7$Li(d,a)$^5$He $^7$Li(d,na)$^4$He | S.Taskaev+(2024), Nucl. Instr. Methods in Physics research B, Vol.554, p. 165460 | Data received from S. Taskaev<br>P. Dimitriou |
| 803 | 05.02.2026 | $^{11}$B(p,a$_{0,1}$)$^8$Be | S.Taskaev+(2024), Nucl. Instr. Methods in Physics research B, Vol.555, p. 165490 | Data received from S. Taskaev<br>P. Dimitriou |
| 802 | 25.09.2024 | $^{10}$B(p,ag$_{1\text{-}0}$)$^7$Be<br>$^{10}$B(p,pg$_{1\text{-}0}$)$^{10}$B<br>$^{11}$B(p,pg$_{1\text{-}0}$)$^{11}$B | H.T.Aslani+(2024), Nucl. Instr. Methods in Physics Research B, Vol.556, p. 165509 | Data received from A. Jokar<br>P. Dimitriou |
| 801 | 25.09.2024 | $^{13}$C($^3$He,p$_{0\text{-}15}$)$^{15}$N<br>$^{13}$C($^3$He,d$_0$)$^{14}$N | L.Hess+(2024), Nucl. Instr. Methods in Physics Research B, Vol. 555, p. 165454 | Data from Supplementary Material published with article<br>P. Dimitriou |
| 800 | 25.09.2024 | $^1$H(p,p$_0$)$^1$H | H.Rafi-Kheiri+(2024), Nucl. Instr. Methods in Physics Research B, Vol.553, p. 165401 | Data received from A. Jokar<br>P. Dimitriou |
| 799 | 27.02.2024 | $^9$Be(p,a$_0$)$^6$Li<br>$^9$Be(p,d$_0$)$^8$Be<br>$^9$Be(p,p$_0$)$^9$Be | F.Maragkos+(2024), Nucl. Instr. Methods in Physics Research B, Vol. 547, p. 165180 | Data received from F. Maragkos<br>P. Dimitriou |
| 798 | 27.02.2024 | $^{12}$C($^3$He,p$_{0,1,2,3,4,5,6}$)$^{14}$N | L.Hess+(2024), Nucl. Instr. Methods in Physics Research B, Vol. 547, p. 165141 | Data from Supplementary Material published with article<br>P. Dimitriou |
| 797 | 18.10.2023 | $^9$Be(d,a$_0$)$^7$Li<br>$^9$Be(d,a$_0$)$^7$Li | A.Saganek+(1971), Acta Physica Polonica, Part B, Vol.2, p.473 | Data calculated using Legendre coefficients from the publication<br>V. Semkova, N. Otsuka |
| 796 | 18.10.2023 | $^7$Li(p,ng$_{1\text{-}0}$)$^7$Be | A.C.Scafes+(2023), European Physical Journal A: Hadrons and Nuclei, Vol.59, p.47 | Data received from A.C.Scafes<br>V. Semkova, N. Otsuka |
| 795 | 18.10.2023 | $^{44}$Ca(p,g$_{3\text{-}1}$)$^{45}$Sc | A.Anttila+(1981),Jour. of Radioanal. and Nuclear Chemistry, Vol.62, p.293 | Duplication deleted<br>V. Semkova, N. Otsuka |
| 794 | 18.10.2023 | $^{19}$F(p,ag$_{3\text{-}0}$)$^{16}$O<br>$^{19}$F(p,ag$_{4\text{-}0}$)$^{16}$O | A.Anttila+(1981),Jour. of Radioanal. and Nuclear Chemistry, Vol.62, p.293 | Deleted. Data not reported in this publication. The same data in A.Z.Kiss+ (1985)<br>V. Semkova, N. Otsuka |
| 793 | 18.10.2023 | | A.Anttila+(1981),Jour. of Radioanal. and Nuclear Chemistry, Vol.62, p.293 | X4Number added<br>V. Semkova, N. Otsuka |
| 792 | 18.10.2023 | | F.Maragkos+(2021), Nucl. Instrum. Methods | X4Number added |

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| | | $^{nat}Si(p,p_0)^{nat}Si$ | in Physics Res., Sect.B, Vol.507, p.20 - 26 | V. Semkova, N. Otsuka |
| 791 | 18.10.2023 | $^{16}O(d,d_0)^{16}O$ | F.Machali+(1968), Annalen der Physik, Vol.21, p.1 | X4Number added<br>V. Semkova, N. Otsuka |
| 790 | 18.10.2023 | $^{nat}Mg(p,p_0)^{nat}Mg$ | F. Maragkos+(2023), Jour. Nucl. Instrum. Methods in Physics Res., Sect.B, Vol.536, p.45 | X4Number added<br>V. Semkova, N. Otsuka |
| 789 | 18.10.2023 | $^{19}F(p,ag_{4-0})^{16}O$<br>$^{19}F(p,ag_{2-0+3-0+4-0})^{16}O$<br>$^{19}F(p,ag_{3-0})^{16}O$<br>$^{19}F(p,ag_{2-0})^{16}O$<br>$^{19}F(p,ag_{1-0})^{16}O$<br>$^{19}F(p,pg_{1-0})^{16}O$ | M.J.Kenny(1981),Australian Journal of Physics, Vol.34, p.35 | X4Number corrected<br>V. Semkova, N. Otsuka |
| 788 | 18.10.2023 | $^{19}F(p,a_0)^{16}O$ | A.Isoya+(1958), Nuclear Physics, Vol.7, Issue.2, p.116 | Replaced with file from EXFOR<br>V. Semkova, N. Otsuka |
| 787 | 18.10.2023 | $^{12}C(^6Li,^6Li_0)^{12}C$ | M. Mayer and M. Schneider, Nucl. Instr. Meth. B183 (2001) 221 | X4Number added<br>V. Semkova, N. Otsuka |
| 786 | 18.10.2023 | $^{10}B(p,ag_{1-0})^7Be$<br>$^{10}B(p,pg_{1-0})^{10}B$ | R.B. Day+(1954), Jour. Physical Review, Vol.95, p.1003. | X4Number added<br>V. Semkova, N. Otsuka |
| 785 | 18.10.2023 | $^{27}Al(p,g)^{28}Si$ | S.Harissopulos+(2000), European Physical Journal A: Hadrons and Nuclei, Vol.9, p.479 | Publication year corrected<br>V. Semkova, N. Otsuka |
| 784 | 23.09.2023 | $^{nat}Si(p,p_0)^{nat}Si$ | Li Gong-Ping+(2002), Nuclear Physics Review, Vol.30, p.385 (2013), China, Vol.19, p.39 | Compiled from S0240<br>V. Semkova, N. Otsuka |
| 783 | 23.09.2023 | $^6Li(p,p_0)^6Li$<br>$^7Li(p,p_0)^7Li$<br>$^7Li(p,pg_{1-0})^7Li$<br>$^{19}F(p,p_0)^{19}F$<br>$^{19}F(p,pg_{1-0})^{19}F$<br>$^{19}F(p,pg_{2-0})^{19}F$<br>$^{19}F(p,pg_{3-1})^{19}F$<br>$^{19}F(p,pg_{4-1+5-2})^{19}F$ | A.C.Scafes+(2023), European Physical Journal A: Hadrons and Nuclei, Vol.59, p.47 | Data received from A.C.Scafes<br>V. Semkova, N. Otsuka |
| 782 | 23.09.2023 | $^9Be(^3He,p_0)^{11}B$<br>$^9Be(^3He,p_1)^{11}B$<br>$^9Be(^3He,p_2)^{11}B$<br>$^9Be(^3He,p_{4+5})^{11}B$<br>$^9Be(^3He,p_6)^{11}B$<br>$^9Be(^3He,p_7)^{11}B$ | V.Foteinou+(2023), Nucl. Instrum. Methods in Physics Res., Sect.B, Vol.542, p.158 | Data received from V. Foteinou<br>V. Semkova, N. Otsuka |
| 781 | 23.09.2023 | $^{11}B(d,pg_{1-0})^{12}B$<br>$^{11}B(d,pg_{2-0})^{12}B$ | H.T.Aslani+(2023), Nucl. Instrum. Methods in Physics Res., Sect.B, Vol.540, p.141 | Data received from A.Jokar<br>V. Semkova, N. Otsuka |
| 780 | 23.09.2023 | $^6Li(d,ng_{1-0})^7Be$<br>$^6Li(d,pg_{1-0})^7Li$<br>$^{19}F(d,pg_{1-0})^{20}F$ | H.T.Aslani+(2023), Nucl. Instrum. Methods in Physics Res., Sect.B, Vol.535, p.96 | Data received from A.Jokar<br>V. Semkova, N. Otsuka |
| 779 | 11.09.2023 | $^{13}C(p,\gamma_0)^{14}N$ | D.F. Hebbard and J.L Vogl, Nucl. Phys. 21 (1960) 652 | P. Dimitriou. Data revised |
| 778 | 24.05.2023 | $^{17}O(p,p_0)^{17}O$ | M.Salimi+(2022), Nucl. Instrum. Methods in Physics Res., Sect.B, Vol.516, p.15 | Duplicated files deleted<br>V. Semkova |
| 777 | 20.05.2023 | $^9Be(^3He,p_0)^{11}B$<br>$^9Be(^3He,p_1)^{11}B$<br>$^9Be(^3He,p_2)^{11}B$<br>$^9Be(^3He,p_{4+5})^{11}B$<br>$^9Be(^3He,p_6)^{11}B$<br>$^9Be(^3He,p_7)^{11}B$<br>$^9Be(^3He,p_8)^{11}B$<br>$^9Be(^3He,p_9)^{11}B$ | C.M.Vitor+(2023), Phys.Scr., Vol.98, p.035306 | Data received from author<br>V. Semkova, N. Otsuka |
| 776 | 20.05.2023 | $^{19}F(d,pg_{1-0})^{20}F$<br>$^6Li(d,ng_{1-0})^7Be$<br>$^6Li(d,pg_{1-0})^7Li$ | E.Taimpiri+(2023), Nucl. Instrum. Methods in Physics Res., Sect.B, Vol.539, p.162 | Data received from author<br>V. Semkova, N. Otsuka |

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| | | $^{7}$Li(d,dg$_{1\text{-}0}$)$^{7}$Li | | |
| 775 | 20.05.2023 | $^{7}$Li(p,pg$_{1\text{-}0}$)$^{7}$Li | A.Ziagkova+(2023), Nucl. Instrum. Methods in Physics Res., Sect.B, Vol.539, p.113 | Data received from author<br>V. Semkova, N. Otsuka |
| 774 | 20.05.2023 | $^{16}$O(p,p$_{0}$)$^{16}$O | M.Kokkoris+(2023), Nucl. Instrum. Methods in Physics Res., Sect.B, Vol.539, p.15 | Data received from author<br>V. Semkova, N. Otsuka |
| 773 | 20.05.2023 | $^{17}$O(p,p$_{0}$)$^{17}$O | M.Salimi+(2022), Nucl. Instrum. Methods in Physics Res., Sect.B, Vol.516, p.15 | Data received from author<br>V. Semkova, N. Otsuka |
| 772 | 20.05.2023 | $^{7}$Li(p,pg$_{1\text{-}0}$)$^{7}$Li | A.Savidou+(1999), Jour. Nucl. Instrum. Methods in Physics Res., Sect.B, Vol.152, p.12 | datum at 1.77 MeV<br>5.0E+08 -> 5.0E+06<br>V. Semkova, N. Otsuka |
| 771 | 20.05.2023 | $^{27}$Al(p,pg$_{1\text{-}0}$)$^{27}$Al<br>$^{27}$Al(p,pg$_{2\text{-}0}$)$^{27}$Al | R.Mateus+(2022), European Physical Journal A: Hadrons and Nuclei, Vol.58, p.209 | Reference updatedr<br>X4Number added<br>V. Semkova, N. Otsuka |
| 770 | 20.05.2023 | | M. Salimi+(2021), Scientific Reports, Vol.11, p.18036 | Reference updated<br>V. Semkova, N. Otsuka |
| 769 | 20.05.2023 | $^{nat}$Mg(p,p$_{0}$)$^{nat}$Mg | F. Maragkos+(2023), Jour. Nucl. Instrum. Methods in Physics Res., Sect.B, Vol.536, p.45 | Data received from author<br>V. Semkova, N. Otsuka |
| 768 | 20.05.2023 | $^{nat}$Si(p,p$_{0}$)$^{nat}$Si | F.Maragkos+(2021), Nucl. Instrum. Methods in Physics Res., Sect.B, Vol.507, p.20 - 26 | Sig- and Enfactors corrected<br>V. Semkova, N. Otsuka |
| 767 | 20.05.2023 | $^{9}$Be(d,a$_{0}$)$^{7}$Li<br>$^{9}$Be(d,a$_{1}$)$^{7}$Li | P.Tsavalas+(2020), Nucl. Instrum. Methods in Physics Res., Sect.B, Vol.479, p.205 | Data from P.Tsavalas<br>V. Semkova, N. Otsuka |
| 766 | 18.01.23 | $^{16}$O(p,p$_{0}$)$^{16}$O | R.W.Harris+(1962), Nuclear Physics, Vol.38, p.259 | Data corrected by EXFOR F1013002 (V.Zerkin) |
| 765 | 29.11.22 | $^{6}$Li(p,$^{3}$He$_{0}$)$^{4}$He | S.Bashkin+(1951), Physical Review, Vol.84, p.1124 | Data corrected by EXFOR A1454003 (V.Zerkin) |
| 764 | 19.09.22 | $^{24}$Mg(p,p$_{0}$)$^{24}$Mg<br>$^{24}$Mg(p,p$_{1}$)$^{24}$Mg | J.R.Duray+(1972), Physical Review, Part C, Nuclear Physics, Vol.6, p.792 | Compiled from the publication<br>V. Semkova, N. Otsuka |
| 763 | 16.09.22 | $^{16}$O(d,d$_{0}$)$^{16}$O | F.Machali+(1974), Annalen der Physik, Vol.21, p.1 | Compiled from the publication<br>V. Semkova, N. Otsuka |
| 762 | 19.09.22 | $^{10}$B(p,p$_{0}$)$^{10}$B | G.Kaur+(2022), Physical Review, Part C, Nuclear Physics, Vol.105, p.024609<br>J.Hoehn+(1981), Jour. of Physics, Part G (Nucl.and Part.Phys.), Vol.7, p.803<br>B.A.Watson+(1969), Physical Review, Vol.182, p.977 | EXFOR D1020002<br>F0288006<br>F0156002<br>V. Semkova, N. Otsuka |
| 761 | 19.09.22 | $^{17}$O(p,p$_{0}$)$^{17}$O | S.,Taskaev+(2021), Nucl., Instrum., Methods in Physics Res.,, Sect.,B, Vol.,525, p.,55 | Compiled from the publication<br>V. Semkova, N. Otsuka |
| 760 | 19.09.22 | $^{17}$O(p,p$_{0}$)$^{17}$O | M.Salimi+(2022), Nucl. Instrum. Methods in Physics Res., Sect.B, Vol.516, p.15 | Compiled from the publication<br>V. Semkova, N. Otsuka |
| 759 | 19.09.22 | $^{33}$S(p,p$_{0}$)$^{33}$S | J.R.Vanhoy+(1989), Physical Review, Part C, Nuclear Physics, Vol.40, p.1959 | EXFOR: C1561005<br>V. Semkova, N. Otsuka |
| 758 | 19.09.22 | $^{20}$Ne(p,g$_{8\text{-}0}$)$^{21}$Na | C.Van Der Leun+(1964), Physica (Utrecht), Vol.30, p.333 | Reaction corrected<br>EXFOR: F0924002<br>V. Semkova, N. Otsuka |
| 757 | 19.09.22 | $^{17}$O(p,a$_{0}$)$^{14}$N | R.E.Brown(1962), Physical Review, Vol.125, p.347 | EXFOR: C2404003<br>V. Semkova, N. Otsuka |
| 756 | 19.09.22 | $^{25}$Mg(p,pg$_{1\text{-}0}$)$^{25}$Mg | J.Cruz+(2022), European Physical Journal , Vol.58, p.128 | Data received from J.Cruz<br>V. Semkova, N. Otsuka |
| 755 | 19.09.22 | $^{7}$Li(p,a$_{0}$)$^{4}$He | W.E.Sweeney Jr+(1969), Jour. Physical Review, Vol.182, p.1007 | Reaction corrected<br>V. Semkova, N. Otsuka |
| 754 | 19.09.22 | $^{6}$Li(p,$^{3}$He$_{0}$)$^{4}$He | F.Bertrand+(1968), Rept. Centre d`Etudes Nucleaires, Saclay Reports, No.3428<br>G.P.Johnston+(1974), Jour. Nuclear Physics, Section A, Vol.224, Issue.2, p.349<br>S.Bashkin+(1951), Jour. Physical Review, Vol.84, Issue.6, p.1124<br>A.J.Elwyn+(1979), Jour. Physical Review, Part C, Nuclear Physics, Vol.20, p.1984<br>Chia-Shou Lin+(1977), Jour. Nuclear Physics, Section A, Vol.275, p.93 | Reaction corrected<br>V. Semkova, N. Otsuka |
| 753 | 19.09.22 | $^{7}$Li(p,a$_{0}$)$^{4}$He | J.M.Freeman+(1958), Nuclear Physics, Vol.5, Issue.1, p.148 | X4Number added<br>V. Semkova, N. Otsuka |
| 752 | 19.09.22 | | A.Z.Kiss+(1985), Jour. of Radioanal. and Nuclear Chemistry, Vol.89, p.123 | Source corrected<br>V. Semkova, N. Otsuka |

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| | | $^{12}C(^{3}He,a_{0})^{11}C$<br>$^{19}F(p,ag_{2\text{-}0+3\text{-}0+4\text{-}0})^{16}O$<br>$^{19}F(p,pg_{2\text{-}0})^{19}F$<br>$^{19}F(p,pg_{1\text{-}0})^{19}F$ | H.-M.Kuan+(1964), Jour. Nuclear Physics, Vol.51, p.481<br><br>A.Fessler +(2000), Nucl. Instrum. Methods in Physics Res., Sect.A, Vol.450, p.353<br><br>J.Cruz+(2021), European Physical Journal Plus, Vol.136, p.969 | |
| 751 | 19.09.22 | $^{13}C(^{3}He,p_{0})^{15}N$<br>$^{13}C(^{3}He,p_{1+2})^{15}N$<br>$^{13}C(^{3}He,p_{3})^{15}N$ | S.Moeller(2017), Nucl. Instrum. Methods in Physics Res., Sect.B, Vol.394, p.134 | EXFOR: O2321<br>V. Semkova, N. Otsuka |
| 750 | 19.09.22 | $^{13}C(^{3}He,a_{0})^{12}C$<br>$^{13}C(^{3}He,a_{1})^{12}C$<br>$^{13}C(^{3}He,p_{0})^{15}N$<br>$^{13}C(^{3}He,p_{1+2})^{15}N$ | S.H.Chew+(1978), Nuclear Physics, Section A, Vol.306, p.125 | EXFOR: O1799002<br>V. Semkova, N. Otsuka |
| 749 | 19.09.22 | $^{6}Li(p,a_{1})^{4}He$<br>$^{6}Li(p,a_{2})^{4}He$ | S.Cavallaro+(1962), Jour. Nuclear Physics, Vol.36, p.597 | Files deleted<br>V. Semkova, N. Otsuka |
| 748 | 19.09.22 | $^{6}Li(^{3}He,p_{1})^{8}Be$<br>$^{10}B(^{3}He,p_{0})^{12}B$<br>$^{10}B(^{3}He,p_{1})^{12}B$ | J.P.Schiffer+(1956), Physical Review, Vol.104, p.1064 | Replaced with files from EXFOR<br>V. Semkova, N. Otsuka |
| 747 | 19.09.22 | $^{3}H(^{3}He,d_{0})^{4}He$ | U.Nocken+(1973), Nuclear Physics, Section A, Vol.213, Issue.1, p.97 | EXFOR: A1053003<br>V. Semkova, N. Otsuka |
| 746 | 19.09.22 | $^{9}Be(^{3}He,p_{4+5})^{11}B$ | G.Provatas+(2020), Nucl. Instrum. Methods in Physics Res., Sect.B, Vol.472, p.36<br>C.S.Lin+(1981), Jour. Chinese Journal of Physics (Taiwan), Vol.19, p.99 | (3He,p5) and (3He,$p_4$) -> (3He,p4+5)<br>V. Semkova, N. Otsuka |
| 745 | 19.09.22 | $^{7}Li(^{3}He,p_{0})^{9}Be$ | D.P.Rath+(1990), Nuclear Physics, Section A, Vol.515, p.338 | R33 files corrected<br>V. Semkova, N. Otsuka |
| 744 | 19.09.22 | $^{7}Li(^{3}He,a_{0})^{6}Li$ | H.Orihara+(1970), Journal of the Physical Society of Japan, Vol.29, Issue.3, p.533 | EXFOR: A1492002<br>V. Semkova, N. Otsuka |
| 743 | 19.09.22 | $^{2}H(a,d)^{4}He$ | V.Quillet+(1993), Nucl. Instrum. Methods in Physics Res., Sect.B, Vol.83, p.47 | Duplication at 30 deg. deleted<br>Data at 20 deg. added<br>EXFOR: O0817003<br>V. Semkova, N. Otsuka |
| 742 | 19.09.22 | $^{2}H(^{3}He,p_{0})^{4}He$ | J.P.Zhu+(2017), Nucl. Instrum. Methods in Physics Res., Sect.B, Vol.412, p.81 | EXFOR: S0089002<br>V. Semkova, N. Otsuka |
| 741 | 19.09.22 | $^{7}Li(p,pg_{1\text{-}0})^{7}Li$ | S.Taskaev+(2021), Nucl. Instrum. Methods in Physics Res., Sect.B, Vol.502, p.85 | R33 file corrected<br>V. Semkova, N. Otsuka |
| 740 | 19.09.22 | $^{2}H(^{3}He,p_{0})^{4}He$ | B.Wielunska+(2016), Nucl. Instrum. Methods in Physics Res., Sect.B, Vol.371, p.41 | X4Number updated<br>(3He,p)-> (3He,$p_0$)<br>V. Semkova, N. Otsuka |
| 739 | 20.12.21 | $^{16}O(d,d_{0})^{16}O$ | E.Ntemou+(2022), Nucl. Instrum. Methods in Physics Res., Sect.B, Vol.510, p.56 | R33 files received from E.Ntemou |
| 738 | 20.12.21 | $^{nat}Y(a,a_{0})^{nat}Y$ | A.F.Gurbich+(2018), Nucl. Instrum. Methods in Physics Res., Sect.B, Vol.420, p.1 | $^{89}Y$ -> $^{nat}Y$ |
| 737 | 20.12.21 | $^{63}Cu(p,g_{1\text{-}0})^{64}Zn$<br>$^{63}Cu(p,g_{2\text{-}1})^{64}Zn$<br>$^{63}Cu(p,g_{2\text{-}0})^{64}Zn$<br>$^{63}Cu(p,g_{4\text{-}1})^{64}Zn$<br>$^{65}Cu(p,g_{1\text{-}0})^{66}Zn$<br>$^{65}Cu(p,g_{2\text{-}1})^{66}Zn$<br>$^{63}Cu(p,pg_{1\text{-}0})^{63}Cu$<br>$^{63}Cu(p,pg_{2\text{-}1})^{63}Cu$<br>$^{65}Cu(p,pg_{1\text{-}0})^{65}Cu$ | T.Tajvidi+(2021), Nucl. Instrum. Methods in Physics Res., Sect.B, Vol.509, p.60 | R33 files received from A. Jokar |
| 736 | 20.12.21 | $^{7}Li(p,ng_{1\text{-}0})^{7}Be$<br>$^{7}Li(p,pg_{1\text{-}0})^{7}Li$ | A.Caciolli+(2006), Nucl. Instrum. Methods in Physics Res., Sect.B, Vol.249, p.98 | X4Number added |
| 735 | 20.12.21 | $^{nat}Si(p,p_{0})^{nat}Si$ | F.Maragkos+(2021), Nucl. Instrum. Methods in Physics Res., Sect.B, Vol.507, p.20 - 26 | R33 files received from F. Maragkos |
| 734 | 20.12.21 | $^{12}C(p,p_{0})^{12}C$ | N.Nikolic+(1963), Physical Review, Vol.132, p.2212 | 4toR33 from EXFOR C2362010 |

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| 733 | 20.12.21 | $^{28}Si(p,p_0)^{28}Si$ | R.Salomonovic(1993), Nucl. Instrum. Methods in Physics Res., Sect.B, Vol.82, p.1 | 4toR33 from EXFOR F0571002 |
| 732 | 20.12.21 | $^{12}C(a,a_0)^{12}C$ | P.Tischhauser+(2009), Physical Review, Part C, Nuclear Physics, Vol.79, p.055803 | 4toR33 from EXFOR C1461014 |
| 731 | 20.12.21 | $^{7}Li(p,pg_{1\text{-}0})^{7}Li$ | J. Raisanen+, Nucl. Instr. and Meth. 28 (1987) 199 | Duplication deleted li7ppgh |
| 730 | 20.12.21 | $^{7}Li(p,pg_{1\text{-}0})^{7}Li$ | S.Taskaev+(2021), Nucl. Instrum. Methods in Physics Res., Sect.B, Vol.502, p.85 | Yield data compiled Units in cross section data corrected. |
| 729 | 20.12.21 | $^{107}Ag(p,pg_{3\text{-}0})^{107}Ag$<br>$^{107}Ag(p,pg_{4\text{-}0})^{107}Ag$<br>$^{109}Ag(p,pg_{3\text{-}0})^{109}Ag$<br>$^{109}Ag(p,pg_{4\text{-}0})^{109}Ag$ | T.Tajvidi+(2021), Nucl. Instrum. Methods in Physics Res., Sect.B, Vol.502, p.18 | Data replaced with data from author X4Number added |
| 728 | 20.12.21 | $^{12}C(^{3}He,p_0)^{14}N$<br>$^{12}C(^{3}He,p_1)^{14}N$ | G.Provatas+(2021), Nucl. Instrum. Methods in Physics Res., Sect.B, Vol.500-501, p.57 | X4Number added |
| 727 | 27.08.21 | $^{7}Li(p,pg_{1\text{-}0})^{7}Li$ | S.Taskaev+(2021), Nucl. Instrum. Methods in Physics Res., Sect.B, Vol.502, p.85 | Compiled from the publication<br>V. Semkova, N. Otsuka |
| 726 | 27.08.21 | $^{7}Li(p,pg_{1\text{-}0})^{7}Li$ | R.Mateus+(2002),Nucl. Instrum. Methods in Physics Res., Sect.B, Vol.190, p.117 | Source corrected X4Number added<br>V. Semkova, N. Otsuka |
| 725 | 27.08.21 | $^{13}C(^{3}He,ag_{8\text{-}0})^{12}C$<br>$^{13}C(^{3}He,ag_{8\text{-}0})^{12}C$ | K.J.Moon+(1978), Journal of the Korean Physical Society, Vol.11, Issue.1, p.6 | x4toR33 from EXFOR C2034<br>V. Semkova, N. Otsuka |
| 724 | 27.08.21 | $^{11}B(^{3}He,p_0)^{13}C$ | B.B.Marsh+(1963), Physical Review, Vol.130, p.2373 | x4toR33 from EXFOR F0282005<br>V. Semkova, N. Otsuka |
| 723 | 27.08.21 | $^{7}Li(^{3}He,d_0)^{9}Be$ | I.I.Bondouk+(1977), Atomkernenergie, Vol.29, p.270 | x4toR33 from EXFOR A1542002<br>V. Semkova, N. Otsuka |
| 722 | 27.08.21 | $^{7}Li(^{3}He,p_0)^{9}Be$<br>$^{7}Li(^{3}He,p_2)^{9}Be$ | I.I.Bondouk+(1975), Revue Roumaine de Physique, Vol.20, p.1095 | x4toR33 from EXFOR A1544002<br>V. Semkova, N. Otsuka |
| 721 | 27.08.21 | $^{14}N(d,a_0)^{12}C$<br>$^{14}N(d,a_1)^{12}C$<br>$^{14}N(d,p_0)^{15}N$ | M.Kokkoris +(2019), Jour. Nucl. Instrum. Methods in Physics Res., Sect.B, Vol.450, p.31 | Source corrected X4Number added<br>V. Semkova, N. Otsuka |
| 720 | 27.08.21 | $^{24}Mg(d,p_0)^{25}Mg$<br>$^{24}Mg(d,p_2)^{25}Mg$ | N. Patronis+(2014), Jour. Nucl. Instrum. Methods in Physics Res., Sect.B, Vol.337, p.97 corrected to N. Patronis+(2014), unpublished | Source corrected<br>V. Semkova, N. Otsuka |
| 719 | 27.08.21 | $^{15}N(p,ag_{1\text{-}0})^{12}C$ | G.Imbriani+(2012),Jour. Physical Review, Part C, Nuclear Physics, Vol.85, p.065810 | Corrected as per C1936009<br>V. Semkova, N. Otsuka |
| 718 | 27.08.21 | $^{11}B(a,a_0)^{11}B$ | W.R.Ott+(1972), Jour. Nuclear Physics, Section A, Vol.198, p.505 | Source and LAB angle corrected in rr file<br>V. Semkova, N. Otsuka |
| 717 | 27.08.21 | | E.Ntemou+(2019), Nucl. Instrum. Methods in Physics Res., Sect.B, Vol.459, p.90<br>E.Ntemou+(2017), Nucl. Instrum. Methods in Physics Res., Sect.B, Vol.407, p.34 | X4Number added<br>V. Semkova, N. Otsuka |
| 716 | 27.08.21 | | A.Anttila+, J. Radioanal. Chem. 62 (1981) 293<br>A.Anttila et al., J. Radioanal. Chem. 62 (1981) 293<br>Barnhard, A.C.L. and Kim, C.C. Nucl. Phys. 28 (1961) 428.<br>H.-S. Cheng, H. Shen, J. Tang, F. Yang, Acta Phys. Sinica 43 (1994) 1569<br>M. Chiari et al., Nucl. Instr. and Meth. B 366 (2016) 77-82<br>M.J. Kenny, Aust. J. Phys. 34 (1981) 35<br>S. Harrisopulos et al., Eur. Phys. J. A 9 (2000) 479<br>C. Chronidou et al., Eur. Phys. J. A 6 (1999) 303<br>E. Ntemou, NIMB Vol. 459 (2019) p. 90<br>E. Ntemou et al., Nucl. Instr. Meth. B407(2017)34<br>W.N. Lennard et al. Nucl. Instr. & Meth. B43 (1989) 187<br>Leung, M.K. Ph.D. dissertation, Univ. of | Source corrected<br>V. Semkova, N. Otsuka |

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| | | | Kentucky (1972)<br>J.B.Marion and G.Weber Phys.Rev. v.102 (1956) 1355 | |
| 715 | 27.08.21 | $^{12}C(^{3}He,p_{0})^{14}N$<br>$^{12}C(^{3}He,p_{1})^{14}N$ | G.Provatas+(2021), Nucl. Instrum. Methods in Physics Res., Sect.B, Vol.500-501, p.57 | R33 files received from G.Provatas<br>V. Semkova, N. Otsuka |
| 714 | 27.08.21 | $^{10}B(^{3}He,p_{0})^{12}C$<br>$^{10}B(^{3}He,p_{1})^{12}C$<br>$^{10}B(^{3}He,d_{0})^{11}C$<br>$^{10}B(^{3}He,d_{1})^{11}C$<br>$^{10}B(^{3}He,d_{2})^{11}C$<br>$^{10}B(^{3}He,d_{3})^{11}C$<br>$^{10}B(^{3}He,a_{0})^{9}B$<br>$^{10}B(^{3}He,a_{2})^{9}B$ | R.A.I.Bell+(1972), Nuclear Physics, Section A, Vol.193, p.385 | x4toR33 from EXFOR F0473<br>V. Semkova, N. Otsuka |
| 713 | 27.08.21 | $^{9}Be(^{3}He,g_{1-0})^{12}C$ | H.D.Shay+(1974), Physical Review, Part C, Nuclear Physics, Vol.9, p.76 | x4toR33 from EXFOR C1201002<br>V. Semkova, N. Otsuka |
| 712 | 27.08.21 | $^{109}Ag(p,pg_{4-0})^{109}Ag$ | M.J.Kenny+, Nucl. Instr. and Meth. 168 (1980) 115 | Egamma corrected<br>V. Semkova, N. Otsuka |
| 711 | 27.08.21 | $^{13}C(^{3}He,p_{0})^{15}N$ | J.P.Schiffer+(1956), Physical Review, Vol.104, p.1064 | x4toR33 from EXFOR A1495010<br>V. Semkova, N. Otsuka |
| 710 | 27.08.21 | $^{7}Li(^{3}He,a_{0})^{6}Li$<br>$^{7}Li(^{3}He,a_{1})^{6}Li$<br>$^{7}Li(^{3}He,a_{2})^{6}Li$ | P.D.Forsyth+(1965), Nuclear Physics, Vol.67, Issue.3, p.517 | x4toR33 from EXFOR A1486<br>V. Semkova, N. Otsuka |
| 709 | 27.08.21 | $^{7}Li(^{3}He,a_{2})^{6}Li$ | H.Orihara+(1970), Journal of the Physical Society of Japan, Vol.29, Issue.3, p.533 | x4toR33 from EXFOR A1492002<br>V. Semkova, N. Otsuka |
| 708 | 27.08.21 | $^{7}Li(^{3}He,p_{0})^{9}Be$<br>$^{7}Li(^{3}He,p_{2})^{9}Be$ | D.P.Rath+(1990), Nuclear Physics, Section A, Vol.515, p.338 | x4toR33 from EXFOR C0139002<br>V. Semkova, N. Otsuka |
| 707 | 27.08.21 | $^{107}Ag(p,pg_{3-0})^{107}Ag$<br>$^{107}Ag(p,pg_{4-0})^{107}Ag$<br>$^{109}Ag(p,pg_{3-0})^{109}Ag$<br>$^{109}Ag(p,pg_{4-0})^{109}Ag$ | T.Tajvidi+(2021), Nucl. Instrum. Methods in Physics Res., Sect.B, Vol.502, p.18 | Compiled from the publication<br>V. Semkova, N. Otsuka |
| 706 | 27.08.21 | $^{7}Li(^{3}He,p_{0})^{9}Be$<br>$^{7}Li(^{3}He,p_{1})^{9}Be$<br>$^{7}Li(^{3}He,p_{2})^{9}Be$<br>$^{7}Li(^{3}He,p_{4})^{9}Be$ | Y.C.Lru(1972), Chinese Journal of Physics (Taiwan), Vol.10, p.76 | x4toR33 from EXFOR O1035002<br>V. Semkova, N. Otsuka |
| 705 | 27.08.21 | $^{16}O(^{3}He,a_{0})^{15}O$ | E.A.Silverstein+(1961), Physical Review, Vol.124, p.868 | x4toR33 from EXFOR C1008006<br>V. Semkova, N. Otsuka |
| 704 | 27.08.21 | $^{9}Be(^{3}He,p_{0})^{11}B$<br>$^{9}Be(^{3}He,p_{1})^{11}B$<br>$^{9}Be(^{3}He,p_{2})^{11}B$<br>$^{9}Be(^{3}He,p_{3})^{11}B$<br>$^{9}Be(^{3}He,p_{5})^{11}B$<br>$^{9}Be(^{3}He,p_{6})^{11}B$ | G.Provatas+(2020), Nucl. Instrum. Methods in Physics Res., Sect.B, Vol.472, p.36 | x4toR33 from EXFOR D0977002<br>V. Semkova, N. Otsuka |
| 703 | 07.07.21 | $^{27}Al(d,d)^{27}Al$ | M. Salimi+(2021) | R33 file removed<br>M. Salimi request |
| 702 | 10.06.21 | $^{27}Al(d,a_{0})^{25}Mg$<br>$^{27}Al(d,a_{1})^{25}Mg$<br>$^{27}Al(d,a_{2})^{25}Mg$<br>$^{27}Al(d,a_{3})^{25}Mg$<br>$^{27}Al(d,a_{4})^{25}Mg$<br>$^{27}Al(d,d)^{27}Al$<br>$^{27}Al(d,p_{0+1})^{28}Al$<br>$^{27}Al(d,p_{2+3})^{28}Al$ | M. Salimi+(2021), Submitted Scientific Reports, Nature | R33 files received from I. Vickridge |

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| | | $^{27}$Al(d,p$_4$)$^{28}$Al<br>$^{27}$Al(d,p$_{5+6}$)$^{28}$Al<br>$^{27}$Al(d,p$_9$)$^{28}$Al<br>$^{27}$Al(d,p$_{10}$)$^{28}$Al<br>$^{27}$Al(d,p$_{11}$)$^{28}$Al<br>$^{27}$Al(d,p$_{12}$)$^{28}$Al | | |
| 701 | 14.05.21 | $^{18}$O(p,p$_0$)$^{18}$O | K.Yagi+(1962), Journal of the Physical Society of Japan, Vol.17, Issue.4, p.595 | EXFOR: E1976<br>V. Semkova, N. Otsuka |
| 700 | 14.05.21 | $^{25}$Mg(p,pg$_{1-0}$)$^{25}$Mg<br>$^{25}$Mg(p,pg$_{2-0}$)$^{25}$Mg<br>$^{25}$Mg(p,pg$_{2-1}$)$^{25}$Mg | K.Preketes-Sigalas+(2016), Nucl. Instrum. Methods in Physics Res., Sect.B, Vol.386, p.4 | X4Number added<br>total syst. unc. added<br>V. Semkova, N. Otsuka |
| 699 | 14.05.21 | $^{23}$Na(p,pg$_{1-0}$)$^{23}$Na<br>$^{23}$Na(p,pg$_{2-1}$)$^{23}$Na | M. Chiari et al., Nucl. Instr. and Meth. B 441 (2019) 108-118 | X4Number added<br>V. Semkova, N. Otsuka |
| 698 | 14.05.21 | $^{11}$B(p,a$_0$)$^{8}$Be | M.Munch+(2020), European Physical Journal A: Hadrons and Nuclei, Vol.56, p.17 | EXFOR: O2472002<br>V. Semkova, N. Otsuka |
| 697 | 14.05.21 | $^{10}$B(a,pg$_{3-0}$)$^{13}$C | T.W.Bonner+(1956), Physical Review, Vol.102, p.1348 | EXFOR: F0464<br>V. Semkova, N. Otsuka |
| 696 | 14.05.21 | $^{10}$B(a,pg$_{1-0}$)$^{13}$C<br>$^{10}$B(a,pg$_{2-0}$)$^{13}$C<br>$^{10}$B(a,pg$_{3-0}$)$^{13}$C | Q.Liu+(2020), Physical Review, Part C, Nuclear Physics, Vol.101, p.025808 | EXFOR: C2511<br>V. Semkova, N. Otsuka |
| 695 | 14.05.21 | $^{16}$O(d,pg$_{1-0}$)$^{17}$O<br>$^{18}$O(d,ng$_{5-2}$)$^{19}$F<br>$^{18}$O(d,pg$_{2-1}$)$^{19}$O | A.Jokar+(2020), Nucl. Instrum. Methods in Physics Res., Sect.B, Vol.482, p.11 | R33 files received from A.Jokar<br>V. Semkova, N. Otsuka |
| 694 | 14.05.21 | $^{6}$Li($^{3}$He,p$_0$)$^{8}$Be<br>$^{7}$Li($^{3}$He,d$_0$)$^{8}$Be<br>$^{7}$Li($^{3}$He,p$_0$)$^{9}$Be<br>$^{7}$Li($^{3}$He,p$_1$)$^{9}$Be<br>$^{7}$Li($^{3}$He,p$_2$)$^{9}$Be<br>$^{7}$Li($^{3}$He,p$_{3+4}$)$^{9}$Be | J.Zhu+(2021), Nucl. Instrum. Methods in Physics Res., Sect.B, Vol.494-495, p.23 | R33 files received from J.Zhu<br>V. Semkova, N. Otsuka |
| 693 | 30.10.20 | $^{10}$B(p,ag$_{1-0}$)$^{7}$Be<br>$^{10}$B(p,pg$_{1-0}$)$^{10}$B<br>$^{11}$B(p,pg$_{1-0}$)$^{11}$B<br>$^{27}$Al(p,ag$_{1-0}$)$^{24}$Mg<br>$^{27}$Al(p,pg$_{1-0}$)$^{27}$Al<br>$^{23}$Na(p,pg$_{1-0}$)$^{23}$Na<br>$^{23}$Na(p,pg$_{2-1}$)$^{23}$Na<br>$^{14}$N(p,pg$_{1-0}$)$^{14}$N<br>$^{44}$Ca(p,g$_{4-0}$)$^{45}$Sc | M.Chiari+(2016), Jour. Nucl. Instrum. Methods in Physics Res., Sect.B, Vol.366, p.77 | "Source" corrected<br>Sample added under Comments<br>V. Semkova, N. Otsuka |
| 692 | 30.10.20 | $^{12}$C($^{3}$He,$^{3}$He$_0$)$^{12}$C | K. Etoh et al. Journal of the Physical Society of Japan 26 (1969) 1335 | Files deleted.<br>triton-induced data<br>V. Semkova, N. Otsuka |
| 691 | 30.10.20 | $^{12}$C(d,p$_0$)$^{12}$C | K. Etoh et al. Journal of the Physical Society of Japan 26 (1969) 1335 | Files deleted.<br>Triton-induced data<br>V. Semkova, N. Otsuka |
| 690 | 30.10.20 | $^{14}$N(p,p$_0$)$^{14}$N<br>$^{14}$N(p,p$_1$)$^{14}$N | M.T.McEllistrem+(1956), Physical Review, Vol.104, p.1008 | (d,p1) data at 167 deg replaced with (d,d0) data; file added from C0993007<br>V. Semkova, N. Otsuka |
| 689 | 30.10.20 | $^{14}$N(a,a$_0$)$^{14}$N | Ding Furong+(1992), Jour. Chinese Physics Letters, Vol.9, p.176 | Replaced with files from S0070002<br>V. Semkova, N. Otsuka |
| 688 | 30.10.20 | $^{14}$N(p,g)$^{15}$O | R. E. Pixley, PhD Thesis, 1957 | "X4Number"; Stat.unc.=4%<br>V. Semkova, N. Otsuka |
| 687 | 30.10.20 | $^{14}$N(a,a$_0$)$^{14}$N | E.A.Silverstein+(1961), Physical Review, Vol.124, p.868 | Replaced with files from C1008002 and C1008003 ERR-S given<br>V. Semkova, N. Otsuka |
| 686 | 30.10.20 | $^{14}$N(a,a$_0$)$^{14}$N | A.F.Gurbich+(2011), Nucl. Instrum. Methods in Physics Res., Sect.B, Vol.269, p.40 | Replaced with files from D0587002 ERR-S |

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| | | | | given<br>V. Semkova, N. Otsuka |
| 685 | 30.10.20 | $^{14}N(a,a_0)^{14}N$ | N.P.Heydenburg+(1953), Jour. Physical Review, Vol.92, p.89 | Replaced with files from F1036002 (ERR-S given)<br>V. Semkova, N. Otsuka |
| 684 | 30.10.20 | $^{14}N(a,p_0)^{17}O$ | N.P.Heydenburg+(1953), Jour. Physical Review, Vol.92, p.89 | "Source" corrected; "X4Number"; Stat.unc.=5%<br>V. Semkova, N. Otsuka |
| 683 | 30.10.20 | $^{16}O(p,p_0)^{16}O$ | R.Amirikas+(1993), Jour. Nucl. Instrum. Methods in Physics Res., Sect.B, Vol.77, p.110 | "mb" data added from D0107003<br>V. Semkova, N. Otsuka |
| 682 | 30.10.20 | $^{16}O(a,a_0)^{16}O$ | C.J.Wetteland+(1998), Rept. Los Alamos Scientific Lab. Reports, No.98-4867 | Replaced with files from C2103006, (ERR-S given)<br>V. Semkova, N. Otsuka |
| 681 | 30.10.20 | $^{16}O(a,a_0)^{16}O$ | Huan-Sheng Cheng et al. Acta Physica Sinica v.2 (1993) 641 | Deleted. Same data as NIM/B 83(1993)449 publication.<br>V. Semkova, N. Otsuka |
| 680 | 30.10.20 | $^{18}O(a,a_0)^{18}O$ | D.Powers+(1964), Physical Review, Vol.134, p.B1237 | Replaced with files from EXFOR (ERR-S given)<br>V. Semkova, N. Otsuka |
| 679 | 30.10.20 | $^{19}F(p,p_0)^{19}F$<br>$^{19}F(p,p_1)^{19}F$ | V.Paneta+(2014), Nucl. Instrum. Methods in Physics Res., Sect.B, Vol.319, p.34 | "Source" corrected<br>Stat. unc. = 5.3%<br>V. Semkova, N. Otsuka |
| 678 | 30.10.20 | $^{19}F(p,a_0)^{16}O$ | W.A.Ranken+(1958), Physical Review, Vol.109, p.1646 | New file added from C1006002<br>V. Semkova, N. Otsuka |
| 677 | 30.10.20 | $^{24}Mg(d,p_0)^{25}Mg$ | D.Bachiller Perea+(2017), Nucl. Instrum. Methods in Physics Res., Sect.B, Vol.406, p.161 | Angle 90->135<br>V. Semkova, N. Otsuka |
| 676 | 30.10.20 | $^{24}Mg(d,p_0)^{25}Mg$<br>$^{24}Mg(d,p_1)^{25}Mg$<br>$^{24}Mg(d,p_2)^{25}Mg$<br>$^{24}Mg(d,p_3)^{25}Mg$<br>$^{24}Mg(d,p_4)^{25}Mg$ | H.Rafi-Kheiri+(2016), Nucl. Instrum. Methods in Physics Res., Sect.B, Vol.373, p.63 | Syst. unc. added.<br>V. Semkova, N. Otsuka |
| 675 | 30.10.20 | $^{26}Mg(a,a_0)^{26}Mg$ | L.V.Namjoshi+(1976), Physical Review, Part C, Nuclear Physics, Vol.13, p.915 | New files added from D6019002<br>V. Semkova, N. Otsuka |
| 674 | 30.10.20 | $^{31}P(p,pg_{1\text{-}0})^{31}P$ | H.Silva+(2019), Nucl. Instrum. Methods in Physics Res., Sect.B, Vol.452, p.26 | New files added<br>data received from J.Cruz<br>V. Semkova, N. Otsuka |
| 673 | 30.10.20 | $^{nat}Mg(d,d_0)^{nat}Mg$ | N.Patronis+(2014), Nucl. Instrum. Methods in Physics Res., Sect.B, Vol.337, p.97 | Source corrected<br>"X4Number" added<br>Syst. unc. = 5%<br>Stat. Unc. = 1%<br>V. Semkova, N. Otsuka |
| 672 | 30.10.20 | $^{28}Si(p,p_0)^{28}Si$ | T.A.Belote+(1961), Physical Review, Vol.122, p.920 | "mb" data added from C2095<br>V. Semkova, N. Otsuka |
| 671 | 30.10.20 | $^{nat}Si(d,d_0)^{nat}Si$ | E.Ntemou+(2019), Nucl. Instrum. Methods in Physics Res., Sect.B, Vol.450, p.24 | "Source" corrected in "mb" files<br>"rr" data added from O2351002<br>V. Semkova, N. Otsuka |
| 670 | 30.10.20 | $^{28}Si(a,a_0)^{28}Si$ | M.K.Leung(1972), Thes. Thesis or dissertation, | "mb" data added from C2100010<br>V. Semkova, N. Otsuka |
| 669 | 30.10.20 | $^{35}Cl(p,a_0)^{32}S$ | R.L.Clarke+(1960), Nuclear Physics, Vol.14, Issue.3, p.472 | File added from C2104006<br>V. Semkova, N. Otsuka |
| 668 | 30.10.20 | $^{nat}Ti(p,p_0)^{nat}Ti$ | E.Rauhala(1989), Nucl. Instrum. Methods in Physics Res., Sect.B, Vol.40/41, p.790 | "mb" data added from O0880003<br>V. Semkova, N. Otsuka |
| 667 | 30.10.20 | $^{nat}Cu(p,p_0)^{nat}Cu$ | A.Nurmela+(1998), Journal of Applied Physics, Vol.84, p.1796 | Files added from A0649002<br>recom. Prof. C. Jeynes<br>V. Semkova, N. Otsuka |
| 666 | 30.10.20 | $^{2}H(^{3}He,p_0)^{4}He$ | B.Wielunska+(2016), Nucl. Instrum. Methods in Physics Res., Sect.B, Vol.371, p.41 | Replaced with file from O1258002<br>new files added |

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| | | | | V. Semkova, N. Otsuka |
| 665 | 30.10.20 | $^{56}Fe(p,p_0)^{56}Fe$ | H.Brandle+(1970), Nuclear Physics, Section A, Vol.151, p.211 | File added<br>from D0130003<br>V. Semkova, N. Otsuka |
| 664 | 30.10.20 | $^{9}Be(p,a_0)^{6}Li$<br>$^{9}Be(p,d_0)^{8}Be$<br>$^{9}Be(p,p_0)^{9}Be$ | S.Krat+(2015), Nucl. Instrum. Methods in Physics Res., Sect.B, Vol.358, p.72 | Replaced with<br>files from O2268<br>V. Semkova, N. Otsuka |
| 663 | 30.10.20 | $^{15}N(a,a_0)^{15}N$ | H.Smotrich+(1961), Physical Review, Vol.122, p.232 | "mb" data added<br>from C0755002<br>V. Semkova, N. Otsuka |
| 662 | 30.10.20 | $^{15}N(a,a_0)^{15}N$ | Mo and Weller, Nucl. Phys. A198 (1972) 153 | Deleted: exp. data from H.Smotrich et al. PR122(1961)232<br>V. Semkova, N. Otsuka |
| 661 | 30.10.20 | $^{26}Mg(p,g_{1-0})^{27}Al$ | M.J.Kenny+, Nucl. Instr. and Meth. 168 (1980) 115 | "mb"->"yield"<br>V. Semkova, N. Otsuka |
| 660 | 30.10.20 | $^{10}B(a,p_0)^{13}C$ | H.Chen+(2003), Nucl. Instrum. Methods in Physics Res., Sect.B, Vol.211, p.1 | "Source" corrected<br>"X4Number" added<br>Syst. unc. = 2%<br>Stat. Unc. = 1.6%<br>V. Semkova, N. Otsuka |
| 659 | 30.10.20 | | 81 R33 files | "X4Number" added<br>V. Semkova, N. Otsuka |
| 658 | 30.10.20 | | 111 R33 files | "Source" corrected<br>"X4Number" added<br>V. Semkova, N. Otsuka |
| 657 | 30.10.20 | | 122 R33 files | "Source" corrected<br>V. Semkova, N. Otsuka |
| 656 | 16.07.20 | $^{34}S(p,g_{1-0})^{35}Cl$<br>$^{41}K(p,g_{1-0})^{42}Ca$<br>$^{41}K(p,g_{2-1})^{42}Ca$<br>$^{41}K(p,g_{3-1})^{42}Ca$<br>$^{41}K(p,pg_{1-0})^{41}K$<br>$^{41}K(p,ag_{1-0})^{38}Ar$<br>$^{43}Ca(p,pg_{1-0})^{43}Ca$<br>$^{44}Ca(p,g_{3-0})^{45}Sc$<br>$^{44}Ca(p,g_{4-0})^{45}Sc$ | A.Anttila+, J. Radioanal. Chem. 62 (1981) 293 | Files k1pge, k1pgd, k1pgc, k1ppgc, k1pagb, ca3ppgb, ca4pge,ca4pgc deleted due to duplications.<br>V. Semkova |
| 655 | 16.07.20 | $^{34}S(p,g_{1-0})^{35}Cl$<br>$^{34}S(p,pg_{1-0})^{34}S$<br>$^{41}K(p,ng_{1-0})^{41}Ca$<br>$^{41}K(p,ng_{2-0})^{41}Ca$<br>$^{41}K(p,ng_{4-0})^{41}Ca$<br>$^{41}K(p,pg_{1-0})^{41}K$<br>$^{41}K(p,ag_{1-0})^{38}Ar$ | A.Z. Kiss+, J. Radioanal. Nucl. Chem. 89 (1985) 123 | Files s4pga, s4ppga, k1pnga, k1pngb, k1pngc, k1pngd, k1ppga, k1paga deleted due to duplications.<br>V. Semkova |
| 654 | 16.07.20 | $^{31}P(p,pg_{1-0})^{31}P$ | A. Savidou et al., Nucl. Instr. and Meth. B152(1999)12 | Duplication deleted<br>V. Semkova |
| 653 | 25.04.20 | $^{27}Al(d,pg_{10-0})^{28}Al$<br>$^{27}Al(d,ng_{2-1})^{28}Si$ | Z.Elekes+,J,NIM/B,168,305,2000 | Deleted<br>EXFOR: O0875024<br>V. Semkova, N. Otsuka |
| 652 | 25.04.20 | $^{14}N(d,p_5)^{15}N$ | N.D.Carstanjen,J,NIM/B,51,12,1990 | Original data in G.Amsel+,J,RPA,4,383,1969<br>V. Semkova, N. Otsuka |
| 651 | 25.04.20 | $^{14}N(d,pg_{1-0})^{15}N$<br>$^{24}Mg(d,ng_{1-0})^{27}Al$<br>$^{32}S(d,ng_{1-0})^{33}Cl$<br>$^{32}S(d,pg_{1-0})^{33}S$ | A.Z.Kiss+,J,NIM/B,85,118,1994 | Data discarded by author<br>V. Semkova, N. Otsuka |
| 650 | 25.04.20 | $^{10}B(p,ag_{1-0})^{7}Be$ | A.Caciolli+,J,EPJ/A,55,171,2019 | Data sent by A.Caciolli<br>V. Semkova, N. Otsuka |
| 649 | 25.04.20 | $^{7}Li(d,d_0)^{7}Li$ | K.Preketes-Sigalas+,J,NIM/B,414,99,2018 | Reference added<br>X4Number added<br>V. Semkova, N. Otsuka |

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| 648 | 25.04.20 | $^{10}$B(p,ag$_{1-0}$)$^{7}$Be<br>$^{10}$B(p,pg$_{1-0}$)$^{10}$B<br>$^{11}$B(p,pg$_{1-0}$)$^{11}$B<br>$^{14}$N(p,pg$_{1-0}$)$^{14}$N<br>$^{23}$Na(p,pg$_{1-0}$)$^{23}$Na<br>$^{27}$Al(p,pg$_{1-0}$)$^{27}$Al<br>$^{27}$Al(p,pg$_{2-0}$)$^{27}$Al<br>$^{27}$Al(p,ag$_{1-0}$)$^{24}$Mg | M.Chiari+,J,NIM/B,366,77,2016 | X4Number added<br>V. Semkova, N. Otsuka |
| 647 | 25.04.20 | $^{11}$B(p,pg$_{1-0}$)$^{11}$B | K.Preketes-Sigalas+,J,NIM/B,368,71,2016 | X4Number added<br>V. Semkova, N. Otsuka |
| 646 | 25.04.20 | $^{10}$B(p,ag$_{1-0}$)$^{7}$Be<br>$^{14}$N(p,pg$_{1-0}$)$^{14}$N<br>$^{28}$Si(p,pg$_{1-0}$)$^{28}$Si<br>$^{29}$Si(p,pg$_{1-0}$)$^{29}$Si | B.Marchand+,J,NIM/B,378,25,2016 | mb->yield<br>X4Number added<br>V. Semkova, N. Otsuka |
| 645 | 25.04.20 | $^{12}$C(p,p$_{0}$)$^{12}$C<br>$^{12}$c(p,p$_{1}$)$^{12}$C | D.Abriola+,J,NIM/B,269,2011,2011 | Replaced with file from EXFOR<br>V. Semkova, N. Otsuka |
| 644 | 25.04.20 | $^{11}$B(p,g$_{0}$)$^{12}$C<br>$^{11}$B(p,g$_{1}$)$^{12}$C | T.Huus+,J,PR,91,599,1953 | Replaced with file from EXFOR<br>V. Semkova, N. Otsuka |
| 643 | 25.04.20 | $^{10}$ B(p,a$_{0}$)$^{7}$Be<br>$^{10}$ B(p,a$_{1}$)$^{7}$Be | J.W.Cronin,J,PR,101,298,1956 | Replaced with files from EXFOR<br>ANG-CM->ANG<br>V. Semkova, N. Otsuka |
| 642 | 25.04.20 | $^{14}$N(p,pg$_{1-0}$)$^{14}$N<br>$^{nat}$N(p,p$_{0}$)$^{nat}$N<br>$^{28}$Si(p,pg$_{1-0}$)$^{28}$Si<br>$^{28}$Si(p,p$_{1}$)$^{28}$Si<br>$^{29}$Si(p,pg$_{1-0}$)$^{29}$Si | L.Csedreki+,J,NIM/B,443,48,2019 | Sigfactors added<br>V. Semkova, N. Otsuka |
| 641 | 25.04.20 | $^{14}$N(d,pg$_{5-0}$)$^{15}$N<br>$^{14}$N(d,pg$_{7-0}$)$^{15}$N<br>$^{14}$N(d,pg$_{4-1}$)$^{15}$N<br>$^{14}$N(d,pg$_{6-1}$)$^{15}$N | A.Jokar+,J,NIM/B,431,25,2018 | X4Number added<br>V. Semkova, N. Otsuka |
| 640 | 25.04.20 | $^{19}$F(p,ag$_{1-0}$)$^{16}$O<br>$^{19}$F(p,ag$_{2-0}$)$^{16}$O<br>$^{19}$F(p,ag$_{3-0}$)$^{16}$O<br>$^{19}$F(p,ag$_{4-0}$)$^{16}$O<br>$^{19}$F(p,ag$_{2-0+3-0+4-0}$)$^{16}$O | M.J.Kenny,J,AUJ,34,35,1981 | Sigfactors corrected<br>V. Semkova, N. Otsuka |
| 639 | 25.04.20 | $^{10}$B(p,ag$_{1-0}$)$^{7}$Be | A.B.Brown+,J,PR,82,159,1951 | EXFOR: A1457<br>V. Semkova, N. Otsuka |
| 638 | 9.04.20 | $^{48}$ Ti(p,p$_{0}$)$^{48}$Ti | Prochnow+, Nucl. Phys. A 194 (1972) p.353 | Added more angles from EXFOR<br>A. Lagoyannis |
| 637 | 9.04.20 | $^{18}$ O(p,a$_{0}$)$^{15}$N | Christensen, NIMB51 (1990) p. 97 | Replaced with files from EXFOR<br>A. Lagoyannis |
| 636 | 9.04.20 | $^{16}$ O(a,a$_{0}$)$^{16}$O | Cameron, Phys. Rev. 90 (1953) p. 839 | Replaced with files from EXFOR<br>A. Lagoyannis |
| 635 | 9.04.20 | $^{16}$ O(a,a$_{0}$)$^{16}$O | Jarjis+, NIMB12 (1985) p. 332 | Replaced with file from EXFOR<br>A. Lagoyannis |
| 634 | 9.04.20 | $^{16}$ O(a,a$_{0}$)$^{16}$O | Davies+, NIMB85 (1994) p. 28 | Added error in cross section<br>A. Lagoyannis |
| 633 | 9.04.20 | $^{16}$ O($^{3}$He,a$_{0}$)$^{15}$O | Alford+, Nuclear Physics 61 (1965) p. 368 | Added error in energy<br>A. Lagoyannis |
| 632 | 9.04.20 | $^{16}$ O($^{3}$He,a$_{0}$)$^{15}$O | Abel+, NIMB45 (1990) p. 100 | Replaced with file from EXFOR<br>A. Lagoyannis |
| 631 | 9.04.20 | $^{16}$ O(d,d$_{0}$)$^{16}$O | Amsel+, Ann. Phys. 9 (1964) p. 297 | New angle added |

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| | | | | A. Lagoyannis |
| 630 | 9.04.20 | $^{16}$O(d,a$_0$)$^{14}$N | Debras+, J. Anal. Chem. 38 (1977) p. 193 | Added error in cross section<br>A. Lagoyannis |
| 629 | 9.04.20 | $^{16}$O(d,a$_0$)$^{14}$N | Picraux, NIM 149 (1978) p.292 | Replaced with file from EXFOR<br>A. Lagoyannis |
| 628 | 9.04.20 | $^{16}$O(d,a$_0$)$^{14}$N | Dietzsch+(1968), Jour. Nuclear Physics, Section A, Vol.114, p.330 | Replaced with file from EXFOR<br>A. Lagoyannis |
| 627 | 9.04.20 | $^{16}$O(d,p$_1$)$^{17}$O<br>$^{16}$O(d,a$_0$)$^{14}$N | Turos Phys.Stat.Sol.(a) 16 (1973) p. 213 | Replaced with file from EXFOR<br>A. Lagoyannis |
| 626 | 9.04.20 | $^{16}$O(d,p$_1$)$^{17}$O | Karabash +,Atomnoy Nauky i Technicky, Yadernie Constanty No.3 (1988) p. 31 (in Russian) | Added error in cross section<br>A. Lagoyannis |
| 625 | 9.04.20 | $^{16}$O(d,p$_1$)$^{17}$O<br>$^{16}$O(d,a$_0$)$^{14}$N | Jiang+, NIMB207 (2003) p. 453 | Added error in cross section<br>A. Lagoyannis |
| 624 | 9.04.20 | $^{16}$O(d,p$_0$)$^{17}$O<br>$^{16}$O(d,p$_1$)$^{17}$O | Debras+, J. Rad. Chem. 38 (1977) p. 193 | Replaced with file from EXFOR<br>A. Lagoyannis |
| 623 | 9.04.20 | $^{16}$O(d,p$_0$)$^{17}$O<br>$^{16}$O(d,p$_1$)$^{17}$O | Debras+, J. Rad. Chem. 38 (1977) p. 193 | Replaced with file from EXFOR<br>A. Lagoyannis |
| 622 | 9.04.20 | $^{16}$O(d,p$_0$)$^{17}$O | Berti+, NIM 201 (1982) p. 476 | Replaced with file from EXFOR<br>A. Lagoyannis |
| 621 | 9.04.20 | $^{16}$O(d,d$_0$)$^{16}$O<br>$^{16}$O(d,p$_0$)$^{17}$O<br>$^{16}$O(d,p$_1$)$^{17}$O<br>$^{16}$O(d,p$_3$)$^{17}$O | Davison+, Canadian Journal of Physics v.48 (1970) p. 2235 | Added error in cross section<br>A. Lagoyannis |
| 620 | 9.04.20 | $^{16}$O(p,p$_0$)$^{16}$O | Ramos+, NIMB190 (2002) p. 95 | Added new angle from EXFOR<br>A. Lagoyannis |
| 619 | 9.04.20 | $^{16}$O(p,p$_0$)$^{16}$O | Braun+, Zeitschrift fur Physik A 311 (1983) p. 173 | Replaced with file from EXFOR<br>A. Lagoyannis |
| 618 | 9.04.20 | $^{16}$O(p,p$_0$)$^{16}$O | Amirikas+, NIMB77 (1993) p. 110 | Replaced with file from EXFOR.<br>Added new angles<br>A. Lagoyannis |
| 617 | 9.04.20 | $^{16}$O(p,p$_0$)$^{16}$O | Yang Guohua+, NIMB61 (1991) p. 175 | Added error in cross section<br>A. Lagoyannis |
| 616 | 9.04.20 | $^{15}$N(d,a$_0$)$^{13}$C | Vickridge+, NIMB108 (1996) pp. 367-370 | Added error in energy<br>A. Lagoyannis |
| 615 | 9.04.20 | $^{15}$N(p,a$_0$)$^{12}$C | Hagedorn+, Phys. Rev. 108 (1957) p. 1015 | Replaced with file from EXFOR<br>A. Lagoyannis |
| 614 | 9.04.20 | $^{14}$N(a,p$_0$)$^{17}$O | Wei+, NIMB 249(2006) p. 85 | Added Energies in the file<br>A. Lagoyannis |
| 613 | 9.04.20 | $^{14}$N(a,p$_0$)$^{17}$O | Giorginis+, NIMB133 (1996) p. 396 | Error in energy added<br>A. Lagoyannis |
| 612 | 9.04.20 | $^{14}$N(a,a$_0$)$^{14}$N | Heydenburg+, Phys. Rev. 92 (1953) p. 89 | Error in cross section added<br>A. Lagoyannis |
| 611 | 9.04.20 | $^{14}$N(a,a$_0$)$^{14}$N<br>$^{14}$N(a,p$_0$)$^{17}$O $^{14}$N(a,p$_1$)$^{17}$O | Terwagne+, J. Appl. Phys. 104 (2008) p. 084909 | Order of energies fixed<br>A. Lagoyannis |
| 610 | 9.04.20 | $^{14}$N(a,a$_0$)$^{14}$N | Artigalas+, NIMB66 (1992) p. 237 | Replaced previous file from EXFOR<br>A. Lagoyannis |
| 609 | 9.04.20 | $^{14}$N(a,a$_0$)$^{14}$N | Qiu+, NIMB71 (1992) p. 324 | Replaced previous file from EXFOR<br>A. Lagoyannis |
| 608 | 9.04.20 | $^{14}$N(d,a$_2$)$^{12}$C | Koenig+, Il Nuovo Cimento A, Vol.39 (1977) p.9 | Added new reactions from EXFOR |

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|---|---|---|---|---|
| | | $^{14}$N(d,a$_3$)$^{12}$C | | A. Lagoyannis |
| 607 | 9.04.20 | $^{14}$N(d,a$_0$)$^{12}$C<br>$^{14}$N(d,a$_1$)$^{12}$C | Gomes Porto+, Nucl. Phys. A 136 (1969) p.385 | Added files with more angles from EXFOR<br>A. Lagoyannis |
| 606 | 9.04.20 | $^{14}$N(d,d$_0$)$^{14}$N | Seiler+, Phys. Rev. 136 (1964) p. 994 | Added files with more angles from EXFOR<br>A. Lagoyannis |
| 605 | 9.04.20 | $^{14}$N(d,a$_0$)$^{12}$C<br>$^{14}$N(d,a$_1$)$^{12}$C | Kim+, Nuclear Data Tables v.A3 (1967) p. 123 | Compilation of existing data.<br>Reference replaced<br>A. Lagoyannis |
| 604 | 9.04.20 | $^{14}$N(d,a$_0$)$^{12}$C<br>$^{14}$N(d,a$_1$)$^{12}$C | Pellegrino+, NIM B 219-220 (2004) p. 140 | Error bars added<br>A. Lagoyannis |
| 603 | 9.04.20 | $^{14}$N(d,p$_5$)$^{15}$N | Berty+, NIM 201 (1982) p. 476 | Replaced previous file from EXFOR<br>A. Lagoyannis |
| 602 | 9.04.20 | $^{14}$N(d,p$_5$)$^{15}$N | Niiler+, NIM 168 (1980) p. 107 | Replaced previous file from EXFOR<br>A. Lagoyannis |
| 601 | 9.04.20 | $^{14}$N(d,p$_4$)$^{15}$N<br>$^{14}$N(d,p$_5$)$^{15}$N<br>$^{14}$N(d,p$_6$)$^{15}$N<br>$^{14}$N(d,p$_7$)$^{15}$N | Csedreki+, NIMB328 (2014) pp. 20-26 | Error bar in energy entered<br>A. Lagoyannis |
| 600 | 9.04.20 | $^{14}$N(d,p$_{1+2}$)$^{15}$N<br>$^{14}$N(d,p$_3$)$^{15}$N<br>$^{14}$N(d,p$_{4+5}$)$^{15}$N<br>$^{14}$N(d,p$_5$)$^{15}$N<br>$^{14}$N(d,a$_0$)$^{12}$C<br>$^{14}$N(d,a$_1$)$^{12}$C | Amsel+, Revue de Physique Appliquee 4 (1969) p. 383 | Replaced previous files from EXFOR<br>A. Lagoyannis |
| 599 | 9.04.20 | $^{14}$N(d,p$_0$)$^{15}$N | Pellegrino+, NIMB 219-220 (2004) p. 140 | Replaced previous file from EXFOR<br>A. Lagoyannis |
| 598 | 9.04.20 | $^{14}$N(d,p$_0$)$^{15}$N | Simpson+, Vacuum 34 (1984) p. 899 | Replaced previous file from EXFOR<br>A. Lagoyannis |
| 597 | 9.04.20 | $^{nat}$N(d,d$_0$)$^{nat}$N | Csedreki+, NIM B328 (2014) pp. 20-26 | Reaction corrected<br>A. Lagoyannis |
| 596 | 9.04.20 | $^{14}$N(d,p$_0$)$^{15}$N<br>$^{14}$N(d,a$_0$)$^{12}$C<br>$^{14}$N(d,a$_1$)$^{12}$C | Kokkoris+ (presented at the IBA-2017 conference) | Updated reference<br>A. Lagoyannis |
| 595 | 9.04.20 | $^{14}$N(p,p$_0$)$^{14}$N | Bolmgren+, Phys. Rev. 105 (1957) p. 210 | Replaced previous files from EXFOR<br>A. Lagoyannis |
| 594 | 9.04.20 | $^{14}$N(p,p$_0$)$^{14}$N | Havranek+, Czech.J.Phys. 41 (1991) No.10, p.921 | Error bar in energy corrected<br>A. Lagoyannis |
| 593 | 9.04.20 | $^{14}$N(p,p$_0$)$^{14}$N | Ramos+, NIM B190 (2002) p. 95 | Replaced previous files from EXFOR<br>A. Lagoyannis |
| 592 | 9.04.20 | $^{32}$S(d,p$_0$)$^{33}$S<br>$^{32}$S(d,p$_1$)$^{33}$S<br>$^{32}$S(d,p$_2$)$^{33}$S<br>$^{32}$S(d,p$_3$)$^{33}$S<br>$^{32}$S(d,p$_7$)$^{33}$S | Lagoyannis+, NIM B 266 (2008) p. 2259 | Updated reference<br>A. Lagoyannis |
| 591 | 9.04.20 | $^{nat}$Si(p,p$_0$)$^{nat}$Si | Salomonovic, NIM B82 (1993) p. 1 | Added files with more angles from EXFOR<br>A. Lagoyannis |
| 590 | 9.04.20 | $^{nat}$Si(p,p$_0$)$^{nat}$Si | Vorona+, Phys. Rev. 116 (1959) p. 1563 | Reaction corrected<br>A. Lagoyannis |
| 589 | 9.04.20 | $^{nat}$Si(p,p$_0$)$^{nat}$Si | Amirikas+, NIM B77 (1993) p. 110 | Replaced previous files from EXFOR<br>A. Lagoyannis |

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| 588 | 9.04.20 | $^{nat}Si(p,p_0)^{nat}Si$ | Ramos+, NIM B190 (2002) p.95 | Error bar in energy corrected<br>A. Lagoyannis |
| 587 | 9.04.20 | $^{28}Si(a,a_0)^{28}Si$ | Kallman+, Jour. Zeitschrift fuer Physik A, Hadrons and Nuclei 356 (1969) p.287 | Added files with more angles from EXFOR<br>A. Lagoyannis |
| 586 | 9.04.20 | $^{28}Si(a,a_0)^{28}Si$ | Coban+, Jour. Nucl. Phys. A678 (2000) p.3 | Replaced previous file from EXFOR<br>A. Lagoyannis |
| 585 | 9.04.20 | $^{27}Al(^{7}Li,^{7}Li_0)^{27}Al$ | Abriola+, Jour. NIM B268 (2010) p.1793 | Added files with more angles from EXFOR<br>A. Lagoyannis |
| 584 | 9.04.20 | $^{27}Al(^{7}Li,^{7}Li_0)^{27}Al$ | Raisanen+, NIM. B73 (1993) p. 439 | Replaced previous file from EXFOR<br>A. Lagoyannis |
| 583 | 9.04.20 | $^{27}Al(^{6}Li,^{6}Li_0)^{27}Al$ | Nurmela+, NIM B155 (1999) p. 211 | Replaced previous file from EXFOR<br>A. Lagoyannis |
| 582 | 9.04.20 | $^{27}Al(d,a_0)^{25}Mg$ | Pellegrino+, NIM B 266 (2008) p. 2268 | Replaced previous file with extended datafile from EXFOR<br>A. Lagoyannis |
| 581 | 9.04.20 | $^{27}Al(p,p_0)^{27}Al$ | Elliott+, Nucl. Phys. 84 (1966) p.209 | Added new files from EXFOR<br>A. Lagoyannis |
| 580 | 9.04.20 | $^{27}Al(p,p_0)^{27}Al$ | Chiari+, NIM B174 (2001) p. 259 | Added new files from EXFOR<br>A. Lagoyannis |
| 579 | 9.04.20 | $^{27}Al(p,p_0)^{27}Al$ | Shahzad+, Nucl. Sci. and Techn. 27 (2016) p. 33 | Error bar in energy added<br>A. Lagoyannis |
| 578 | 9.04.20 | $^{27}Al(p,p_0)^{27}Al$ | Rauhala+, NIM B40/41 (1989) p. 790 | Error bar in energy added<br>A. Lagoyannis |
| 577 | 9.04.20 | $^{27}Al(p,p_0)^{27}Al$ | Ramos+, NIM B190 (2002) p. 95 | Error bar in cross section added<br>A. Lagoyannis |
| 576 | 9.04.20 | $^{nat}Mg(d,d_0)^{nat}Mg$ | Patronis+, NIM B 337 (2014) pp. 97-101 | Reaction corrected<br>A. Lagoyannis |
| 575 | 9.04.20 | $^{nat}Mg(p,p_0)^{nat}Mg$ | Rauhala+, NIM B33 (1988) p. 628 | Error bar in energy added<br>A. Lagoyannis |
| 574 | 9.04.20 | $^{26}Mg(p,p_0)^{26}Mg$ | Romanovskij+, Izvestiya Akademii Nauk v.50 (1986) 129 | Corrected angles<br>A. Lagoyannis |
| 573 | 9.04.20 | $^{24}Mg(a,a_0)^{24}Mg$ | Cseh+, Nucl. Phys. A385 (1982) p. 43 | Replaced file from EXFOR<br>A. Lagoyannis |
| 572 | 9.04.20 | $^{24}Mg(a,a_0)^{24}Mg$ | Ikossi+, Nucl. Phys. A319 (1979) p. 109 | Corrected existing files<br>A. Lagoyannis |
| 571 | 9.04.20 | $^{24}Mg(d,p_0)^{25}Mg$<br>$^{24}Mg(d,p_1)^{25}Mg$<br>$^{24}Mg(d,p_2)^{25}Mg$<br>$^{24}Mg(d,p_3)^{25}Mg$<br>$^{24}Mg(d,p_4)^{25}Mg$ | Rafi-kheiri+, NIMB 373 (2016) pp. 63-69 | Corrected existing files<br>A. Lagoyannis |
| 570 | 9.04.20 | $^{24}Mg(p,p_1)^{24}Mg$ | Lazarev+, Izv. Rossiiskoi Akademii Nauk, Ser.Fiz., Vol.56, p.197 | Corrected existing files<br>A. Lagoyannis |
| 569 | 9.04.20 | $^{13}C(^{3}He,p_0)^{15}N$<br>$^{13}C(^{3}He,p_0)^{15}N$<br>$^{13}C(^{3}He,a_2)^{12}C$ | Illsley+, Phys. Rev. 107 (1957) p. 538<br>Weller+, Nucl. Phys. A122 (1968) p. 529 | Replaced with data from EXFOR<br>Corrected energy range<br>A. Lagoyannis |
| 568 | 9.04.20 | $^{13}C(d,a_0)^{11}B$ | Marsh+, Phys. Rev. 130 (1963) p. 2373<br>Marion+, Phys. Rev. 102 (1956) p. 1355 | Corrected data<br>Corrected reference<br>A. Lagoyannis |
| 567 | 9.04.20 | $^{12}C(a,a_0)^{12}C$ | Somatri+, Nucl. Instr. Meth. B113 (1996) p. 284<br>Wetteland+, LA-UR-98-4867 (1998) | Corrected energies<br>Included stat. errors<br>A. Lagoyannis |
| 566 | 9.04.20 | $^{12}C(a,a_0)^{12}C$ | Kovash+, Phys. Rev. C 31 (1985) p. 1065 | Corrected data at 156º and added two more angles<br>A. Lagoyannis |
| 565 | 9.04.20 | $^{12}C(^{3}He,_3He_0)^{12}C$<br>$^{12}C(^{3}He,p_3)^{14}N$ | Kuan+, Nucl. Phys. 51 (1964) p. 481 | New datafiles added<br>A. Lagoyannis |

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|---|---|---|---|---|
| 564 | 9.04.20 | $^{12}C(^{3}He,p_0)^{14}N$<br>$^{12}C(^{3}He,p_1)^{14}N$<br>$^{12}C(^{3}He,p_2)^{14}N$ | Bromley+, Phys.Rev. 105 (1957) p. 957 | Replaced previous files with EXFOR data<br>A. Lagoyannis |
| 563 | 9.04.20 | $^{12}C(^{3}He,p_0)^{14}N$<br>90° | Tong+, NIMB45 (1990) p. 91 | Multiple files removed<br>A. Lagoyannis |
| 562 | 9.04.20 | $^{12}C(^{3}He,{}_3He_0)^{12}C$<br>$^{12}C(^{3}He,a_0)^{11}C$ | Schwarz+, Nuclear Physics v.88 (1966) 539 | Corrected files<br>A. Lagoyannis |
| 561 | 9.04.20 | $^{12}C(^{3}He,p_2)^{14}N$<br>$^{12}C(^{3}He,p_3)^{14}N$<br>$^{12}C(^{3}He,a_0)^{11}C$ | Terwagne+, NIMB122 (1997)p. 1-7 | Replaced previous files with EXFOR data<br>A. Lagoyannis |
| 560 | 9.04.20 | $^{12}C(d,d_0)^{12}C$<br>$^{12}C(d,p_0)^{13}C$ | Kim+, Nuclear Data Tables v.A2 (1966) p. 353<br>Balin+, Preprint FEI-1341, Obninsk (1982) | Compilation of existing data<br>Multiple file<br>Files removed (A. Lagoyannis) |
| 559 | 9.04.20 | $^{12}C(d,p_0)^{13}C$ | McEllistrem+, Phys. Rev. 104 (1956) p. 1008<br>Huez+, NIMB105 (1972) p. 197 | Added new files from EXFOR<br>A. Lagoyannis |
| 558 | 9.04.20 | $^{12}C(d,p_0)^{13}C$ | Lambert+, Jour. Comptes Renus, Serie B, Physique, Vol. 262 (1966) p. 1459 | Corrected existing files and added more angles<br>A. Lagoyannis |
| 557 | 9.04.20 | $^{12}C(p,p_0)^{12}C$ | Liu+, Nucl. Phys. 74 (1993) p. 439 | Corrected reference<br>A. Lagoyannis |
| 556 | 9.04.20 | $^{12}C(p,p_0)^{12}C$ | Amirikas+, NIMB77 (1993) p. 110<br>Mazzoni+, NIMB136-138 (1998) p. 86 | Included energy errorbars in datafiles from EXFOR files<br>A. Lagoyannis |
| 555 | 9.04.20 | $^{12}C(p,p_0)^{12}C$ | Salomonevic+, NIMB82(1993) p. 1<br>Jackson+, Phys. Rev. 89 (1953) p. 365 | Replaced old files and added more angles from EXFOR<br>A. Lagoyannis |
| 554 | 9.04.20 | $^{23}Na(d,p_0)^{24}Na$<br>$^{23}Na(d,p_3)^{24}Na$ | Behay+, Nucl. Phys. 74 (1965) p. 225 | New datafiles added from EXFOR<br>A. Lagoyannis |
| 553 | 9.04.20 | $^{23}Na(p,p_0)^{23}Na$ | Dearnaley+, Philos. Mag. ser. 8, 1 (1956) p. 821 | Replace previous files with datafiles from EXFOR<br>A. Lagoyannis |
| 552 | 9.04.20 | $^{11}B(a,a_0)^{11}B$<br>130°, 140°, 150° | Ott+, Nucl. Phys. A (1972) p.505 | New datafiles from EXFOR<br>A. Lagoyannis |
| 551 | 9.04.20 | $^{11}B(p,p_0)^{11}B$ | Dejneko+, Izv. Rossiiskoi Akademii Nauk, Ser. Fiz., Vol. 38 (1974) p.1694 | Replace with new datafile from EXFOR<br>A. Lagoyannis |
| 550 | 9.04.20 | $^{11}B(p,p_1)^{11}B$ | Boni+, NIMB35 (1988) p. 80 | Previous entry #32 removed<br>Authors measure gammas not alphas<br>A. Lagoyannis |
| 549 | 9.04.20 | $^{10}B(d,a_0)^{8}Be$<br>$^{10}B(d,a_1)^{8}Be$ | Becker+, Phys. Rev. 119 (1960) p. 1076<br>Debras+, J. Radioanalytical Chemistry 38 (1977) p.193,<br>Comsan+, Atomkernenergie 13 (1968) p.415 | New data files from EXFOR<br>A. Lagoyannis |
| 548 | 9.04.20 | $^{10}B(d,p_0)^{11}B$<br>160° | Kokkoris+, NIMB263 (2007) p.357 | Added missing angle and re-compiled all files from EXFOR<br>A. Lagoyannis |
| 547 | 9.04.20 | $^{10}B(d,a_0)^{8}Be$<br>30°, 50°, 70°, 90°, 110°, 130°, 150°, 170° | Buck+, Nucl. Phys. A297 (1978) p.231 | Replaced previous files with data from EXFOR<br>A. Lagoyannis |
| 546 | 9.04.20 | $^{10}B(p,a_1)^{7}Be$ | Boni+, NIMB35 (1988) p. 80 | Previous entry #34 removed<br>Authors measure gammas not alphas<br>A. Lagoyannis |
| 545 | 9.04.20 | $^{10}B(p,p_0)^{10}B$ | Overley+, Phys. Rev. 128 (1962) p.315 | Replace file with EXFOR data that extend to lower energies |

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| | | | | Sent by A. Lagoyannis |
| 544 | 9.04.20 | $^{9}$Be(a,a$_{0}$)$^{9}$Be | Taylor+, Phys. Rev. C7 (1973) p.1837 | Replace file for 136$^{o}$, include 2 more angles: 99$^{o}$, 117$^{o}$ from EXFOR<br>Sent by A. Lagoyannis |
| 543 | 9.04.20 | $^{9}$Be(a,a$_{0}$)$^{9}$Be | Goss+, Phys. Rev. C7 (1973) p.1837 | Replace file for 157.3$^{o}$, include 3 more angles: 90$^{o}$, 129$^{o}$, 139$^{o}$ from EXFOR<br>Sent by A. Lagoyannis |
| 542 | 9.04.20 | $^{9}$Be(a,a$_{0}$)$^{9}$Be<br>170.5$^{o}$ | Leavitt+, NIMB85 (1994) p.37 | Replace previous file/include energy errorbars<br>A. Lagoyannis |
| 541 | 9.04.20 | $^{9}$Be($^{3}$He,p$_{0}$)$^{11}$B<br>$^{9}$Be($^{3}$He,p$_{1}$)$^{11}$B | Wolicki+, Phys. Rev. 116(1959) p.1585 | New files from EXFOR<br>A. Lagoyannis |
| 540 | 9.04.20 | $^{9}$Be(d,a$_{1}$)$^{7}$Li<br>90$^{o}$, 165$^{o}$ | Biggerstaff+, Nucl. Phys. 36 (1962) p.631 | Replaced with corrected files<br>A. Lagoyannis |
| 539 | 9.04.20 | $^{9}$Be(p,p$_{0}$)$^{9}$Be | Allab+, Journal de Physique 44 (1983) p.579 | Replaced with corrected files<br>A. Lagoyannis |
| 538 | 9.04.20 | $^{7}$Li(a,a$_{1}$)$^{7}$Li<br>121$^{o}$ | Bohlen+, Nucl. Phys. A179 (1972) p.504 | New file from EXFOR<br>A. Lagoyannis<br>EXFOR A1505004 |
| 537 | 9.04.20 | $^{7}$Li(p,a$_{0}$)$^{4}$He | Sweeney+, Phys. Rev. 182 (1969) p.1007 | Replaced with corrected files<br>A. Lagoyannis |
| 536 | 9.04.20 | $^{7}$Li(p,a$_{0}$)$^{4}$He<br>156.8$^{o}$ | Warters+, Nucl. Phys. 36 (1962) p.597 | Multiple entries of the same data removed<br>A. Lagoyannis |
| 535 | 9.04.20 | $^{7}$Li(p,a$_{1}$)$^{4}$He<br>$^{7}$Li(p,a$_{2}$)$^{4}$He | Cavallaro+, Nucl. Phys. 36 (1962) p.597 | Files replace previously incorrect<br>(p,a$_{0}$) and (p,a$_{1}$) files, respectively<br>A. Lagoyannis |
| 534 | 9.04.20 | $^{6}$Li($^{3}$He,p$_{0}$)$^{8}$Be | Vignon+, Journ. Physique 30 (1969) p. 913 | New file from A. Lagoyannis<br>EXFOR D0601003 |
| 533 | 9.04.20 | $^{6}$Li(d,a$_{0}$)$^{4}$He | Cai+, private communication | New file from A. Lagoyannis<br>EXFOR S0017005 |
| 532 | 9.04.20 | $^{6}$Li(p,p$_{0}$)$^{6}$Li<br>89$^{o}$, 118$^{o}$, 156$^{o}$ | Mccray+, Phys. Rev. 130 (1963) p. 2034 | New file from EXFOR<br>A. Lagoyannis |
| 531 | 9.04.20 | $^{4}$He(p,p$_{0}$)$^{4}$He | Freier+, Phys. Rev. 75 (1949) p. 1345 | New file from EXFOR<br>A. Lagoyannis |
| 530 | 9.04.20 | $^{3}$H(p,p$_{0}$)$^{3}$H | Cai+, NIMB 268 (2010) p. 3373 | New file from EXFOR<br>A. Lagoyannis |
| 529 | 9.04.20 | $^{2}$H(p,p$_{0}$)$^{2}$H | Sherr+, Phys. Rev. 72 (1947) p.662 Kocher+, Nucl. Phys. A132 (1969) p. 455 | New files from EXFOR<br>A. Lagoyannis |
| 528 | 9.04.20 | $^{1}$H(p,p$_{0}$)$^{1}$H | Worthington+, Phys. Rev. 90 (1953) p.899 Blair+, Phys. Rev. 74 (1948) p. 533 | Statistical errors corrected<br>A. Lagoyannis |
| 527 | 9.04.20 | Links to EXFOR datafiles have been included where they were missing | | A. Lagoyannis |
| 526 | 23.01.20 | $^{9}$Be(d,d$_{0}$)$^{9}$Be,<br>$^{23}$Na(d,d$_{0}$)$^{23}$Na,<br>$^{31}$P(d,d$_{0}$)$^{31}$P | NIMB 459 (2019) p.90 NIMB 461 (2019) p.124 NIMB 459 (2019) p.84 | R33 files sent by E. Ntemou |
| 525 | 11.02.19 | $^{nat}$Cu(a,a$_{0}$)$^{nat}$Cu,<br>$^{nat}$Ni(a,a$_{0}$)$^{nat}$,<br>$^{89}$Y(a,a$_{0}$)$^{89}$Y | | EXFOR: F1345003 F1345002 F1345004 |
| 524 | 08.02.19 | Corrected reaction for $^{25}$Mg(d,a$_{4+5}$)$^{23}$Na, $^{25}$Mg(d,a$_{8+9}$)$^{23}$Na | | A. Lagoyannis |

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| 523 | 08.02.19 | $^{17}O(p,a)^{14}N$ | | EXFOR: T0411002 |
| 522 | 07.02.19 | References updated and corrected for over 80 data files. | | IAEA-NDS |
| 521 | 07.02.19 | $^{14}N(d,pg_{4-1})^{15}N$, $14N(d,pg_{5-1})^{15}N$, $14N(d,pg_{6-1})^{15}N$, $14N(d,pg_{7-0})^{15}N$ | A. Jokar+, Nucl. Instr. and Meth. B 431 (2018) 25. | R33 files sent by A. Jokar |
| 520 | 05.02.19 | $^{14}N(p,pg_{1-0})^{14}N$, $^{nat}N(p,p_0)^{nat}N$, $^{28}Si(p,p_1)^{28}Si$, $^{28}Si(p,pg_{1-0})^{28}Si$, $^{29}Si(p,pg_{1-0})^{29}Si$, $^{nat}Si(p,p_0)^{nat}Si$ | L. Csedreki+, Nucl. Instr. and Meth. B 443 (2019) 48-56. | R33 files sent by L. Csedreki |
| 519 | 30.01.19 | Updated references for PIGE data for: $^{19}F$,$^{23}Na$, $^{27}Al$, $^{28}Si$ | M. Chiari, I. Bogdanovic, A. Kiss | IAEA-NDS |
| 518 | 18.01.19 | Ranges of $E_g$ for (p,g) reactions have been included in the R33 files for: $^{6,7}Li$, $^{10,11}B$, $^{9}Be$, $^{12,13}C$, $^{14,15}N$, $^{16}O$, $^{20}Ne$, $^{23}Na$, $^{27}Al$ | Advised by M. Mayer | R33 files sent by A. Lagoyannis |
| 517 | 02.03.18 | $^{nat}Si(d,d_0)^{nat}Si$ | E. Ntemou+, Nucl. Instr. and Meth. B (IBA 2017) | R33 files sent by E. Ntemou |
| 516 | 22.11.17 | Ratio-to-Rutherford cross sections for $(?,?_0)$ on $^{nat}Ge$ | M. Bozoian+, Nucl. Instr. and Meth. B 51 (1990) 311-319 | IAEA-NDS |
| 515 | 22.11.17 | Ratio-to-Rutherford cross sections for $(?,?_0)$ on $^{nat}Cu$, $^{nat}Y$ | J.A. Martin+, Appl. Phys. Lett. 52 (1988) 2177 | IAEA-NDS |
| 514 | 14.11.17 | Ratio-to-Rutherford cross sections for $(?,?_0)$ on $^{12}C$, $^{16}O$, $^{23}Na$, $^{27}Al$, $^{28}Si$ $^{nat}Cl$, $^{40}Ca$, $^{48}Ti$, $^{56}Fe$, $^{59}Co$, $^{nat}Ni$, $^{nat}Cu$, $^{nat}Ge$ | J. Tang+, Nucl. Instr. and Meth. B 74 (1993) 491-495 | IAEA-NDS |
| 513 | 13.11.17 | $^{nat}N(d,d_0)^{nat}N$ | M. Kokkoris+, Nucl. Instr. and Meth. B 410 (2017) 29-36 | R33 files sent by M. Kokkoris |
| 512 | 13.11.17 | $^{14}N(d,p_0,a_0,a_1)$ | M. Kokkoris+, presented at the IBA-2017 conference | Updated R33 files sent by M. Kokkoris |
| 511 | 26.09.17 | $^{7}Li(p,ng_{1-0})^{7}Be$ $^{7}Li(p,pg_{1-0})^{7}Li$ $^{19}F(p,pg_{1-0})^{19}F$ $^{19}F(p,pg_{2-0})^{19}F$ $^{19}F(p,pg_{3-1})^{19}F$ $^{19}F(p,pg_{4-1+5-2})^{19}F$ | D. Bachiller Perea+, Nucl. Instr. and Meth. B 406 (2017) 161 | Data sent by A. Zucchiatti for the PIGE CRP |
| 510 | 21.08.17 | $^{6}Li(d,d_0)^{6}Li$ | E. Ntemou et al., Nucl. Instr. Meth. B407(2017)34 | R33 files sent by E. Ntemou |
| 509 | 21.08.17 | $^{7}Li(d,d_0)^{7}Li$ | K. Preketes-Sigalas et al. (submitted to NIM B) | R33 files sent by K. Preketes-Sigalas |
| 508 | 05.04.17 | $^{27}Al(p,p_0)^{27}Al$ | K. Shahzad+, Nucl. Sci. Technol. 27(2016)33 | EXFOR D0834002 |
| 507 | 16.03.17 | $^{19}F(d,d_0)^{19}F$ | V. Foteinou+, Nucl. Instr. Meth. B396 (2017) 1-4 | R33 files sent by T. Lagoyannis |
| 506 | 09.03.17 | PIGE data for: $^{6}Li$, $^{11}B$, $^{12}C$, $^{14}N$, $^{16}O$, $^{23}Na$, $^{24}Mg$, $^{27}Al$, $^{28}Si$, $^{32}S$, $^{35}Cl$, $^{39}K$ | A.Z. Kiss et al., Nucl. Instr. & Meth. B85 (1994) 118 | References sent by T. Lagoyannis for the PIGE CRP |
| 506 | 09.03.17 | PIGE data for:$^{51}V$, $^{50}Cr$, | G.A.Krivonosov+, Sov. J. Nucl. Phys. 24 (1976) 461 | References sent by T. Lagoyannis for the PIGE |

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|---|---|---|---|---|
| | | $^{52}$Cr, $^{53}$Cr, $^{54}$Cr, $^{57}$Fe, $^{58}$Fe, $^{59}$Co, $^{71}$Ni, $^{72}$Ni, $^{63}$Cu, $^{65}$Cu, $^{64}$Zn, $^{66}$Zn, $^{67}$Zn, $^{68}$Zn, $^{74}$Se, $^{76}$Se, $^{77}$Se, $^{78}$Se, $^{80}$Se | | CRP |
| 506 | 09.03.17 | PIGE data for: $^{48}$Ti, $^{51}$V, $^{55}$Mn, $^{56}$Fe, $^{63}$Cu, $^{65}$Cu, $^{68}$Zn, $^{107}$Ag, $^{109}$Ag, $^{197}$Au | M.J.Kenny+, Nucl. Instr. and Meth. 168 (1980) 115 | References sent by T. Lagoyannis for the PIGE CRP |
| 506 | 09.03.17 | $^{52}$Cr(p,pg$_{1\text{-}0}$)$^{52}$Cr $^{56}$Fe(p,pg$_{1\text{-}0}$)$^{56}$Fe | S.R.Kennett+, Nucl. Phys. A 363 (1981) 233 | References sent by T. Lagoyannis for the PIGE CRP |
| 506 | 09.03.17 | $^{12}$C(p,p$_0$)$^{12}$C | S.M.Duvanov+, JINR-P15-96-69, 1996 | Data sent by S.M.Duvanov |
| 505 | 16.02.17 | $^{9}$Be(p,p$_0$)$^{9}$Be $^{9}$Be(p,d$_0$)$^{8}$Be $^{9}$Be(p,a$_0$)$^{6}$Li | N. Catarino et al., Nucl. Instr. Meth. B371(2016)50 | References sent by T. Lagoyannis for the PIGE CRP |
| 505 | 16.02.17 | $^{2}$H(a,d)$^{4}$He | Z. Han et al., Nucl. Instr. Meth. B375(2016)13 | References sent by T. Lagoyannis for the PIGE CRP |
| 505 | 16.02.17 | PIGE data for: $^{7}$Li, $^{9}$Be, $^{10}$B, $^{11}$B, $^{12}$C, $^{14}$N, $^{16}$O, $^{19}$F, $^{23}$Na, $^{24}$Mg, $^{26}$Mg, $^{27}$Al, $^{28}$Si, $^{30}$Si, $^{31}$P, $^{32}$S, $^{35}$Cl, $^{39}$K, $^{40}$Ca, $^{46}$Ti, $^{48}$Ti, $^{51}$V, $^{52}$Cr, $^{54}$Fe, $^{56}$Fe, $^{51}$V, $^{108}$Pd, $^{107}$Ag, $^{109}$Ag, $^{194}$Pt, $^{195}$Pt and $^{197}$Au | J. Raisanen+, Nucl. Instr. and Meth. 28 (1987) 199 | References sent by T. Lagoyannis for the PIGE CRP |
| 504 | 23.01.17 | $^{4}$He(p,p$_0$)$^{4}$He | V. Godinho+(2016), ACS Omega , Vol.1, Issue.6, p.1229 | Data sent by F.J. Ferrer |
| 504 | 23.01.17 | $^{12}$C($^{3}$He,p$_0$)$^{14}$N | S. Moeller, Nucl. Instrum. Methods B394 (2017) 134 | Data sent by S. Moeller |
| 503 | 18.01.17 | $^{44}$Ca(p,g$_{2\text{-}1}$)$^{45}$Sc $^{44}$Ca(p,g$_{3\text{-}1}$)$^{45}$Sc $^{44}$Ca(p,g$_{4\text{-}0}$)$^{45}$Sc $^{31}$P(p,g$_{1\text{-}0}$)$^{32}$S $^{33}$S(p,pg$_{1\text{-}0}$)$^{33}$S | A.Anttila+, J. Radioanal. Chem. 62 (1981) 293 | Files sent by T. Lagoyannis for the PIGE CRP |
| 503 | 18.01.17 | $^{48}$Ca(p,ng$_{7\text{-}4}$)$^{48}$Sc $^{44}$Ca(p,pg$_{1\text{-}0}$)$^{44}$Ca $^{42}$Ca(p,pg$_{1\text{-}0}$)$^{42}$Ca $^{41}$K(p,ng$_{5\text{-}0}$)$^{41}$Ca $^{41}$K(p,ng$_{4\text{-}0}$)$^{41}$Ca $^{41}$K(p,ng$_{2\text{-}0}$)$^{41}$Ca $^{41}$K(p,ng$_{1\text{-}0}$)$^{41}$Ca $^{41}$K(p,ag$_{1\text{-}0}$)$^{38}$Ar $^{41}$K(p,pg$_{2\text{-}0}$)$^{41}$K $^{41}$K(p,pg$_{1\text{-}0}$)$^{41}$K $^{33}$S(p,pg$_{1\text{-}0}$)$^{33}$S | A.Z. Kiss+, J. Radioanal. Nucl. Chem. 89 (1985) 123 | Files sent by T. Lagoyannis for the PIGE CRP |
| 503 | 18.01.17 | $^{28}$Si(p,pg$_{1\text{-}0}$)$^{28}$Si $^{29}$Si(p,pg$_{1\text{-}0}$)$^{29}$Si | M.J. Kenny+, Nucl. Instr. and Meth. 168 (1980) 115 | Files sent by T. Lagoyannis for the PIGE CRP |
| 503 | 18.01.17 | $^{29}$Si(p,pg$_{2\text{-}1}$)$^{29}$Si $^{29}$Si(p,pg$_{1\text{-}0}$)$^{29}$Si $^{29}$Si(p,pg$_{2\text{-}0}$)$^{29}$Si $^{30}$Si(p,pg$_{1\text{-}0}$)$^{30}$Si | A. Savidou et al, NIMB,152,12 (1999) | Files sent by T. Lagoyannis for the PIGE CRP |
| 502 | 22.12.16 | $^{23}$Na(p,ag$_{1\text{-}0}$)$^{20}$Ne $^{23}$Na(p,pg$_{1\text{-}0}$)$^{23}$Na $^{24}$Mg(p,pg$_{1\text{-}0}$)$^{24}$Mg | A.Z. Kiss+, J. Radioanal. Nucl. Chem. 89 (1985) 123 A.Anttila+, J. Radioanal. Chem. 62 (1981) 293 M.J.Kenny+, Nucl. Instr. and Meth. 168 (1980) 115 M.J.Kenny+, Nucl. Instr. and | New entries EXFOR: D0384020 EXFOR: D0384018 EXFOR: D0384019 T. Lagoyannis |

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| | | $^{25}$Mg(p,pg$_{2\text{-}0}$)$^{24}$Mg $^{25}$Mg(p,pg$_{2\text{-}1}$)$^{24}$Mg $^{26}$Mg(p,g$_{1\text{-}0}$)$^{27}$Al $^{26}$Mg(p,g$_{2\text{-}0}$)$^{27}$Al | Meth. 168 (1980) 115 | for the PIGE CRP |
| 501 | 20.12.16 | PIGE data for: $^{14}$N, $^{15}$N and $^{16}$O $^{14}$N(d,ng$_{2\text{-}0}$)$^{15}$O $^{15}$N(p,g$_{0}$)$^{16}$O | A.Anttila+, J. Radioanal. Chem. 62 (1981) 293 A.Z. Kiss+, J. Radioanal. Nucl. Chem. 89 (1985) 123 Z.Elekes et al., Nucl. Instr. and Meth. B168 (2000) 305 C.Rolfs et al., Nuclear Physics A235(1974)450 W.J.O`connell+, Phys. Rev. C17 (1978) 892 | Corrected for the isotopic composition Reaction corrected EXFOR C0647004 EXFOR C1223002 T. Lagoyannis for the PIGE CRP |
| 500 | 19.12.16 | $^{19}$F(p,pg)$^{19}$F $^{19}$F(p,ag)$^{16}$O $^{19}$F(p,ag$_{2\text{-}0}$)$^{16}$O $^{19}$F(p,ag$_{3\text{-}0}$)$^{16}$O $^{19}$F(p,ag$_{4\text{-}0}$)$^{16}$O $^{19}$F(p,pg$_{3\text{-}1}$)$^{19}$F $^{19}$F(p,pg$_{4\text{-}1+5\text{-}2}$)$^{19}$F | D.Grambole+, Journ. Radioan. Nucl. Chemistry 83 (1984) 107 M.J. Kenny, Aust. J. Phys. 34 (1981) 35 M.J.Kenny+, Nucl. Instr. Meth. 168(1980)11 A.Ranken+, Phys. Rev. 109, 1646 (1958) | Theta corrected Theta corrected EXFOR D0384002 EXFOR C1006005 T. Lagoyannis for the PIGE CRP |
| 499 | 09.12.16 | $^{6}$Li(p,p$_{0}$)$^{6}$Li $^{6}$Li(p,a$_{0}$)$^{3}$He | J.A.Mccray(1963), Jour. Physical Review, Vol.130, p.2034 S.Bashkin and H.T.Richards Phys.Rev. 84 (1951) 1124 W.D.Warters,W.A.Fowler and C.C.Lauristen Phys.Rev 91 (1953) 917 J.A.Mccray(1963), Jour. Physical Review, Vol.130, p.2034 J.B.Marion,G.Weber and F.S.Mozer Phys.Rev. 104 (1956) 140 | File corrected Files deleted P. Dimitriou |
| 498 | 01.12.16 | $^{7}$Li(p,a$_{0}$)$^{4}$He | V.Paneta+(2012), Jour. Nucl. Instrum. Methods in Physics Res., Sect.B, Vol.288, p.53 | P. Dimitriou Files corrected |
| 497 | 02.11.16 | $^{7}$Li(p,p$_{0}$)$^{7}$Li | W.D.Warters+(1953), Jour. Physical Review, Vol.91, Issue.4, p.917 | EXFOR: A1404002, A1404004 |
| 496 | 19.09.16 | $^{19}$F(p,p$_{1}$)$^{19}$F $^{19}$F(p,p$_{2}$)$^{19}$F $^{19}$F(p,p$_{3}$)$^{19}$F $^{19}$F(p,p$_{4}$)$^{19}$F $^{19}$F(p,p$_{5}$)$^{19}$F | M.Chiari et al. Nucl. Intr. and Meth. B 384 (2016) 37-41 | R33 files sent by M.Chiari |
| 495 | 26.08.16 | $^{31}$P(p,pg$_{1\text{-}0}$)$^{31}$P | A. Jokar et al.,Nucl.Instr.and Meth.B 383(2016)152 | R33 files sent by O. Kakuee for the PIGE CRP |
| 494 | 06.07.16 | $^{19}$F(p,ag$_{2\text{-}0+3\text{-}0+4\text{-}0}$)$^{16}$O | P. Cabanelas et al., Nucl. Instr. Meth. B Nucl. Instr. Meth. B 381(2016)110 | Data sent by P. Cabanelas for the PIGE CRP |
| 493 | 10.05.16 | $^{24}$Mg(p,pg$_{1\text{-}0}$)$^{24}$Mg $^{25}$Mg(p,pg$_{2\text{-}0}$)$^{25}$Mg $^{25}$Mg(p,pg$_{2\text{-}1}$)$^{25}$Mg $^{25}$Mg(p,pg$_{1\text{-}0}$)$^{25}$Mg $^{26}$Mg(p,pg$_{2\text{-}0}$)$^{26}$Mg | N. Sharifzadeh et al., Nucl. Instr. Meth. B Nucl. Instr. Meth. B 372(2016)109 | R33 files sent by O. Kakuee for the PIGE CRP |
| 492 | 03.05.16 | $^{35}$Cl(d,pg$_{1\text{-}0}$)$^{36}$Cl $^{35}$Cl(d,pg$_{2\text{-}0}$)$^{36}$Cl $^{37}$Cl(d,pg$_{1\text{-}0}$)$^{38}$Cl $^{37}$Cl(d,pg$_{2\text{-}0}$)$^{38}$Cl | A. Jokar et al., Nucl.Instr.and Meth.B 377(2016)37 | R33 files sent by O. Kakuee for the PIGE CRP |
| 491 | 03.05.16 | $^{14}$N(p,p$_{1\text{-}0}$)$^{14}$N $^{28}$Si(p,p$_{1\text{-}0}$)$^{28}$Si $^{29}$Si(p,p$_{1\text{-}0}$)$^{29}$Si $^{10}$B(p,a$_{1\text{-}0}$)$^{7}$Be $^{10}$B(p,p$_{1\text{-}0}$)$^{10}$B $^{11}$B(p,p$_{1\text{-}0}$)$^{11}$B | B. Marchand et al, Nucl. Instr. Meth. B Nucl. Instr. Meth. B 378 (2016) 25 | R33 files sent by J. Raisanen for the PIGE CRP |
| 490 | 31.03.16 | $^{16}$O(d,d$_{0}$)$^{16}$O | H. Rafi-kheiri+, Nucl. Instr. Meth. B 373 (2016) 40-43 | R33 files sent by O. Kakuee for the PIGE CRP |
| 489 | 31.03.16 | $^{24}$Mg(d,p$_{0}$)$^{25}$Mg $^{24}$Mg(d,p$_{1}$)$^{25}$Mg $^{24}$Mg(d,p$_{2}$)$^{25}$Mg $^{24}$Mg(d,p$_{3}$)$^{25}$Mg $^{24}$Mg(d,p$_{4}$)$^{25}$Mg | H. Rafi-kheiri+, Nucl. Instr. Meth. B 373 (2016) 63 | R33 files sent by O. Kakuee for the PIGE CRP |

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| 488 | 08.03.16 | $^{32}S(p,p_0)^{32}S$ | M. Chiari, unpublished data | R33 files sent by M. Chiari for the PIGE CRP |
| 487 | 25.02.16 | $^{27}Al(p,g_{1-0})^{28}Si$ | M. Chiari, unpublished data | R33 files sent by M. Chiari for the PIGE CRP |
| 486 | 24.02.16 | $^{11}B(p,g_0)^{12}C$ $^{11}B(p,g_1)^{12}C$ | T.Huus et al., Phys. Rev. 91 (1953) 599 | EXFOR F0363002 sent by T. Lagoyannis for the PIGE CRP |
| 485 | 24.02.16 | $^{11}B(p,g_0)^{12}C$ $^{11}B(p,g_1)^{12}C$ | R. E. Segel, Phys. Rev. 139B, 818 (1965) | EXFOR F0332002 sent by T. Lagoyannis for the PIGE CRP |
| 484 | 24.02.16 | $^{11}B(p,g_0)^{12}C$ $^{11}B(p,g_1)^{12}C$ | R. E. Segel, Phys. Rev. 139B, 818 (1965) | EXFOR F0332002 sent by T. Lagoyannis for the PIGE CRP |
| 483 | 24.02.16 | $^{10}B(p,g_{2-0})^{11}C$ $^{10}B(p,g_{3-0})^{11}C$ $^{10}B(p,g_{5-0})^{11}C$ | M. Wiescher et al. Phys. Rev C28 (1983) 1431 | EXFOR C1274003 sent by T. Lagoyannis for the PIGE CRP |
| 482 | 24.02.16 | $^{13}C(p,g_{2-1})^{14}N$ $^{13}C(p,g_{3-0})^{14}N$ $^{13}C(p,g_{4-0})^{14}N$ $^{13}C(p,g_{5-1})^{14}N$ $^{13}C(p,g_{6-4})^{14}N$ | J.D.King+(1994), Jour. Nuclear Physics, Section A, Vol.567, p.354 | EXFOR C0872005 sent by T. Lagoyannis for the PIGE CRP |
| 481 | 24.02.16 | $^{12}C(p,g_0)^{13}N$ | C.Rolfs+(1974), Jour. Nuclear Physics, Section A, Vol.227, p.291 | EXFOR A0191003 sent by T. Lagoyannis for the PIGE CRP |
| 480 | 21.01.16 | $^{16}O(d,p_0)^{17}O$ $^{16}O(d,p_1)^{17}O$ | H.Rafi-kheiri+, Nucl. Instr. and Meth. B 371 (2016) 46 | R33 files sent by O. Kakuee for the PIGE CRP |
| 479 | 18.12.15 | $^{16}O(p,pg_{2-0})^{16}O$ $^{16}O(p,pg_{3-0})^{16}O$ | J. Raisanen+, Nucl. Instr. and Meth. 28 (1987) 199 | V. Semkova |
| 478 | 18.12.15 | $^{16}O(p,pg_{1-0})^{16}O$ $^{12}C(p,pg_{2-0})^{12}C$ | P.Dyer+, Physical Review C23(1981)1865 | V. Semkova |
| 477 | 16.12.15 | $^{27}Al(p,a_0)^{24}Mg$ $^{27}Al(p,p_1)^{27}Al$ $^{27}Al(p,p_2)^{27}Al$ | R.O.Nelson+, Physical Review C29(1984)1656 | P. Dimitriou, EXFOR: C0686003 |
| 476 | 16.12.15 | $^{16}O(d,a_0)^{14}N$ $^{16}O(d,p_0)^{17}O$ $^{16}O(d,p_1)^{17}O$ | N. Longequeue+, Le Journal de Physique 26 (1965) 367 | V. Semkova |
| 475 | 08.12.15 | $^{28}Si(p,pg_{1-0})^{28}Si$ $^{29}Si(p,pg_{1-0})^{29}Si$ | A. Jokar et al.,accepted for publication in NIMB | R33 files sent by A. Jokar for the PIGE CRP |
| 474 | 08.12.15 | $^{12}C(d,d_0)^{12}C$ $^{12}C(d,p_0)^{13}C$ $^{12}C(d,p_1)^{13}C$ | M.Allab+, Jour. Journal de Physique, 31(1970)919 | V. Semkova |
| 473 | 30.11.15 | $^{9}Be(p,g_{1-0})^{10}B$ $^{9}Be(p,g_{2-1})^{10}B$ $^{9}Be(p,g_{3-1})^{10}B$ | W.E. Meyerhof et al. Phyical Review v.115 (1959) 1227 | V. Semkova |
| 472 | 30.11.15 | $^{25}Mg(p,pg_{2-1})^{25}Mg$ $^{25}Mg(p,pg_{2-0})^{25}Mg$ $^{25}Mg(p,pg_{1-0})^{25}Mg$ | K. Preketes-Sigalas et al, Unpublished Data | R33 files sent by T. Lagoyannis for the PIGE CRP |
| 471 | 30.11.15 | $^{32}S(p,p_0)^{32}S$ | L.K. Fifield et al., Nuclear Physics A220(1974) 183 | V. Semkova |
| 470 | 30.11.15 | $^{6}Li(p,g_0)^{7}Li$ $^{6}Li(p,g_1)^{7}Li$ | Sweeney, W. E. (1969). Bull. Am. Phys. Soc. 14, 487 | R33 files sent by T. Lagoyannis for the PIGE CRP |
| 469 | 30.10.15 | $^{4}He(p,p_0)^{4}He$ | A.Nurmela+, Journal of Applied Physics 82(1997)1983 T.Cai+, Nucl. Instrum. Methods in Physics Res. B268(2010)3373 P.Pusa+, Nucl. Instrum. Methods in Physics Res. B222(2004)686 L.Kraus+, Nuclear Physics A224(1974)45 P.D.Miller+, Physical Review 112(1958)2043 G.Freier+, Physical Review 75(1949)1345 | EXFOR O0915002 EXFOR S0071003 EXFOR D0197002 EXFOR A1080002 EXFOR A1140002 EXFOR A1106002 P. Dimitriou |
| 468 | 19.10.15 | $^{16}O(d,pg_{1-0})^{17}O$ | C.Rolfs et al., Z.Phys A353 (1995) 127 | V. Semkova |

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| | | | I.Vickridge et al., Nucl. Instr. & Meth. B85 (1994) 95 | |
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| 466 | 28.09.15 | $^{27}Al(p,pg_{1\text{-}0})^{27}Al$<br>$^{27}Al(p,pg_{2\text{-}0})^{27}Al$<br>$^{25}Mg(p,pg_{1\text{-}0})^{25}Mg$ | A.P.Jesus et al., to be published | R33 files sent by A.P.Jesus for the PIGE CRP |
| 465 | 19.07.15 | $^{23}Na(p,pg_{1\text{-}0})^{23}Na$<br>$^{23}Na(p,pg_{2\text{-}1})^{23}Na$ | N. Sharizadeh et al., to be published | R33 files sent by O. Kakuee for the PIGE CRP |
| 464 | 2.07.15 | $^{15}N(p,ag_{1\text{-}0})^{12}C$ | C.Rolfs et al., Nuclear Physics A235(1974)450 | EXFOR C0547003 |
| 463 | 2.07.15 | $^{14}N(p,pg_{1\text{-}0})^{14}N$<br>$^{14}N(p,pg_{2\text{-}1})^{14}N$ | P.Dayer et al., PRC 23, 1865, 1981 | |
| 462 | 25.06.15 | $^{14}N(a,a_0)^{14}N$ | K.Yasuda et al. Nucl. Instrum. Meth. B343 (2015) 1 | EXFOR E2469003 |
| 461 | 25.06.15 | $^{nat}Mg(d,d_0)^{nat}Mg$ | N. Patronis et al., Nucl. Instr. Meth B 337 (2014) 97 | |
| 460 | 25.06.15 | $^{1}H(a,p)^{4}He$ | Hongliang Zhang et al. Nucl. Instr. Meth. B335 (2014) 85 | |
| 459 | 25.06.15 | $^{nat}Cu(a,a_0)^{nat}Cu$<br>$natZn(a,a_0)^{nat}Zn$ | A.Climent-Font et al., Nucl. Instrum. Meth. B332 (2014) 191 | EXFOR O2227002, O2227003<br>EXFOR O2227004, O2227005 |
| 458 | 25.06.15 | $^{9}Be(p,a_0)^{6}Li$<br>$^{9}Be(p,d_0)^{8}Be$<br>$^{9}Be(p,p_0)^{9}Be$ | S. Krat et al., Nucl. Instr. Meth. B358 (2015) 72 | |
| 457 | 25.06.15 | $^{9}Be(^{3}He,p_0)^{11}B$<br>$^{9}Be(^{3}He,p_1)^{11}B$<br>$^{9}Be(^{3}He,p_2)^{11}B$<br>$^{9}Be(^{3}He,p_2)^{11}B$ | N.P.Barradas et al., Nucl. Instrum. Methods B346 (2015) 21 | EXFOR O2259002 |
| 456 | 25.06.15 | $^{9}Be(a,a_0)^{9}Be$ | I.Lombardo et al., Nucl. Instrum. Methods B302 (2013) 19 | EXFOR O2137003 |
| 455 | 25.06.15 | $^{10}B(p,a_0+a_1)^{7}Be$ | A.Kafkarkou et al., Nucl. Instrum. Methods B316 (2013) 48 | EXFOR C2072002 |
| 454 | 3.06.15 | $^{27}Al(p,ag_{1\text{-}0})^{24}Mg$<br>$^{27}Al(p,pg_{1\text{-}0})^{27}Al$<br>$^{27}Al(p,pg_{2\text{-}0})^{27}Al$ | A. Jokar et al.,to be published | R33 files sent by A. Jokar |
| 453 | 3.06.15 | $^{12}C(d,pg_{1\text{-}0})^{13}C$<br>$^{12}C(d,pg_{2\text{-}0})^{13}C$<br>$^{12}C(d,pg_{3\text{-}0})^{13}C$ | S.Tryti et al., Nucl.Phys. A251 (1975) 206 and Nucl.Phys. A201 (1973) 135<br>F.Papillon et al., Nucl. Instr. Meth. B 132 (1997) 468 | V. Semkova EXFOR: O2121, O1066 |
| 452 | 7.05.15 | $^{19}F(p,pg_{1\text{-}0})^{19}F$<br>$^{19}F(p,pg_{2\text{-}0})^{19}F$ | M. Chiari et al., to be published | R33 files sent by M. Chiari for the PIGE CRP |
| 451 | 30.03.15 | $^{23}Na(p,pg_{1\text{-}0})^{23}Na$ | L. Csedreki et al., to be published | R33 files sent by L. Csedreki for the PIGE CRP |
| 450 | 30.03.15 | $^{10}B(p,ag_{1\text{-}0})^{7}Be$<br>$^{10}B(p,pg_{1\text{-}0})^{10}B$<br>$^{11}B(p,pg_{1\text{-}0})^{11}B$<br>$^{14}N(p,pg_{1\text{-}0})^{14}N$<br>$^{23}Na(p,pg_{1\text{-}0})^{23}Na$<br>$^{28}Si(p,pg_{1\text{-}0})^{28}Si$<br>$^{29}Si(p,pg_{1\text{-}0})^{29}Si$ | M. Chiari (unpublished data) | R33 files sent by M. Chiari for the PIGE CRP |
| 449 | 4.03.15 | $^{28}Si(a,a_0)^{28}Si$ | K.-M.Kallman(1996), Jour. Zeitschrift fuer Physik A, Hadrons and Nuclei, Vol.356, p.287 H.Yonezawa+(1994), Jour. Nucl. Instrum. Methods in Physics Res., Sect.B, Vol.88, p.207 M.K.Leung(1972), Thes. Thesis or dissertation, Name.Leung J.J.Lawrie+(1986), Jour. Zeitschrift fuer Physik A, Hadrons and Nuclei, Vol.325, p.175 A.Coban+(2000), Jour. Nuclear | EXFOR O0966002 EXFOR F0117004 EXFOR C2100007-C2100010 EXFOR O0950002 EXFOR F0602002 |

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| | | | Physics, Section A, Vol.678, p.3 | |
| 448 | 4.03.15 | $^{23}Na(p,pg_{1\text{-}0})^{23}Na$<br>$^{23}Na(p,pg_{2\text{-}1})^{23}Na$ | I. Zamboni et al., to be published | R33 files sent by I. Zamboni for the PIGE CRP |
| 447 | 23.01.15 | $^{27}Al(p,ag_{1\text{-}0})^{24}Mg$<br>$^{27}Al(p,pg_{1\text{-}0})^{27}Al$<br>$^{27}Al(p,pg_{2\text{-}0})^{27}Al$ | I. Zamboni et al., to be published | R33 files sent by I. Zamboni for the PIGE CRP |
| 446 | 19.01.15 | $^{11}B(p,pg_{1\text{-}0})^{11}B$ | K. Preketes-Sigalas et al, Nucl. Instr. Meth. B 368 (2016) 71 | Data sent by A. Lagoyannis for the PIGE CRP |
| 445 | 14.01.15 | $^{27}Al(p,pg_{1\text{-}0})^{27}Al$ | Jyrki Raisanen, Univ. Helsinki, Finland | Data sent by Jyrki Raisanen for the PIGE CRP |
| 444 | 14.01.15 | $^{7}Li(p,pg_{1\text{-}0})^{7}Li$<br>$^{19}F(p,pg_{1\text{-}0})^{19}F$<br>$^{19}F(p,pg_{1\text{-}0})^{19}F$<br>$^{19}F(p,pg_{2\text{-}0})^{19}F$ | M. Fonseca et al, unpublished data A. P. Jesus et al., data to be published A. P. Jesus et al., Nucl. Instrum. Meth. B 161-163 (2000) 186 | R33 files sent by A. P. Jesus for the PIGE CRP |
| 443 | 6.01.15 | $^{23}Na(p,pg_{1\text{-}0})^{23}Na$<br>$^{23}Na(p,pg_{2\text{-}1}))^{23}Na$ | M. Chiari et al., to be published | R33 files sent by M. Chiari for the PIGE CRP |
| 442 | 30.12.14 | $^{23}Na(p,pg_{1\text{-}0})^{23}Na$<br>$^{23}Na(p,pg_{2\text{-}1})^{23}Na$ | A. P. Jesus et al, data to be published | R33 files sent by A. P. Jesus for the PIGE CRP |
| 441 | 30.12.14 | $^{19}F(p,pg_{3\text{-}1})^{19}F$<br>$^{19}F(p,pg_{4\text{-}1+5\text{-}2})^{19}F$<br>$^{19}F(p,pg_{1\text{-}0})^{19}F$<br>$^{19}F(p,pg_{2\text{-}0})^{19}F$<br>$^{24}Mg(p,pg_{1\text{-}0})^{24}Mg$<br>$^{25}Mg(p,pg_{2\text{-}1})^{25}Mg$<br>$^{25}Mg(p,pg_{2\text{-}0})^{25}Mg$<br>$^{25}Mg(p,pg_{1\text{-}0})^{25}Mg$ | I. Zamboni+(2015), Nucl. Instr. and Meth. B 342 (2015) 266-270 | R33 files sent by I. Zamboni for the PIGE CRP |
| 440 | 8.12.14 | $^{27}Al(p,pg_{2\text{-}0})^{27}Al$<br>$^{10}B(p,ag_{1\text{-}0})^{7}Be$<br>$^{10}B(p,pg_{1\text{-}0})^{10}B$<br>$^{23}Na(p,pg_{1\text{-}0})^{23}Na$<br>$^{25}Mg(p,pg_{1\text{-}0})^{25}Mg$<br>$^{28}Si(p,pg_{1\text{-}0})^{28}Si$<br>$^{31}P(p,pg_{1\text{-}0})^{31}P$ | Savidou+(1999), Jour. Nucl. Instrum. Methods in Physics Res., Sect.B, Vol.152, p.12 | Reference sent by M. Chiari for the PIGE CRP |
| 439 | 8.12.14 | $^{24}Mg(d,p_0)^{25}Mg$<br>$^{24}Mg(d,p_1)^{25}Mg$<br>$^{24}Mg(d,p_2)^{25}Mg$ | N. Patronis+(2014), Jour. Nucl. Instrum. Methods in Physics Res., Sect.B, Vol.337, p.97 | Data sent by V. Paneta |
| 438 | 8.12.14 | $^{7}Li(p,a_0)^{4}He$ $^{7}Li(p,p_0)^{7}Li$<br>$^{7}Li(p,p_1)^{7}Li$<br>$^{19}F(p,a_0)^{16}O$<br>$^{19}F(p,a_{1,2})^{16}O$<br>$^{19}F(p,p_0)^{19}F$ | V.Paneta+(2012), Jour. Nucl. Instrum. Methods in Physics Res., Sect.B, Vol.288, p.53 | Data sent by V. Paneta |
| 437 | 18.11.14 | $^{27}Al(p,ag_{1\text{-}0})^{24}Mg$<br>$^{27}Al(p,pg_{2\text{-}0})^{27}Al$<br>$^{27}Al(p,pg_{1\text{-}0})^{27}Al$ | M. Chiari (unpublished data) for the PIGE CRP | R33 files sent by M. Chiari for the PIGE CRP |
| 436 | 18.11.14 | $^{10}B(p,ag)^{7}Be$<br>$^{10}B(p,pg)^{10}B$ | A. Lagoyannis et al, NIMB 342 (2015) 271-276 | R33 files sent by A. Lagoyannis for the PIGE CRP |
| 435 | 03.11.14 | $^{10}B(p,p_{1\text{-}0})^{10}B$ $^{10}B(p,p_{2\text{-}1})^{10}B$ $^{10}B(p,p_{3\text{-}1})^{10}B$<br>$^{10}B(p,p_{4\text{-}1})^{10}B$<br>$^{10}B(p,p_{74\text{-}1})^{10}B$ | R.B. Day+, J. Phys. Rev. 95, p.1003 (1957) S.E. Hunt+, J. Phys. Rev., Vol.106, p.1012 (1957) R.E. Segel+, J. Phys. Rev 145 (1966) 736 | Units corrected as per comments of T. Lagoyannis |
| 434 | 28.10.14 | $^{nat}Mg(d,d_0)^{nat}Mg$ | N. Patronis et al., Nucl. Instr. & Meth B337 (2014) 97-101 | Data sent by N. Patronis |
| 433 | 28.10.14 | $^{27}Al(p,ag_{1\text{-}0})^{24}Mg$ | L. Csedreki et al., to be published | Data sent by A.Z. Kiss for |

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| | | $^{27}Al(p,pg_{2-0})^{27}Al$<br>$^{27}Al(p,pg_{1-0})^{27}Al$ | | the PIGE CRP |
| 432 | 15.10.14 | $^{37}Cl(p,ag_{1-0})^{34}S$<br>$^{37}Cl(p,g_{1-0+4-0})^{38}Ar$ | R.O. Weber+(1985), Nucl. Phys. A439 (1985) 176 | Data sent by Adelaide Pedro de Jesus for PIGE CRP |
| 431 | 15.10.14 | $^{27}Al(p,a_{0})^{24}Mg$<br>$^{27}Al(p,a_{1-0})^{24}Mg$ | K.L. Warsh+(1963), Nucl. Phys. 44 (1963) 329 | Data sent by Adelaide Pedro de Jesus for PIGE CRP |
| 430 | 30.09.14 | $^{nat}Cr(p,g_{1-0})$ | Data corrected by A. Goncharov | Data sent by A. Goncharov for the PIGE CRP |
| 429 | 30.09.14 | $^{12}C(d,d_{0})^{12}C$<br>$^{12}C(d,p_{0})^{13}C$<br>$^{12}C(d,p_{1})^{13}C$<br>$^{12}C(d,pg_{1-0})^{13}C$<br>$^{12}C(d,pg_{2-0})^{13}C$<br>$^{12}C(d,pg_{3-0})^{13}C$ | L. Csedreki+(2014), Nucl. Instr. & Meth., B328 (2014)59-64. | Data sent by A.Z. Kiss for the PIGE CRP |
| 428 | 30.09.14 | $^{14}N(d,d_{0})^{15}N$<br>$^{14}N(d,p_{4})^{15}N$<br>$^{14}N(d,p_{5})^{15}N$<br>$^{14}N(d,p_{6})^{15}N$<br>$^{14}N(d,pg_{4-1})^{15}N$<br>$^{14}N(d,pg_{6-1})^{15}N$<br>$^{14}N(d,pg_{5-0})^{15}N$<br>$^{14}N(d,pg_{7-0})^{15}N$ | L. Csedreki+(2014), Nucl. Instr. & Meth., B328 (2014)20-26. | Data sent by A.Z. Kiss for the PIGE CRP |
| 427 | 10.07.14 | $^{nat}Cr(p,g_{1-0})$ | A. Goncharov for PIGE CRP, unpublished | Data sent by A. Goncharov for the PIGE CRP |
| 426 | 10.07.14 | $^{7}Li(p,ng_{1-0})^{7}Be$<br>$^{7}Li(p,pg_{1-0})^{7}Li$ | J.Raisanen+(1987), Nucl. Instr. & Meth. B28 (1987) 199 Savidou+(1999), Nucl. Instr. & Meth. B152 (1999) 12 | References were sent by A. Zucchiatti the PIGE CRP |
| 425 | 27.06.14 | $^{19}F(p,pg_{7-0})^{19}F$ | Kuan et al, Nucl. Phys. A193 (1972) 497 | References were sent by A. Zucchiatti the PIGE CRP |
| 424 | 27.06.14 | $^{19}F(p,pg_{1-0})^{19}F$<br>$^{19}F(p,pg_{2-0})^{19}F$ | D.Grambole et al, Journ. Radioan. Nucl. Chemistry 83 (1984) 107 A.Ranken+, Phys. Rev. 109 (1958) 1646 Savidou et al., Nucl. Instr. & Meth. B152 (1999) 12 | References were sent by A. Zucchiatti the PIGE CRP |
| 423 | 27.06.14 | $^{19}F(p,ag_{2-0})^{16}O$ | Kuan et al, Nucl. Phys. A193 (1972) 497 J.P.Ribeiro, A.P.Jesus, et al, Nucl. Phys. A688, (2001) 468 | References were sent by A. Zucchiatti the PIGE CRP |
| 422 | 27.06.14 | $^{19}F(p,ag_{2-0+2-1+4-0})^{16}O$ | Jarjis, Nucl. Instr. & Meth. 154 (1978) 383 Dieumegard et al, Nucl. Instr. & Meth. 168 (1980) 93 Grambole et al., J. Radioan. Nucl. Chemistry, 83 107 (1984) 93 | References were sent by A. Zucchiatti the PIGE CRP |
| 421 | 27.06.14 | $^{14}N(p,pg_{6-4})^{14}N$<br>$^{14}N(p,pg_{2-1})^{14}N$<br>$^{14}N(p,pg_{1-0})^{14}N$<br>$^{14}N(p,pg_{4-1})^{14}N$<br>$^{14}N(p,pg_{7-1})^{14}N$<br>$^{14}N(p,pg_{4-0})^{14}N$<br>$^{14}N(p,pg_{9-0})^{14}N$ | H.Benhabiles-Mezhoud et al, Phys. Rev. C 83, 024603, 2011 | References were sent by J. Raisanen for the PIGE CRP |
| 420 | 27.06.14 | $^{14}N(p,pg_{0})^{15}O$<br>$^{14}N(p,pg_{1-0})^{14}N$ | G.W.Phillips+, Phys. Rev. C5 (1972) 297 | References were sent by J. Raisanen for the PIGE CRP |
| 419 | 23.06.14 | $^{24}Mg(d,p_{0})^{25}MG$<br>$^{24}Mg(d,p_{1})^{25}MG$<br>$^{24}Mg(d,p_{2})^{25}MG$ | V. Paneta et al., Nucl. Instr. and Meth. B 319 (2014) 34 | R33 files sent by M. Kokkoris |
| 418 | 23.06.14 | $^{10}B(p,ag)^{7}Be$ | A. Lagoyannis, Unpublished Data | R33 files sent by A. Lagoyannis for the PIGE CRP |
| 417 | 23.06.14 | Proton-induced thick target gamma-ray yields | A.Anttila+, J. Radioanal. Chem. 62 (1981) 293 | Reference sent by A.Z. Kiss for the PIGE CRP |

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| | | for the elemental analysis of the Z=3-9, 11-21 elements | | |
| 416 | 17.06.14 | Relative thick target yields for PIGE analysis from proton (2.4-4.2 MeV) induced reactions on $^{7}$Li, $^{9}$Be, $^{10}$B, $^{11}$B, $^{12}$C, $^{13}$C, $^{14}$N, $^{15}$N, $^{16}$O, $^{17}$O, $^{18}$O, $^{19}$F, $^{23}$Na, $^{25}$Mg, $^{26}$Mg, $^{27}$Al, $^{28}$Si, $^{29}$Si, $^{30}$Si, $^{32}$S, $^{33}$S, $^{34}$S, $^{35}$Cl, $^{37}$Cl, $^{39}$K, $^{41}$K, $^{42}$Ca, $^{43}$Ca, $^{44}$Ca, $^{48}$Ca, $^{45}$Sc. | A.Z. Kiss+, J. Radioanal. Nucl. Chem. 89 (1985) 123 | Reference sent by A.Z. Kiss for the PIGE CRP |
| 415 | 12.03.14 | $^{56}$Fe(p,pg$_{1\text{-}0}$)$^{56}$Fe | G.A.Krivonosov+, Sov. J. Nucl. Phys. 24 (1976) 461 | Files corrected. |
| 414 | 12.03.14 | $^{23}$Na(p,ag$_{1\text{-}0}$)$^{20}$Ne | F.Bodart et al., J. Radioanal. Chem. 35 (1977) 95<br>P.Trocellier, Ch.Engelmann. J. Radioanal. Chem. 67 (1981) 135 | Files corrected. |
| 413 | 12.03.14 | $^{15}$N(p,ag)$^{12}$C | G. Imbriani et al. Phys Rev. C 85, 065810 (2012) | Sent by H.-W. Becker for PIGE CRP. |
| 412 | 12.03.14 | $^{27}$Al(p,g)$^{28}$Si | S. Harrisopulos et al., Eur. Phys. J. A 9 (2000) 479 | Sent by H.-W. Becker for PIGE CRP. |
| 411 | 12.03.14 | $^{19}$F(p,ag$_{4\text{-}0}$)$^{16}$O<br>$^{19}$F(p,ag$_{3\text{-}0}$)$^{16}$O<br>$^{19}$F(p,ag$_{2\text{-}0}$)$^{16}$O<br>$^{19}$F(p,ag$_{2\text{-}0+3\text{-}0+4\text{-}0}$)$^{16}$O | A. Fessler et al., NIM A450 (2000) 353 | Files renamed and corrected. |
| 410 | 12.03.14 | $^{19}$F(p,ag$_{2\text{-}0+3\text{-}0+4\text{-}0}$)$^{16}$O<br>$^{19}$F(p,ag$_{4\text{-}0}$)$^{16}$O<br>$^{19}$F(p,ag$_{3\text{-}0}$)$^{16}$O<br>$^{19}$F(p,ag$_{2\text{-}0}$)$^{16}$O | M.J. Kenny, Aust. J. Phys. 34 (1981) 35 | Files renamed and corrected. |
| 409 | 11.03.14 | $^{12}$C(d,p$_{0}$)$^{13}$C | M.Lambert+(1966), Jour. Comptes Rendus, Serie B, Physique, Vol.262, p.1459 | EXFOR: D0679004 updated |
| 408 | 11.03.14 | $^{28}$Si(d,pg$_{1\text{-}0}$)$^{29}$Si<br>$^{28}$Si(d,pg$_{2\text{-}0}$)$^{29}$Si<br>$^{28}$Si(d,pg$_{3\text{-}0}$)$^{29}$Si<br>$^{28}$Si(d,pg$_{10\text{-}0}$)$^{29}$Si | L. Csedreki et al., accepted for publication in NIMB | Data sent by A.Z. Kiss for the PIGE CRP |
| 407 | 26.02.14 | $^{nat}$Ti(p,g$_{1\text{-}0}$) | A. Goncharov et al. unpublished | Sent by A. Goncharov for PIGE CRP |
| 406 | 26.02.14 | $^{48}$Ti(p,g$_{1\text{-}0}$)$^{49}$V | A. Goncharov et al. unpublished | Revised ti8pga.r33 file sent by A. Goncharov |
| 405 | 26.02.14 | $^{7}$Be(p,g)$^{8}$B | F. Hammache, Ph.D. thesis | Data from NACRE Compilation corrected |
| 404 | 26.02.14 | $^{14}$N(p,g)$^{15}$O | Schroeder et al., Nucl. Phys. A467 (1987) 240. | Data from NACRE Compilation corrected |
| 403 | 26.02.14 | $^{13}$C(p,g$_{0\text{-}5}$)$^{14}$N | J.D.King+(1994), Jour. Nuclear Physics, Section A, Vol.567, p.354 | Data replaced with partial cross-section data from EXFOR: C0872.004 (feedback from A. Gurbich) |
| 402 | 26.02.14 | $^{15}$N(p,g$_{0}$)$^{16}$O | Rolfs and Rodney, Nucl. Phys. A235 (1974) 450. | Data from EXFOR: C0647.004: replace previous file n5pgb.r33 containing NACRE cross sections (feedback from A. Gurbich) |
| 401 | 26.02.14 | $^{9}$Be(p,g$_{0,1,6}$)$^{10}$B | L.N.Generalov+, Jour. Izv. Ross. Akad. Nauk, Ser.Fiz., Vol.69, p.1629 | Files renamed accordingly (feedback from A. Gurbich) |
| 400 | 26.02.14 | $^{9}$Be(p,g$_{0\text{-}7}$)$^{10}$B | W.E.Meyerhof+, Jour. Physical Review, Vol.115, p.1227 | Files renamed accordingly (feedback from A. Gurbich) |
| 399 | 13.02.14 | $^{27}$Al(p,pg$_{1\text{-}0}$)$^{27}$Al<br>$^{27}$Al(p,pg$_{2\text{-}0}$)$^{27}$Al | M. Chiari et al., accepted for publication in NIMB | Data sent by M. Chiari for the PIGE CRP |

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| 398 | 05.02.14 | $^{12}C(p,g_0)^{13}N$ | C.Rolfs+(1974), Jour. Nuclear Physics, Section A, Vol.227, p.291<br>N.Burtebaev+(2008), Jour. Physical Review, Part C, Nuclear Physics, Vol.78, p.035802<br>F.C.Young+(1963), Jour. Nuclear Physics, Vol.44, p.486 | P. Dimitriou, EXFOR: A0191002, A0819002, F0232002 |
| 397 | 05.02.14 | $^{12}C(p,g)^{13}N$ | C.L. Bailey and W.R. Stratton, Phys. Rev. 77 (1950) 194<br>R.N. Hall and W.A. Fowler, Phys. Rev. 77 (1950) 197<br>W.A.S. Lamb and R.E. Hester, Phys. Rev. 107 (1957) 550.<br>J.L. Vogl, unpublished. NACRE Compilation Nucl. Phys. A 656 (1999) 3-183<br>A.I.Vojtov+(1977), Rept. Joint Inst. for Nucl. Res., Dubna Reports, No.11087 | NACRE Compilation Tassos Lagoyannis for PIGE CRP<br>P. Dimitriou, EXFOR: A0191002, A0019002, C0872006 |
| 396 | 05.02.14 | $^{13}C(p,g_0)^{14}N$<br>$13C(p,g_1)^{14}N$<br>$13C(p,g_2)^{14}N$<br>$13C(p,g_3)^{14}N$<br>$13C(p,g_4)^{14}N$<br>$13C(p,g_5)^{14}N$ | J.D.King+(1994), Jour. Nuclear Physics, Section A, Vol.567, p.354 | P. Dimitriou, EXFOR: C0872006 |
| 395 | 05.02.14 | $^{13}C(p,g)^{14}N$ | D.F. Hebbard+, Nucl. Phys. 21 (1960) 652. NACRE Compilation Nucl. Phys. A 656 (1999) 3-183 | NACRE Compilation Tassos Lagoyannis for PIGE CRP |
| 394 | 05.02.14 | $^{14}N(p,g_0)^{15}O$ | N.T.Burtebaev+(2004), Jour. Izv. Rossiiskoi Akademii Nauk, Ser.Fiz., Vol.68, p.1554 | P. Dimitriou, EXFOR: F0763002 |
| 393 | 05.02.14 | $^{14}N(p,g)^{15}O$ | W. A. S. Lamb+, Phys. Rev. 108 (1957) 1304. NACRE Compilation Nucl. Phys. A 656 (1999) 3-183<br>R. E. Pixley, PhD Thesis, 1957. NACRE Compilation Nucl. Phys. A 656 (1999) 3-183<br>Schroeder et al., Nucl. Phys. A467 (1987) 240. NACRE Compilation Nucl. Phys. A 656 (1999) 3-183 | NACRE Compilation Tassos Lagoyannis for PIGE CRP |
| 392 | 05.02.14 | $^{15}N(p,g_0)^{16}O$ | K.A.Snover+(1983), Jour. Physical Review, Part C, Nuclear Physics, Vol.27, Issue.5, p.183<br>W.J.O`connell+(1978), Jour. Physical Review, Part C, Nuclear Physics, Vol.17, p.892 | P. Dimitriou, EXFOR: C1919002, C1223002 |
| 391 | 05.02.14 | $^{15}N(p,g)^{16}O$ | D.F. Hebbard, Nucl. Phys. 15 (1960) 289. NACRE Compilation Nucl. Phys. A 656 (1999) 3-183<br>Rolfs and Rodney, Nucl. Phys. A235 (1974) 450. NACRE Compilation Nucl. Phys. A 656 (1999) 3-183 | NACRE Compilation Tassos Lagoyannis for PIGE CRP |
| 390 | 05.02.14 | $^{15}N(p,pg_{2\text{-}1})^{15}N$ | K.A.Snover+(1983), Jour. Physical Review, Part C, Nuclear Physics, Vol.27, Issue.5, p.1837 | P. Dimitriou, EXFOR: C1919007 |
| 389 | 05.02.14 | $^{20}Ne(p,g)^{21}Na$ | Rolfs and Rodney, Nucl. Phys. A241 (1975) 460. NACRE Compilation Nucl. Phys. A 656 (1999) 3-183 | NACRE Compilation Tassos Lagoyannis for PIGE CRP |
| 388 | 05.02.14 | $^{16}O(p,g_0)^{17}F$<br>$16O(p,g_1)^{17}F$ | C.Rolfs(1973), Jour. Nuclear Physics, Section A, Vol.217, p.29 | P. Dimitriou, EXFOR: C0175002 |
| 387 | 05.02.14 | $^{16}O(p,g)^{17}F$ | Hester te al., Phys. Rev. 111 (1958) 1604. NACRE Compilation Nucl. Phys. A 656 (1999) 3-183<br>N. Tanner, Phys. Rev. 114 (1959) 1060. NACRE Compilation Nucl. Phys. A 656 (1999) 3-183 | NACRE Compilation Tassos Lagoyannis for PIGE CRP |
| 386 | 30.01.14 | $^{11}B(p,g_0)^{12}C$<br>$^{11}B(p,g_1)^{12}C$ | L.N.Generalov+(2005), Jour. Izv. Rossiiskoi Akademii Nauk, Ser.Fiz., Vol.69, p.85 | P. Dimitriou, EXFOR: F0716002 |
| 385 | 30.01.14 | $^{11}B(p,g_0)^{12}C$<br>$^{11}B(p,g_1)^{12}C$ | R.G.Allas+(1964), Jour. Nuclear Physics, Vol.58, p.122 | P. Dimitriou, EXFOR: F0344002 |
| 384 | 30.01.14 | $^{11}B(p,g_0)^{12}$ | M.T.Collins+(1982), Jour. Physical Review, Part C, Nuclear Physics, Vol.26, p.332 | P. Dimitriou, EXFOR: T0038002 |
| 383 | 30.01.14 | $^{11}B(p,g_0)^{12}C$ | M.C.Wright+(1982), Jour. Physical Review, Part C, Nuclear Physics, Vol.25, p.2823 | P. Dimitriou, EXFOR: C1254002 |

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| | | $^{11}B(p,g_1)^{12}C$<br>$^{11}B(p,g_3)^{12}C$ | | |
| 382 | 21.01.14 | $^{9}Be(p,g)^{10}B$ | D. Zahnow+, Nucl. Phys. A589 (1995) 95<br>W. F. Hornyak+, Nucl. Phys. 50 (1953) 424 | Sent by Tassos Lagoyannis (NCSR "Demokritos") for the PIGE CRP |
| 381 | 21.01.14 | $^{9}Be(p,g_0)^{10}B$<br>$^{9}Be(p,g_1)^{10}B$<br>$^{9}Be(p,g_2)^{10}B$<br>$^{9}Be(p,g_3)^{10}B$<br>$^{9}Be(p,g_4)^{10}B$<br>$^{9}Be(p,g_7)^{10}B$ | W.E.Meyerhof+, Jour. Physical Review, Vol.115, p.1227 | P. Dimitriou, EXFOR: F0122002 |
| 380 | 21.01.14 | $^{9}Be(p,g_0)^{10}B$<br>$^{9}Be(p,g_1)^{10}B$<br>$^{9}Be(p,g_6)^{10}B$ | L.N.Generalov+, Jour. Izv. Ross. Akad. Nauk, Ser.Fiz., Vol.69, p.1629 | P. Dimitriou, EXFOR: F0760004 |
| 379 | 21.01.14 | $^{7}Be(p,g)^{8}B$ | R.W.Kavanagh , Nucl. Phys. 15 (1960) 411<br>R.W.Kavanagh+, Bull. Am. Phys. Soc.14 (1969) 1209<br>F.J. Vaughn+, Phys. Rev. C2 (1970) 1657<br>P. D. Parker+, Astroph. J. 153 (1968) L85<br>B.W. Filippone+, Phys. Rev. C28 (1983) 2222<br>F. Hammache+, Phys. Rev. Lett. 80 (1998) 928<br>M. Hass+, Phys. Lett. B462, 1999, 237<br>F. Hammache, Ph.D. thesis | Sent by Tassos Lagoyannis (NCSR "Demokritos") for the PIGE CRP |
| 378 | 21.01.14 | $^{10}B(p,g)^{11}C$ | A.P.Tonchev+, Phys. Rev C28 (1983) 1431 | Sent by Tassos Lagoyannis (NCSR "Demokritos") for the PIGE CRP |
| 377 | 21.01.14 | $^{10}B(p,g_0)^{11}C$<br>$^{10}B(p,g_2)^{11}C$<br>$^{10}B(p,g_5)^{11}C$ | M. Wiescher+, Phys. Rev C28 (1983) 1431 | Sent by Tassos Lagoyannis (NCSR "Demokritos") for the PIGE CRP |
| 376 | 8.01.14 | $^{7}Li(p,g)^{8}Be$ | V. Riech, Phys. Lett. 6 (1963) 267<br>R. R. Perry+(1963), Nucl. Phys. 45 (1963) 586<br>D. Zahnow+(1995), Z. Phys. A351 (1995) 229 | Sent by Tassos Lagoyannis (NCSR "Demokritos") for the PIGE CRP |
| 375 | 8.01.14 | $^{6}Li(p,g)^{7}Be$ | F.E. Cecil et al, Nucl. Phys. A539 (1992) 75<br>F.C.Barker, Aust. J. Phys. 33 (1980) 159<br>Z. E. Switkowski+, Nucl. Phys. A331 (1979) 50 | Q-value corrected |
| 374 | 20.12.13 | $^{6}Li(p,g)^{7}Be$ | F.E. Cecil et al, Nucl. Phys. A539 (1992) 75<br>F.C.Barker, Aust. J. Phys. 33 (1980) 159<br>Z. E. Switkowski et al, Nucl. Phys. A331 (1979) 50 | Sent by Tassos Lagoyannis (NCSR "Demokritos") for the PIGE CRP |
| 373 | 19.12.13 | $^{48}Ti(p,g_{1-0})^{49}V$ | A. Goncharov for PIGE CRP, unpublished | A. Goncharov |
| 372 | 12.12.13 | | Corresponding EXFOR subentry No. were added to 1371 IBANDL files by V.Semkova, IAEA.<br>Programmatic quality checking vs. EXFOR database by V.Zerkin, IAEA. | |
| 371 | 17.10.13 | $^{18}O(a,a_0)^{18}O$ | S.Gorodetzky+(1968), Jour. Journal de Physique, Vol.29, p.271 | A.Gurbich, EXFOR: D0272002 |
| 370 | 17.10.13 | $^{18}O(a,a_0)^{18}O$ | V.Z.Goldberg+(2004), Jour. Physical Review, Part C, Nuclear Physics, Vol.69, p.024602 | A.Gurbich, EXFOR: D0352002 and D0352003 |
| 369 | 17.10.13 | $^{18}O(p,p_0)^{18}O$ | D.L.Sellin+(1969), Jour. Annals of Physics (New York), Vol.51, Issue.3, p.461 | A.Gurbich, EXFOR: C1918002 |
| 368 | 16.10.13 | $^{9}Be(d,ng_{1-0})^{10}B$ | G.A.Sziki+(2006), Jour. Nucl. Instrum. Methods in Physics Res., Sect.B, Vol.251, p.343 | Request received from A. Kiss |
| 367 | 14.10.13 | $^{11}B(d,pg_{1-0})^{12}B$<br>$^{11}B(d,pg_{2-0})^{12}B$<br>$^{19}F(d,pg_{1-0})^{20}F$<br>$^{6}Li(d,pg_{1-0})^{7}Li$<br>$^{16}O(d,pg_{1-0})^{17}O$ | G.A.Sziki+(2006), Jour. Nucl. Instrum. Methods in Physics Res., Sect.B, Vol.251, p.343 | P.Dimitriou: Egamma added |
| 366 | 04.10.13 | $^{12}C(^{3}He,p_0)^{14}N$ | R.L.Johnston et al. Physical Review v.109 (1958) 884 | Corrected data received from S.Taova inserted in file c2hp0i.r33 |
| 365 | 03.10.13 | $^{12}C(^{3}He,p_0)^{14}N$ | R.L.Johnston et al. Physical Review v.109 (1958) 884 | A.Gurbich: data in file c2hp0i.r33 deleted |

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| 363 | 30.09.13 | $^{16}O(a,a_0)^{16}O$ | J.R.Cameron, Phys. Rev. 90 (1953) 839 | A.Gurbich: file o6aa0h.r33 deleted |
| 362 | 18.09.13 | $^{15}N(a,a_0)^{15}N$ | S.Wilmes+(2002), Jour. Physical Review, Part C, Vol.66, p.065802 | A.Gurbich, EXFOR: F0748002 |
| 361 | 18.09.13 | $^{14}N(p,p_0)^{14}N$ | M.L.West+(1969), Jour. Physical Review, Vol.179, p.1047 | A.Gurbich, EXFOR: C1513002 |
| 360 | 18.09.13 | $^{56}Fe(p,pg_{1-0})^{56}Fe$ | G.A.Krivonosov+, Sov. J. Nucl. Phys. 24 (1976) 461 | A.Gurbich, EXFOR: F0920010 |
| 359 | 16.09.13 | $^{107}Ag(p,pg_{3-0})^{107}Ag$<br>$^{107}Ag(p,pg_{4-0})^{107}Ag$<br>$^{109}Ag(p,pg_{3-0})^{109}Ag$<br>$^{109}Ag(p,pg_{4-0})^{109}Ag$<br>$^{197}Au(p,pg_{2-1})^{197}Au$<br>$^{197}Au(p,pg_{3-0})^{197}Au$<br>$^{110}Pd(p,pg_{1-0})^{110}Pd$<br>$^{104}Pd(p,pg_{1-0})^{104}Pd$<br>$^{108}Pd(p,pg_{1-0})^{108}Pd$<br>$^{194}Pt(p,pg_{1-0})^{194}Pt$<br>$^{195}Pt(p,pg_{1-0})^{195}Pt$<br>$^{195}Pt(p,pg_{4-0})^{195}Pt$<br>$^{195}Pt(p,pg_{6-0})^{195}Pt$<br>$^{196}Pt(p,pg_{1-0})^{196}Pt$<br>$^{198}Pt(p,pg_{1-0})^{198}Pt$<br>$^{103}Rh(p,pg_{3-0})^{103}Rh$<br>$^{103}Rh(p,pg_{4-0})^{103}Rh$ | G.Deconninck+, J. Radioanal. Chem. 24 (1975) 437 | Gurbich |
| 358 | 20.08.13 | $^{7}Li(p,pg_{1-0})^{7}Li$ | B.Ja.Guzhovskij+(1984), Jour. Izv.Kaz.Akad.Nauk,Ser.Fiz.-Mat., Vol.4, p.24 | EXFOR: F0024002 |
| 357 | 20.08.13 | $^{16}O(d,p_1)^{17}O$ | H.C.Kim+(1964), Jour. Nuclear Physics, Vol.57, p.526 | Correction of the file o6dp1o.r33 requested by A.Gurbich, EXFOR: C1438002 |
| 356 | 20.08.13 | $^{7}Li(p,pg_{1-0})^{7}Li$ | Aslam et al., Jour. of Radioanal. and Nucl. Chem.,254,533,2003 | Thanks to M. Chiari for the reference, EXFOR: C0901003 |
| 355 | 14.08.13 | $^{52}Cr(p,g)^{53}Mn$<br>$^{54}Fe(p,g)^{55}Co$ | S.R.Kennett et. al. NP/A,363,233,1981 | EXFOR: D0682002<br>EXFOR: D0682004 |
| 354 | 19.07.13 | $^{32}S(p,p_0)^{32}S$ | A. Li-Scholz et al. Appl. Phys. Lett. 56 (1990) 2696 | Gurbich |
| 353 | 19.07.13 | $^{23}Na(p,ag_{1-0})^{20}Ne$<br>$^{23}Na(p,pg_{1-0})^{23}Na$ | P.Trocellier, Ch.Engelmann. J. Radioanal. Chem. 67 (1981) 135 | Gurbich |
| 352 | 28.06.13 | $^{32}S(a,a_0)^{32}S$ | A.Coban+(1999), Jour. Nuclear Physics, Section A, Vol.645, p.3 | EXFOR: O0963002 |
| 351 | 28.06.13 | $^{16}O(d,p_0)^{16}O$ | V.Gomes+(1965), Jour. Nuclear Physics, Vol.68, p.417 | EXFOR: D0323003 |
| 350 | 28.06.13 | $^{16}O(d,a_1)^{14}N$ | O.Dietzsch+(1968), Jour. Nuclear Physics, Section A, Vol.114, p.330 | EXFOR: F0727006 |
| 348 | 20.06.13 | $^{16}O(d,p_0)^{17}O$ | H.C.Kim+(1964), Jour. Nuclear Physics, Vol.57, p.526 | EXFOR: C1438002 |
| 347 | 20.06.13 | $^{14}N(d,pg_{7-0})^{15}N$<br>$^{14}N(d,pg_{5-0})^{15}N$ | H.Van Bebber + Nucl. Instr. & Meth. B136-138 (1998) 72 | EXFOR: O0856000<br>EXFOR: O0856000 |
| 346 | 20.06.13 | $^{6}Li(a,d_0)^{8}Be$ | H.R.Blieden+(1963), Jour. Nuclear Physics, Vol.49, p.209 | EXFOR: F0167013 |
| 345 | 20.06.13 | $^{19}F(p,ag_{4-0})^{16}O$<br>$^{19}F(p,ag_{3-0})^{16}O$<br>$^{19}F(p,ag_{2-0})^{16}O$<br>$^{19}F(p,ag_{2-0+3-0+4-0})^{16}$ | A. Fessler et al., NIM A450 (2000) 353 | Gurbich |
| 344 | 20.06.13 | $^{19}F(p,ag_{2-0+3-0+4-0})^{16}O$<br>$^{19}F(p,ag_{4-0})^{16}O$<br>$^{19}F(p,ag_{3-0})^{16}O$ | M.J. Kenny, Aust. J. Phys. 34 (1981) 35 | Gurbich |

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| | | $^{19}F(p,ag_{2\text{-}0})^{16}O$<br>$^{19}F(p,pg_{2\text{-}0})^{19}F$<br>$^{19}F(p,pg_{1\text{-}0})^{19}F$ | | |
| 343 | 20.06.13 | $^{9}Be(p,p_{2})^{9}Be$<br>$^{9}Be(p,p_{0})^{9}Be$<br>$^{9}Be(p,a_{1})^{6}Li$<br>$^{9}Be(p,a_{0})^{6}Li$ | H.R.Blieden+(1963), Jour. Nuclear Physics, Vol.49, p.209 | EXFOR: F0167004<br>EXFOR: F0167003<br>EXFOR: F0167002<br>EXFOR: F0167002 |
| 342 | 20.06.13 | $^{16}O(d,pg_{0})^{17}O$<br>$^{19}F(d,pg_{0})^{20}F$<br>$^{6}Li(d,pg)^{7}Li$<br>$^{11}B(d,pg)^{12}B$ | G.A.Sziki+(2006), Jour. Nucl. Instrum. Methods in Physics Res., Sect.B, Vol.251, p.343 | EXFOR: D4155005<br>EXFOR: D4155006<br>EXFOR: D4155002<br>EXFOR: D4155004 |
| 341 | 14.06.13 | $^{nat}Si(p,p_{0})^{nat}Si$ | T.A.Belote+(1961), Jour. Physical Review Vol.122 (1961) p.920 | A.Gurbich (sinpp0h.r33, sinpp0i.r33) |
| 340 | 27.05.13 | $^{13}C(p,p_{0})^{13}C$ | N.P.Barradas et al., Nucl. Instr. and Meth. B 316 (2013) 81 | A.Gurbich (c3pp0k.r33, c3pp0l.r33) |
| 339 | 20.02.13 | | Web-interface was relocated and redesigned by V.Zerkin, IAEA | |
| 338 | 31.01.12 | $^{7}Li(d,a)^{5}He$ | E.Friedland+(1971), Jour. Zeitschrift fuer Physik, Vol.243, Issue.2, p.12 | EXFOR: A142000 |
| 337 | 31.01.12 | $^{7}Li(d,a)^{5}He$ | J.M.Delbrouck-Habaru+(1969), Jour. Bull.Societe Royale des Sciences de Liege, Vol.38, p.24 | EXFOR: A152900 |
| 336 | 30.01.12 | $^{26}Mg(d,a_{7})^{24}Na$ | E.K.Lin+(1970), Jour. Chinese Journal of Physics (Taiwan), Vol.8, p.3 | EXFOR: O102900 |
| 335 | 30.01.12 | $^{26}Mg(p,p_{0})^{26}Mg$ | W.N.Wang+(1972), Jour. Chinese Journal of Physics (Taiwan), Vol.10, p. | EXFOR: O103000 |
| 334 | 30.01.12 | $^{26}Mg(p,p_{0})^{26}Mg$ | G.Doukellis+(1987), Jour. Nuclear Physics, Section A, Vol.467, p.51 | EXFOR: C073300 |
| 333 | 30.01.12 | $^{25}Mg(p,p_{0})^{25}Mg$ | G.Adams+(1984), Jour. Jour. of Physics, Part G (Nucl.and Part.Phys.), Vol.10, p.174 | EXFOR: C125600 |
| 332 | 30.01.12 | $^{24}Mg(a,a_{0})^{24}Mg$ | S.S.So+(1966), Jour. Nuclear Physics, Vol.84, p.64 | EXFOR: T025300 |
| 331 | 30.01.12 | $^{24}Mg(d,p_{8})^{25}Mg$ | Omar+(1964), Jour. Nuclear Physics, Vol.56, p.9 | EXFOR: F092201 |
| 330 | 30.01.12 | $^{24}Mg(d,p_{0})^{25}Mg$ | S.M.Lee+(1968), Jour. Nuclear Physics, Section A, Vol.122, Issue.1, p.9 | EXFOR: E220100 |
| 329 | 30.01.12 | $^{24}Mg(d,d_{0})^{24}Mg$ | Omar+(1964), Jour. Nuclear Physics, Vol.56, p.9 | EXFOR: F092200 |
| 328 | 30.01.12 | $^{24}Mg(p,p_{1})^{24}Mg$ | A.Ruh+(1970), Jour. Nuclear Physics, Section A, Vol.151, p.47 | EXFOR: F090500 |
| 327 | 30.01.12 | $^{24}Mg(p,p_{1})^{24}Mg$ | O.B.Chubinskiy+(1984), Jour. Izv. Rossiiskoi Akademii Nauk, Ser.Fiz., Vol.48, Issue.5, p.97 | EXFOR: F024500 |
| 326 | 30.01.12 | $^{24}Mg(p,p_{1})^{24}Mg$ | V.V.Lazarev+(1992), Jour. Izv. Rossiiskoi Akademii Nauk, Ser.Fiz., Vol.56, p.19 | EXFOR: F076200 |
| 325 | 30.01.12 | $^{24}Mg(p,p_{0})^{24}Mg$ | R.M.Prior+(1991), Jour. Nuclear Physics, Section A, Vol.533, p.41 | EXFOR: C007500 |
| 324 | 30.01.12 | $^{24}Mg(p,p_{0})^{24}Mg$ | E.A.Romanovskij+(1984), Jour. Izv. Rossiiskoi Akademii Nauk, Ser.Fiz., Vol.48, p.97 | EXFOR: F057800 |
| 323 | 20.01.12 | $^{nat}Si(p,p_{0})^{nat}Si$ | M.J.F.Healy+(2000), Jour. Nucl. Instrum. Methods in Physics Res., Sect.B, Vol.161, p.136 | EXFOR: O0853002 |
| 322 | 20.01.12 | $^{7}Li(p,p_{1})^{7}Li$ | R.Gleyvod+(1965), Jour. Nuclear Physics, Vol.63, Issue.4, p.650 | EXFOR: A1471003 |
| 321 | 20.01.12 | $^{7}Li(p,p_{1})^{7}Li$ | M.Laurat(1969), Rept. Centre d`Etudes Nucleaires, Saclay Reports, No.3727 | EXFOR: A1508009 |
| 320 | 20.01.12 | $^{7}Li(p,p_{1})^{7}Li$ | K.Kilian+(1969), Jour. Nuclear Physics, Section A, Vol.126, Issue.3, p.529 | EXFOR: A1443005 |
| 319 | 20.01.12 | $^{7}Li(p,p_{1})^{7}Li$ | K.Kilian+(1969), Jour. Nuclear Physics, Section A, Vol.126, Issue.3, p.529 | EXFOR: A1443003 |
| 318 | 20.01.12 | $^{7}Li(p,p_{0})^{7}Li$ | S.Bashkin+(1951), Jour. Physical Review, Vol.84, Issue.6, p.1124 | EXFOR: A1454004 |
| 317 | 20.01.12 | $^{7}Li(p,p_{0})^{7}Li$ | R.Gleyvod+(1965), Jour. Nuclear Physics, Vol.63, Issue.4, p.650 | EXFOR: A1471002 |
| 316 | 20.01.12 | $^{7}Li(p,p_{0})^{7}Li$ | K.Kilian+(1969), Jour. Nuclear Physics, Section A, Vol.126, Issue.3, p.529 | EXFOR: A1443004 |
| 315 | 20.01.12 | $^{7}Li(p,p_{0})^{7}Li$ | K.Kilian+(1969), Jour. Nuclear Physics, | EXFOR: A1443002 |

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| | | | Section A, Vol.126, Issue.3, p.529 | |
| 314 | 20.01.12 | $^{7}Li(p,a_{1})^{4}He$ | S.Cavallaro+(1962), Jour. Nuclear Physics, Vol.36, p.597 | EXFOR: F0025003 |
| 313 | 20.01.12 | $^{7}Li(p,a_{0})^{4}He$ | I.Golicheff+(1974), Jour. Journal of Radioanalytical Chemistry, Vol.22, p.113 | EXFOR: D0388003 |
| 312 | 20.01.12 | $^{7}Li(p,a)^{4}He$ | S.Cavallaro+(1962), Jour. Nuclear Physics, Vol.36, p.597 | EXFOR: F0025002 |
| 311 | 20.01.12 | $^{7}Li(p,a)^{4}He$ | N.Sarma+(1963), Jour. Nuclear Physics, Vol.44, Issue.2, p.205 | EXFOR: A1473002 |
| 310 | 20.01.12 | $^{7}Li(p,a)^{4}He$ | J.B.Marion+(1966), Jour. Nuclear Physics, Vol.77, Issue.1, p.129 | EXFOR: A1461002 |
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| 178 | 1.09.11 | $^{10}B(p,a_0)^{7}Be$ | M.Youn+(1991), Jour. Nuclear Physics, Section A, Vol.533, p.321 | EXFOR: F0421003 |
| 177 | 31.08.11 | $^{9}Be(^{3}He,a_1)^{8}Be$ | N.Sakamoto+(1974), Jour. Bull.of Inst.Chemical Research, Kyoto Univ., Vol.52, p.146 | EXFOR: F0233003 |
| 176 | 31.08.11 | $^{9}Be(^{3}He,a_0)^{8}Be$ | B.Bilwes+(1978), Jour. Journal de Physique, Vol.39, p.805 | EXFOR: F0210002 |
| 175 | 31.08.11 | $^{9}Be(^{3}He,p_0)^{11}B$ | C.S.Lin+(1981), Jour. Chinese Journal of Physics (Taiwan), Vol.19, p.99 | EXFOR: O1032002 |
| 174 | 31.08.11 | $^{9}Be(^{3}He,^{3}He_0)^{9}Be$ | I.Bondouk+(1974), Jour. Revue Roumaine de Physique, Vol.19, p.653 | EXFOR: F0455003 |
| 173 | 30.08.11 | $^{16}O(a,a_0)^{16}O$ | J.John+(1969), Jour. Physical Review, Vol.181, p.1455 | EXFOR: C0751016 |
| 172 | 30.08.11 | $^{16}O(a,a_0)^{16}O$ | J.John+(1969), Jour. Physical Review, Vol.181, p.1455 | EXFOR: C0751014 |
| 171 | 30.08.11 | $^{16}O(a,a_0)^{16}O$ | J.John+(1969), Jour. Physical Review, Vol.181, p.1455 | EXFOR: C0751013 |
| 170 | 30.08.11 | $^{16}O(a,a_0)^{16}O$ | J.John+(1969), Jour. Physical Review, Vol.181, p.1455 | EXFOR: C0751012 |
| 169 | 30.08.11 | $^{16}O(a,a_0)^{16}O$ | J.John+(1969), Jour. Physical Review, Vol.181, p.1455 | EXFOR: C0751011 |
| 168 | 30.08.11 | $^{16}O(a,a_0)^{16}O$ | J.John+(1969), Jour. Physical Review, Vol.181, p.1455 | EXFOR: C0751010 |
| 167 | 30.08.11 | $^{9}Be(a,a_0)^{9}Be$ | Z.A.Saleh+(1974), Jour. Annalen der Physik, Vol.31, p.76 | EXFOR: F0317002 |
| 166 | 30.08.11 | $^{9}Be(a,a_0)^{9}Be$ | J.Liu+(1996), Jour. Nucl. Instrum. Methods in Physics Res., Sect.B, Vol.108, p.247 | EXFOR: C0160002 |
| 165 | 29.08.11 | $^{13}C(p,p_0)^{13}C$ | V.A.Latorre+(1966), Jour. Physical Review, Vol.144, p.891 | EXFOR: F1073005 |
| 164 | 23.08.11 | $^{nat}Si(p,p_0)^{nat}Si$ | J.Vorona et al. Phys. Rev. 116 (1959) 1563 | Thanks to D.Abriola |
| 163 | 16.08.11 | $^{12}C(d,p_0)^{13}C$ | G.Debras+(1977), Jour. Journal of Radioanalytical Chemistry, Vol.38, p.193 | EXFOR: D0144003 |
| 162 | 15.08.11 | $^{nat}Si(p,p_0)^{nat}Si$ | A.R.Ramos+(2002), Jour. Nucl. Instrum. Methods in Physics Res., Sect.B, Vol.190, p.95 | EXFOR: D0078006 |
| 161 | 12.08.11 | $^{9}Be(d,p_0)^{10}Be$ | L.Bondouk+(1974), Jour. Revue Roumaine de Physique, Vol.19, p.551 | EXFOR: F0130002 |
| 160 | 12.08.11 | $^{9}Be(d,p)^{10}Be$ | I.I.Bondouk+(1974), Jour. Atomkernenergie, Vol.24, p.62 | EXFOR: F0180002 |
| 159 | 12.08.11 | $^{9}Be(d,d_0)^{9}Be$ | J.H.Renken(1963), Jour. Physical Review, Vol.132, p.2627 | EXFOR: F0141002 |
| 158 | 12.08.11 | $^{9}Be(d,d_0)^{9}Be$ | F.Machali+(1968), Jour. Atomkernenergie, Vol.13, p.29 | EXFOR: F0186002 |
| 157 | 12.08.11 | $^{9}Be(d,d_0)^{9}Be$ | J.M.Lombaard+(1972), Jour. Zeitschrift fuer Physik, Vol.249, p.349 | EXFOR: F0152002 |
| 156 | 12.08.11 | $^{9}Be(d,a_1)^{7}Li$ | E.Friedland+(1974), Jour. Zeitschrift fuer Physik, Vol.267, p.97 | EXFOR: F0153007 |
| 155 | 11.08.11 | $^{9}Be(p,d)^{8}Be$ | F.Bertrand+(1968), Rept. Centre d`Etudes Nucleaires, Saclay Reports, No.3575 | EXFOR: F0107004 |
| 154 | 11.08.11 | $^{9}Be(p,a)^{6}Li$ | F.Bertrand+(1968), Rept. Centre d`Etudes Nucleaires, Saclay Reports, No.3575 | EXFOR: F0107005 |
| 153 | 6.08.11 | $^{9}Be(p,p_1)^{9}Be$ | M.Yasue(1974), Jour. Journal of the Physical Society of Japan, Vol.36, p.1254 | EXFOR: F0135002 |
| 152 | 6.08.11 | $^{9}Be(p,d)^{8}Be$ | F.Bertrand+(1968), Rept. Centre d`Etudes Nucleaires, Saclay Reports, No.3575 | EXFOR: F0107003 |
| 151 | 6.08.11 | $^{9}Be(p,a_1)^{6}Li$ | G.M.Temmer(1963), Conf. Nucl. Reaction Mechanisms Conf., Padua 1962, p.1013 | EXFOR: F0010002 |

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| 150 | 6.08.11 | $^{9}$Be(p,a)$^{6}$Li | F.Bertrand+(1968), Rept. Centre d`Etudes Nucleaires, Saclay Reports, No.3575 | EXFOR: F0107002 |
| 149 | 6.08.11 | $^{9}$Be(p,p$_0$)$^{9}$Be | F.H.Read+(1961), Jour. Proceedings of the Physical Society (London),, Vol.77, p.65 | EXFOR: F0090003 |
| 148 | 6.08.11 | $^{9}$Be(p,p$_0$)$^{9}$Be | Tsan Mo+(1969), Jour. Physical Review, Vol.187, p.1220 | EXFOR: F0123002 |
| 147 | 6.08.11 | $^{9}$Be(p,p$_0$)$^{9}$Be | M.Yasue+(1972), Jour. Journal of the Physical Society of Japan, Vol.33, p.265 | EXFOR: F0134004 |
| 146 | 6.08.11 | $^{9}$Be(p,p$_0$)$^{9}$Be | Ju.G.Mashkarov+(1976), Jour. Izv. Rossiiskoi Akademii Nauk, Ser.Fiz., Vol.40, p.1218 | EXFOR: F0474002 |
| 145 | 6.08.11 | $^{9}$Be(p,p$_0$)$^{9}$Be | M.Allab+(1983), Jour. Journal de Physique, Vol.44, p.579 | EXFOR: F0333002 |
| 144 | 6.08.11 | $^{9}$Be(p,p$_0$)$^{9}$Be | A.Kiss+(1977), Jour. Nuclear Physics, Section A, Vol.282, p.44 | EXFOR: F0192003 |
| 143 | 5.08.11 | $^{36}$Ar(p,p$_0$)$^{36}$Ar | N.I.Kaloskamis+(1997), Jour. Physical Review, Part C, Nuclear Physics, Vol.55, p.640 | EXFOR: C0162004 |
| 142 | 5.08.11 | $^{36}$Ar(p,p$_0$)$^{36}$Ar | N.I.Kaloskamis+(1997), Jour. Physical Review, Part C, Nuclear Physics, Vol.55, p.640 | EXFOR: C0162003 |
| 141 | 5.08.11 | $^{36}$Ar(p,p$_0$)$^{36}$Ar | N.I.Kaloskamis+(1997), Jour. Physical Review, Part C, Nuclear Physics, Vol.55, p.640 | EXFOR: C0162002 |
| 140 | 5.08.11 | $^{36}$Ar(p,p$_0$)$^{36}$Ar | J.F.Wilkerson+(1992), Jour. Nuclear Physics, Section A, Vol.549, p.223 | EXFOR: T0066014 |
| 139 | 5.08.11 | $^{40}$Ar(p,p$_0$)$^{40}$Ar | A.C.L.Barnard+(1961), Jour. Nuclear Physics, Vol.28, p.428 | EXFOR: C0859003 |
| 138 | 4.08.11 | $^{27}$Al($^{7}$Li,$^{7}$Li$_0$)$^{27}$Al | M.Mayer+(2003), Jour. Nucl. Instrum. Methods in Physics Res., Sect.B, Vol.207, p.263 | EXFOR: A0521005 |
| 137 | 4.08.11 | $^{27}$Al($^{6}$Li,$^{6}$Li$_0$)$^{27}$Al | M.Mayer+(2003), Jour. Nucl. Instrum. Methods in Physics Res., Sect.B, Vol.207, p.263 | EXFOR: A0521004 |
| 136 | 4.08.11 | $^{27}$Al(a,a$_0$)$^{27}$Al | Cheng Huan-Sheng+(1991), Jour. Nucl. Instrum. Methods in Physics Res., Sect.B, Vol.56, p.749 | EXFOR: O0882002 |
| 135 | 4.08.11 | $^{27}$Al(d,a$_0$)$^{25}$Mg | G.Corleo+(1968), Jour. Nuovo Cimento B, Vol.56, p.83 | EXFOR: D0389004 |
| 134 | 3.08.11 | $^{27}$Al(p,p$_0$)$^{27}$Al | R.O.Nelson+(1984), Jour. Physical Review, Part C, Nuclear Physics, Vol.29, p.1656 | EXFOR: C0686002 |
| 133 | 2.08.11 | $^{27}$Al(p,p$_0$)$^{27}$Al | R.V.Elliott+(1966), Jour. Nuclear Physics, Vol.84, p.209 | EXFOR: D0391002 |
| 132 | 31.05.11 | $^{nat}$Si(p,p$_0$)$^{nat}$Si | B.Becker+(2008), Priv. Private communication, Name.Becker | EXFOR: D0554002 |
| 131 | 31.05.11 |   | A bug in sorting PIGE data was eliminated. The gamma-ray energy was displayed in the table. | |
| 130 | 15.03.11 | $^{16}$O(a,a$_0$)$^{16}$O | A.J.Ferguson+(1940), Jour. Physical Review, Vol.58, p.666 | EXFOR: C0916004 |
| 129 | 15.03.11 | $^{16}$O(a,a$_0$)$^{16}$O | J.R.Cameron(1953), Jour. Physical Review, Vol.90, p.839 | EXFOR: T0243002 |
| 128 | 15.03.11 | $^{16}$O(a,a$_0$)$^{16}$O | W.E.Hunt+(1967), Jour. Physical Review, Vol.160, p.782 | EXFOR: T0242002 |
| 127 | 15.03.11 | $^{16}$O(a,a$_0$)$^{16}$O | E.Berthoumieux+(1998), Jour. Nucl. Instrum. Methods in Physics Res., Sect.B, Vol.136-138, p.55 | EXFOR: F0587002 |
| 126 | 24.01.11 | $^{12}$C(p,p$_0$)$^{12}$C | V.Paneta et al. to be published | Thanks to V.Paneta |
| 125 | 20.01.11 | $^{16}$O(a,a$_0$)$^{16}$O | L.C.Mcdermott+(1960), Jour. Physical Review, Vol.118, p.175 | EXFOR: T0244002 |
| 124 | 29.12.10 |   | The file system was reorganized to meet security requirements | |
| 123 | 20.12.10 | $^{27}$Al($^{7}$Li,$^{7}$Li$_0$)$^{27}$Al | D.Abriola+(2010), Jour. Nucl. Instrum. Methods in Physics Res., Sect.B, Vol.268, p.1793 | EXFOR: D0609002 |
| 122 | 20.12.10 | $^{19}$F(p,p$_0$)$^{19}$F | P.Cuzzocrea+(1980), Jour. Lettere al Nuovo Cimento, Vol.28, p.515 | EXFOR: A0189005 |
| 121 | 27.09.10 | $^{27}$Al(d,a$_0$)$^{25}$Mg | S. Pellegrino et al., Nucl. Instr. and Meth. B 266 (2008) 2268 | Correction of the file al7da0a.r33 |
| 120 | 09.09.10 | $^{35}$Sc(p,p$_0$)$^{35}$Sc | G. Provatas et al. Submitted to NIMB | Thanks to G. Provatas |
| 119 | 08.09.10 | $^{6}$Li(d,a$_0$)$^{4}$He | V. Foteinou et al. Submitted to NIMB | Thanks to V.Foteinou |
| 118 | 19.07.10 | $^{11}$B(p,p$_0$)$^{11}$B | M.Kokkoris et al. Submitted to NIMB | Thanks to M.Kokkoris |

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| | | $^{11}B(p,a_0)^{11}B$ | | |
| 117 | 21.04.10 | $^{27}Al(d,a_{0-4})^{25}Mg$<br>$^{27}Al(d,p_{0-5})^{28}Al$<br>$^{27}Al(p,p_0)^{27}Al$<br>$^{12}C(^3He,^3He_0)^{12}C$<br>$^{12}C(^3He,a_0)^{11}C$<br>$^{12}C(d,d_0)^{12}C$<br>$^{12}C(p,p_0)^{12}C$<br>$^{13}C(d,d_0)^{13}C$<br>$^{13}C(d,g_0)^{15}N$<br>$^{13}C(d,p_1)^{14}C$<br>$^{13}C(p,p_0)^{13}C$<br>$^{19}F(a,p_{0,1})^{22}Ne$<br>$^{54}Fe(p,p_0)^{54}Fe$<br>$^{56}Fe(p,p_0)^{56}Fe$<br>$^{58}Fe(p,p_0)^{58}Fe$<br>$^{25}Mg(d,a_{0-8})^{23}Na$<br>$^{14}N(a,a_0)^{14}N$<br>$^{16}O(^3He,^3He_0)^{16}O$ | J.H.Williamson Nuclear Physics v.69 (1965) 481<br>A.Osman et al. Annalen der Physik v.36 (1979) 64<br>M.K.Mehta et al. Nuclear Physics v.89 (1966) 22<br>R.L.A. Cottrell et al. Nuclear Physics A 109 (1968) 288<br>J.J.Schwartz et al. Nuclear Physics v.88 (1966) 539<br>J.C. Armstrong et al. Physical Review v. 144 (1966) 823<br>W. Del Bianco et al. Canadian Journal of Physics v. 56 (1978) 3<br>V.A.Latorre et al. Physical Review v.144 (1966) 891<br>J.Kuperus, Physica v. 31 (1965) 1603<br>D.P. Lindstrom et al. Nuclear Physics A v.168 (1971) 37<br>W.N.Wang et al. Nuclear Physics A v.102 (1967) 537<br>Ding Furong et al. Chinese Physics Letters v.9 (1992) 176<br>H.Ropke et al. Nuclear Physics A97 (1967) 609 | Thanks to S.Taova |
| 116 | 15.03.10 |   | A new feature was developed to select a type of retrieved data. | |
| 115 | 02.02.10 | $^{11}B(p,p_0)^{11}B$<br>$^{12}C(d,d_0)^{12}C$<br>$^{12}C(d,p_{0-3})^{13}C$<br>$^{12}C(^3He,^3He_0)^{12}C$<br>$^{40}Ca(a,p_{0-5})^{43}Sc$<br>$^{40}Ca(d,p_1)^{41}Ca$<br>$^{40}Ca(^3He,a_0)^{39}Ca$<br>$^{40}Ca(^3He,^3He_0)^{40}Ca$<br>$^{41}Ca(a,a_0)^{41}Ca$<br>$^{48}Ca(a,a_0)^{48}Ca$<br>$^{39}K(a,a_0)^{39}K$<br>$^{39}K(p,p_{0,1,4})^{39}K$<br>$^{18}O(p,a_0)^{15}N$<br>$^{18}O(p,p_{0,1})^{18}O$<br>$^{32}S(d,p_{0-7})^{33}S$<br>$^{30}Si(d,p_0)^{31}Si$ | Y. Rihet et al. Le Jurnal De Fhysique v.38 (1977) 17<br>D.G. Gerke et al. Nuclear Physics v.75 (1966) 609<br>J.P. Schapira et al. Nuclear Physics v.80 (1966) 565<br>D.Frekers et al. Nuclear Physics A v.394 (1983) 189<br>I.Fodor et al. Nuclear Physics v.73 (1965) 155<br>D. Cline et al. Nuclear Physics v.73 (1965) 33<br>S.S.Vasilyev et al. Izvestiya Akademii Nauk v.32 (1968) 2013<br>S.N.Wosu et al. Nuclear Instruments and Methods in Physics Research B v.237 (2005) 631<br>J. Stevens et al. Nuclear Physics v.76 (1966) 129<br>H.R.Saad et al. Nuclear Physics v.84 (1966) 629<br>A.K.Valter et al. Eksperimental'noi i Teoret. Fiziki v.43 (1962) 2038 | Thanks to S.Taova |
| 114 | 22.01.10 |   |   | Files for proton elastic scattering from cromium (W.R.Wylie et al.) were corrected. |
| 113 | 19.11.09 | $^{nat}K(p,p_0)^{nat}K$<br>$^{39}K(p,a_0)^{36}Ar$ | M.Kokkoris et al. To be published | Thanks to M.Kokkoris |
| 112 | 18.11.09 |   | New feature was developed to select a range of interest in the plots. | |
| 111 | 12.10.09 | $^{27}Al(p,a_{0-5})^{24}Mg$<br>$^{10}B(p,p_0)^{10}B$<br>$^{11}B(a,a_0)^{11}B$<br>$^{11}B(^3He,a_{0-4})^{10}B$<br>$^{12}C(d,d_0)^{12}C$<br>$^{13}C(d,g_0)^{15}N$<br>$^{19}F(p,a_{0-4})^{16}O$<br>$^{19}F(p,p_{0,1,3-5})^{19}F$<br>$^{25}Mg(d,a_{0-4,6})^{23}Na$<br>$^{14}N(a,a_0)^{14}N$ | G.P. Lawrence et al. Nuclear Physics v.65 (1965) 275<br>G.B. Andreev et al. Izvestiya Akademii Nauk v.26 (1962) 1134<br>N. Moncoffre et al. Nucl. Instr. and Meth. B v.45 (1990) 81<br>J.A. Lonergan et al. Nuclear Physics, v. 88 (1966) 465<br>H.Cords et al. Nuclear Physics A v.127 (1969) 95<br>W.Del Bianco et al. Nuclear Physics A v.270 (1976) 45<br>H.M. Kuan et al. Nuclear Physics A v.193 (1972) 497<br>V.Y.Gontchar et al. Nuclear Physics v.62 | Thanks to S.Taova |

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| | | $^{15}N(d,a_{0,1})^{13}C$<br>$^{16}O(d,a_0)^{14}N$<br>$^{16}O(^3He,a_0)^{15}O$ | (1965) 410<br>N.P. Heydenburg et al. Physical Review v.92 (1953) 89<br>N.A.Mansour et al. Nuclear Physics v.65 (1965) 433<br>W.P. Alford et al. Nuclear Physics 61 (1965) 368 | |
| 110 | 21.08.09 | $^{14}N(a,p_0)^{17}O$ | D.F.Herring et al. Phys. Rev. 112 (1958) 1210 | EXFOR C1363003 |
| 109 | 03.08.09 | $^{36}Ar(p,p_0)^{36}Ar$<br>$^{10}B(p,a_0)^{13}C$<br>$^{11}B(a,p_0)^{14}C$<br>$^{12}C(a,ag)^{12}C$<br>$^{12}C(a,g)^{16}O$<br>$^{13}C(d,p_{1,3,6})^{14}C$<br>$^{52}Cr(p,p_{0,1})^{52}Cr$<br>$^{19}F(a,p_0)^{22}Ne$<br>$^{19}F(d,a_{0,1,2,3})^{17}O$<br>$^{19}F(p,p_0)^{19}F$<br>$^{16}O(a,gp_0)^{20}Ne$<br>$^{16}O(d,d_0)^{16}O$<br>$^{16}O(d,p_{0,1,3})^{17}O$<br>$^{16}O(p,g_0)^{17}O$<br>$^{16}O(p,p_0)^{16}O$ | V.E.Storizhko et al. Izvestiya Academii Nauk v.28 (1964) 1145<br>L.C.McIntyre et al. Nicl. Instr. and Meth. B66 (1992) 221<br>I.V.Mitchell, et al. Nucl. Phys. 58 (1964) 529<br>J.D.Larson, et al. Nucl. Phys. 56 (1964) 497<br>J.M. Lacambra, et al. Nucl. Phys. 68 (1965) 273<br>Y.Ozawa et al. Nucl. Phys. A440 (1985) 13<br>W.A. Schier, et al. Nucl. Phys. A226 (1976) 16<br>P.R Chagnon, Nucl. Phys. 59 (1964) 257<br>Y. Takeuchi, et al. Nucl. Phys. A109 (1968) 105<br>M.F. Jahns, et al. Nucl. Phys. 59 (1964) 314<br>W.J. Thompson, et al. Nucl. Phys. A105 (1967) 678<br>C.Van Der Leun et al. Phys. Let. 18 (1965) 134<br>N.E.Davison, et al. Can. J. Phys. 48 (1970) 2235<br>R.Morlock et al. Phys. Rev. Let. 79 (1997) 3837 | Thanks to S.Taova |
| 108 | 21.05.09 | $^{23}Na(d,a_{3+4})^{21}Ne$ | G.L. Vysotskij et al. Izvestiya Academii Nauk v.34 (1970) 1761 | Files na3da3a.r33, na3da3b.r33, na3da3c.r33, na3da3d.r33 were corrected. Thanks to M.Mayer for pointing out the mistake. |
| 107 | 15.05.09 | $^{20}Ne(p,p_0)^{20}Ne$ | A.K. Valter et al. Izvestiya Academii Nauk v.24 (1960) 884 | File ne0pp0d.r33 was corrected. Thanks to M.Mayer for pointing out the mistake. |
| 106 | 11.05.09 |   | Error bars were added in plots according to the 2nd and 4th columns of the R33 files | |
| 105 | 05.05.09 | $^{14}N(a,a_0)^{14}N$ | I.Bogdanovic et al., to be published | Thanks to Iva Bogdanovic. |
| 104 | 04.05.09 | $^{12}C(^3He,p_0)^{14}N$ | R.L.Johnston et al. Physical Review v.109 (1958) 884 | File c2hp0i.r33 was corrected. Thanks to M.Mayer for pointing out the mistake. |
| 103 | 21.04.09 | $^{27}Al(p,a_0)^{24}Mg$<br>$^{27}Al(p,a_1)^{24}Mg$<br>$^{12}C(a,a_0)^{12}C$<br>$^{12}C(d,d_0)^{12}C$<br>$^{12}C(d,p_{0,1,2,3})^{13}C$<br>$^{12}C(p,p_1)^{12}C$<br>$^{40}Ca(a,a_0)^{40}Ca$<br>$^{35}Cl(p,a_0)^{32}S$<br>$^{35}Cl(p,p_0)^{35}Cl$<br>$^{37}Cl(p,a_0)^{34}S$<br>$^{52}Cr(p,p_{0,1})^{52}Cr$<br>$^{26}Mg(p,a_{0,1})^{23}Na$<br>$^{26}Mg(p,g)^{27}Al$<br>$^{14}N(^3He,p_0)^{16}O$<br>$^{20}Ne(d,p_{5,6,7})^{21}Ne$<br>$^{22}Ne(p,p_{0,1})^{22}Ne$<br>$^{16}O(d,a_0)^{14}N$<br>$^{16}O(d,p_{0,1})^{17}O$ | M.A.Abuzeid et al. Nuclear Physics v.45 (1963) 123<br>K.L.Warsh et al. Nuclear Physics v.44 (1963) 329<br>J.M. Morris et al. Nuclear Physics A v.112 (1968) 97<br>G.G.Ohlsen et al. Nuclear Physics v.45 (1963) 523<br>J.M.F.Jeronymo et al. Nuclear Physics v.43 (1963) 417<br>F.C.Barker et al. Nuclear Physics v.45 (1963) 449<br>D.Frekers et al. Nuclear Physics A v.394 (1983) 189<br>K.V.Karadzhev et al. Yadernaya Fizika v.4 (1966) 909<br>J.H. Bjerregaard et al. Nuclear Physics v.51 (1964) 641<br>L.G.Lawrence et al. Nuclear Physics v.44 (1963) 518<br>A.S.Kachan et al. Izvestiya Akademii Nauk v.72 (2008) 1630<br>S.Gorodetzky et al. Nuclear Physics v.43 (1963) 92<br>M.Lambert et al. Nuclear Physics A v.112 (1968) 161 | Thanks to S.Taova |

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| | | $^{18}O(p,a_0)^{15}N$<br>$^{28}Si(d,p_{0,1,2,3,4})^{29}Si$<br>$^{29}Si(p,p_0)^{29}Si$<br>$^{10}B(p,ag)^{7}Be$<br>$^{10}B(p,pg)^{10}B$ | M.A.Abuzeid et al. Nuclear Physics v.50 (1964) 106<br>G.Calvi, et al. Nuclear Physics v.38 (1962) 436<br>S.Gorodetzky et al. Nuclear Physics v.42 (1963) 462<br>C.C.Hsu et al. Physical Review C v.7 (1973) 1425<br>V.E. Storizhko et al. Izvestiya Akademii Hauk v. 28 (1964)<br>T.R.Ophel et al. Nuclear Physics v.33 (1962) 198 | |
| 102 | 02.02.09 | $^{16}O(a,a_0)^{16}O$ | H.Yonezawa et al. Nucl. Instrum. Methods in Physics Res., Sect.B, Vol.88, p.207 | EXFOR: F0117003 |
| 101 | 21.01.09 | $^{27}Al(d,a_0)^{25}Mg$<br>$^{27}Al(p,a_0)^{24}Mg$<br>$^{36}Ar(p,p_0)^{36}Ar$<br>$^{9}Be(p,g)^{10}B$<br>$^{12}C(d,p_0)^{13}C$<br>$^{12}C(^{3}He,p_{0,1,2})^{14}N$<br>$^{13}C(d,a_0)^{11}B$<br>$^{40}Ca(p,p_{1,2,3})^{40}Ca$<br>$^{35}Cl(p,a_0)^{32}S$<br>$^{6}Li(a,a_0)^{6}Li$<br>$^{7}Li(p,p_0)^{7}Li$<br>$^{16}O(a,a_0)^{16}O$<br>$^{16}O(p,p_0)^{16}O$<br>$^{31}P(a,p_0)^{34}S$<br>$^{31}P(d,p)^{32}P$<br>$^{31}P(p,a_0)^{28}Si$<br>$^{31}P(p,p_0)^{31}P$<br>$^{29}Si(p,p_0)^{29}Si$<br>$^{51}V(p,p_0)^{51}V$ | M.A.Abuzeid et al. Izvestiya Academii Nauk v.28 (1964) 1717<br>S.S.Vasilyev et al. Izvestiya Academii Nauk v.30 (1966) 1399<br>V.E.Storizhko et al. Izvestiya Academii Nauk v.28 (1964) 1146<br>R.J.Ellison et al. Nuclear Physics v.35 (1962) 606<br>A.P. Klyucharev et al. Izvestiya Akademii Nauk v.30 (1966) 224<br>R.L.Johnston et al. Physical Review v.109 (1958) 884<br>A.P. Klyucharev et al. Izvestiya Akademii Nauk v.30 (1966) 435<br>Yu.Kasanyu et al. Izvestiya Academii Nauk v.32 (1968) 1650<br>B.Bosnjakovic et al. Nuclear Physics A v.110 (1968) 17<br>G.Dearnaley et al. Nuclear Physics v.36 (1962) 71<br>G.B.Andreev et al. Izvestiya Akademii Nauk v.29 (1965) 210<br>Cheng Huan-Sheng et al. Acta Physica Sinica v.2 (1993) 641<br>R.W.Harris et al. Nuclear Physics v.38 (1962) 259<br>W.A. Schier et al. Nuclear Physics A v.254 (1975) 80<br>A.T. Baranik et al. Nuclear Physics v.72 (1965) 49<br>J. Vernotte et al. Nuclear Physics A v.212 (1973) 493<br>M.Cenja et al. Izvestiya Academii Nauk v.47 (1983) 2143<br>A.N.L'vov et al. Izvestiya Akademii Nauk v.30 (1966) 439<br>C.M.Lamba et al. Nuclear Physics A v.110 (1968) 111 | Thanks to S.Taova. |
| 100 | 12.12.08 | $^{14}N(a,a)^{14}N$<br>$^{14}N(a,p_{0,1})^{17}O$ | G. Terwagne, G. Genard, M.Yedji and G.G. Ross, J. Appl. Phys. 104, 084909, 2008 | Thanks to Guy Terwagne |
| 99 | 07.12.08 |   | New complete CD version of IBANDL has been prepared. | |
| 98 | 20.10.08 | $^{31}P(p,p)^{31}P$<br>$^{19}F(p,p)^{19}F$<br>$^{19}F(p,a)^{16}O$<br>$^{31}P(d,a)^{29}Si$<br>$^{12}C(d,p)^{13}C$<br>$^{50}Cr(p,p)^{50}Cr$<br>$^{36}Ar(p,p)^{36}Ar$<br>$^{12}C(a,a)^{12}C$<br>$^{23}Na(d,a)^{21}Ne$ | D.F. Fang, Phys. Rev. C37 (1988) 28<br>S.Ouichaoui et al. Nuovo Cim.A v.86 (1985) 170<br>M.A.Abuzeid et al. Nuclear Physics v.60 (1964) 264<br>T.W.Bonner, Phys. Rev, 101 (1956) 209<br>E.A Romanovskij et al. Izvestiya Akademii Nauk v.41 (1977) 211<br>C.C.Kim et al. Nuclear Physics v.28 (1961) 438<br>Ph. Martin et al. Nuclear Physics A v.202 (1973) 257<br>G.L. Vysotskij et al. Izvestiya Academii Nauk v.34 (1970) 1761 | Thanks to S.Taova. |
| 97 | 13.10.08 | $^{14}N(d,p_3)^{15}N$ | Nucl.Instrum.Methods B51 (1990) 152 | Mistakes were corrected in the reference and in the Q-value. Thanks to Frans Munnik. |
| 96 | 01.09.08 | $^{31}P(p,p)^{31}P$ | D.F. Fang et al. Physical Review C v.37 (1988) 28 | Thanks to S.Taova |

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|---|---|---|---|---|
| | | $^{31}$P(p,a)$^{28}$Si | | |
| 95 | 25.08.08 | $^{14}$N(p,p)$^{14}$N | To be published in J. Appl. Phys. | Thanks to Iva Bogdanovic Radovic |
| 94 | 28.07.08 | $^{29}$F(p,p)$^{19}$F | A.P. Jesus et al. Nucl. Instr. and Meth. B 174 (2001) 229 |   |
| 93 | 16.07.08 | $^{19}$F(p,p)$^{19}$F<br>$^{19}$F(p,a)$^{16}$O<br>$^{23}$Na(p,p)$^{23}$Na<br>$^{23}$Na(p,a)$^{20}$Ne<br>$^{27}$Al(d,p)$^{28}$Al<br>$^{27}$Al(d,a)$^{25}$Mg<br>$^{28}$Si(d,p)$^{29}$Si | S.Ouichaoui et al. Nuovo Cim.A v.94 (1986) 133<br>M.Cortni et al. Nuclear Physics v.77 (1966) 625<br>S.S.Vasilyev et al. Izvestiya Academii Nauk v.32 (1968) 587<br>B.H.Wildenthal et al. Physical Review v.135 (1964) B680 | Thanks to S.Taova. |
| 92 | 14.07.08 | $^{11}$B(d,a)$^{9}$Be<br>$^{11}$B(d,p)$^{12}$B<br>$^{28}$Si(d,p)$^{29}$Si | to be published | Thanks to M.Kokkoris. |
| 91 | 27.05.08 | $^{14}$N(d,$a_{0,1}$)$^{12}$C<br>$^{14}$N(d,$p_0$)$^{15}$N | to be published | Thanks to M.Kokkoris. |
| 90 | 25.03.08 | $^{9}$Be(p,$p_0$)$^{9}$Be<br>$^{13}$C(a,$a_0$)$^{13}$C<br>$^{13}$C(a,$d_0$)$^{15}$N<br>$^{14}$N(p,$p_0$)$^{14}$N<br>$^{14}$N(a,$p_{0,1,2,3}$)$^{17}$O<br>$^{20}$Ne(p,$p_0$)$^{20}$Ne<br>$^{23}$Na(p,$p_0$)$^{23}$Na<br>$^{25}$Mg(p,$p_0$)$^{25}$Mg<br>$^{26}$Mg(p,$p_0$)$^{26}$Mg<br>$^{28}$Si(p,$p_0$)$^{28}$Si<br>$^{34}$S(p,$p_0$)$^{34}$S<br>$^{35}$Cl(p,$p_0$)$^{35}$Cl<br>$^{38}$Ar(p,g)$^{39}$K<br>$^{40}$Ar(p,$p_0$)$^{40}$Ar<br>$^{40}$Ca(p,$p_0$)$^{40}$Ca<br>$^{39}$K(p,$p_0$)$^{39}$K<br>$^{46}$Ti(p,$pg_{1-0}$)$^{46}$Ti<br>$^{48}$Ti(p,$pg_{1-0}$)$^{48}$Ti<br>$^{48}$Ti(p,$p_0$)$^{48}$Ti<br>$^{50}$Ti(p,$p_0$)$^{50}$Ti | P.M. Gopych et al. Yadernaya Fizika v. 14 (1971) 247<br>V.M. Lebedev et al. Izvestiya Akademii Nauk v. 37 (1973) 2663<br>A.K.Valter et al. Izvestiya Academii Nauk v.23 (1959) 839<br>N.S.Zelenskaya et al. Yadernaya Fizika v.11, (1970) 722<br>A.K. Valter et al. Izvestiya Academii Nauk v.24 (1960) 884<br>M.Cenja et al. Nuclear Physics A v.307 (1978) 65<br>S.S.Vasilyev et al. Izvestiya Academii Nauk v.31 (1967) 295<br>S.S.Vasilyev et al. Yadernaya Fizika v.6 (1967) 1134<br>A.V. Spassky et al. Yadernaya Fizika v. 7 (1968) 251<br>K.V. Karadzhev et al. Yadernaya Fizika v.22 (1975) 886<br>A.K. Valter et al. Izvestiya Academii Nauk v.23 (1959) 846<br>E. Koltay et al. Nuclear Physics A249 (1975) 173<br>R.J.De Meijer et al. Nuclear Physics A v.155 (1970) 109<br>A.I. Popov et al. Izvestiya Academii Nauk v.26 (1962) 1074<br>G.A.Krivonosov et al. Yadernaya Fizika v.24 (1976) 461<br>S. Maripuu Nuclear Physics A v. 153 (1970) 183<br>S. Maripuu Nuclear Physics A v. 151 (1970) 465 | Thanks to S.Taova |
| 89 | 08.03.08 | $^{7}$Li(p,$p_0$)$^{7}$Li<br>$^{19}$F(p,$p_0$)$^{19}$F<br>$^{23}$Na(p,$p_0$)$^{23}$Na | U. Fasoli et al., Nuovo Cimento 34 (1964) 542<br>P.R. Malmberg, Phys. Rev. 101 (1956) 114<br>S. Bashkin and H.T. Richards, Phys. Rev. 84 (1951) 1124<br>G. Dearnaley, Philos. Mag. ser. 8, 1 (1956) 821 | Thanks to M.Chiari |
| 88 | 07.03.08 | $^{23}$Na(p,p'g)$^{23}$Na<br>$^{23}$Na(p,$p_0$)$^{23}$Na | Nucl. Instr. and Meth. in press | Thanks to M.Chiari |
| 87 | 29.01.08 | $^{9}$Be(d,$a_0$)$^{7}$Li<br>$^{12}$C($^{3}$He,$^{3}$He$_0$)$^{12}$C<br>$^{13}$C($^{3}$He,$a_{0,1,2}$)$^{12}$C<br>$^{13}$C($^{3}$He,$^{3}$He$_0$)$^{13}$C<br>$^{13}$C($^{3}$He,$p_{0,1}$)$^{15}$N<br>$^{14}$N(d,$a_{0,1}$)$^{12}$C<br>$^{19}$F(a,$a_0$)$^{19}$F<br>$^{19}$F(a,$p_0$)$^{22}$Ne | A.S. Dejneko et al. Izvestiya Akademii Nauk v.44 (1980) 2382<br>H.R. Weller et al. Nuclear Physics A122 (1968) 529<br>L. Zolnai et al. Izvestiya Akademii Nauk v.44 (1980) 2281<br>J.R. Vanhoy et al. Physical Review C v.36 (1987) 920<br>H.M. Omar et al. Nuclear Physics v.56 (1964) 97<br>G.K. Khomyakov et al. Yadernaya Fizika v.20 | Thanks to S.Taova |

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| | | $^{26}$Mg(p,p$_0$)$^{26}$Mg<br>$^{26}$Mg(d,p$_{6+7}$)$^{27}$Mg<br>$^{23}$Na(p,a$_0$)$^{20}$Ne<br>$^{20}$Ne(p,g)$^{21}$Na<br>$^{30}$Si(d,p$_{0,1}$)$^{31}$Si<br>$^{46}$Ti(p,p$_{0-3}$)$^{46}$Ti<br>$^{48}$Ti(p,p$_{0,1}$)$^{48}$Ti | (1974) 846<br>C.Van Der Leun et al. Physica v.30 (1964) 333<br>K.F. Ustimenkov et al. Izvestiya Academii Nauk v.45 (1981) 2150<br>E.A. Antropov et al. Izvestiya Akademii Nauk v.40 (1976) 807 | |
| 86 | 17.01.08 |   | Automatic conversion from EXFOR to R33 (by V.Zerkin) is open for testing. | |
| 85 | 04.01.08 | $^{12}$C(d,p$_0$)$^{13}$C | to be published | Thanks to Jose A. R. Pacheco de Carvalho |
| 84 | 16.11.07 | $^{9}$Be(p,p$_0$)$^{9}$Be<br>$^{12}$C(d,d$_0$)$^{12}$C<br>$^{19}$F(d,p$_{0,2,5}$)$^{20}$F<br>$^{23}$Na(d,p$_{0,8,9}$)$^{24}$Na<br>$^{23}$Na(p,p$_{0,1}$)$^{23}$Na<br>$^{23}$Na(p,a$_{0,1}$)$^{20}$Ne | V.S.Siksin et al. Yadernaya Fizika v.12 (1970) 15<br>V.I.Borovlev et al. Yadernaya Fizika v.14 (1971) 47<br>B.M. Zabegaj et al. Yadernaya Fizika v.11 (1970) 277<br>J.R. Vanhoy et al. Physical Review C v.36 (1987) 920 | Thanks to S.Taova |
| 83 | 31.10.07 | $^{12}$C(a,p$_0$)$^{15}$N<br>$^{24}$Mg(p,p$_0$)$^{24}$Mg<br>$^{26}$Mg(p,p$_0$)$^{26}$Mg<br>$^{27}$Al(p,p$_0$)$^{27}$Al<br>$^{30}$Si(d,p$_{0,1}$)$^{31}$Si<br>$^{31}$P(d,a$_0$)$^{29}$Si<br>$^{48}$Ca(d,d$_0$)$^{48}$Ca | Izvestiya Academii Nauk v.35 (1971) 193<br>Izvestiya Academii Nauk v.38 (1974) 868<br>Izvestiya Academii Nauk v.38 (1974) 868<br>Izvestiya Academii Nauk v.38 (1974) 130<br>Atomkernenergie v.11-39 (1966) 238<br>Izvestiya Academii Nauk v.37 (1973) 1633<br>Nuclear Physics A160 (1971) 289 | Thanks to S.Taova |
| 82 | 20.09.07 | $^{12}$C(d,p$_1$)$^{13}$C | Izvestiya Academii Nauk v.35 (1971) 1670 | Thanks to S.Taova |
| 81 | 21.08.07 | $^{12}$C(p,p$_0$)$^{12}$C | H.L.Jackson et al. Phys. Rev. 89 (1953) 365 | 146.2o EXFOR C1000002 |
| 80 | 25.07.07 | $^{10}$B(d,a$_0$)$^{8}$Be | M.N.H.Comsan+(1968), Atomkernenergie, Vol.13, p.415<br>J.B.Marion+(1956), Physical Review, Vol.103, p.1408<br>G.Debras+(1977), J.of Radioanalytical Chemistry, Vol.38, p.193 | EXFOR |
| 79 | 21.07.07 | $^{nat}$Si(a,a$_0$)$^{nat}$Si | K.-M.Kallman, Zeitschrift fuer Physik, A356 (1996) 287 | EXFOR O0966002 |
| 78 | 19.07.07 |   | IBANDL and SigmaCalc were combined. Now theoretical and experimental data can be easily compared | |
| 77 | 15.07.07 | $^{nat}$Si(a,a$_0$)$^{nat}$Si | A. Coban et al. Nucl. Phys. A678 (2000) 3 | EXFOR F0602001 |
| 76 | 10.07.07 | $^{16}$O($^{3}$He,a$_0$)$^{15}$O<br>$^{16}$O($^{3}$He,p$_{0-6}$)$^{18}$F | W.N. Lennard et al. Nucl. Instr. & Meth. B43 (1989) 187 | Differential cross sections for 4 angles measured at one energy point are listed in "comment" section of R33 files |
| 75 | 06.07.07 |   | New graphics design | |
| 74 | 26.06.07 | $^{10}$B(a,a)$^{10}$B<br>$^{10}$B(p,p)$^{10}$B<br>$^{11}$B(a,a)$^{11}$B<br>$^{11}$B(p,p)$^{11}$B | EXFOR | Thanks to M.Mayer |
| 73 | 10.06.07 | $^{16}$O(d,a)$^{14}$N | G.Amsel, Thesis, Ann.Phys., 9(1964), 297 | Theta values 165 and 135 were interchanged |
| 72 | 22.05.07 | $^{14}$N(d,p)$^{15}$N<br>$^{14}$N(d,a)$^{12}$C | To be published |   |
| 71 | 13.04.07 | $^{32}$S(d,p$_{0-3,7}$)$^{33}$S | Submitted to Nucl. Instrum. Meth. | Thanks to M.Kokkoris. |
| 70 | 04.04.07 | $^{27}$Al(p,p$_0$)$^{27}$Al | To be published in CAARI-06 Proceedings | Thanks to Z.Siketic |
| 69 | 20.03.07 | $^{10}$B(d,a$_{0,1}$)$^{8}$Be<br>$^{10}$B(d,p$_{0-6}$)$^{11}$B | Submitted to Nucl. Instrum. Meth. | Thanks to M.Kokkoris. |
| 68 | 20.03.07 | $^{16}$O(d,p$_1$)$^{17}$O | Il Nuovo Cimento 14A (1973) 692 | EXFOR SUBENT D0152001 |
| 67 | 13.02.07 | $^{13}$C(d,a$_{0,1}$)$^{11}$B<br>$^{13}$C(d,p$_0$)$^{14}$C | Nucl. Instrum. Meth. B254 (2007) 25 | The (d,a) cross sections are new and the (d,p$_0$) data are corrected. Thanks to |

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| | | | | Tristan Thome. |
| 66 | 30.01.07 | $^{nat}$B(a,a)$^{nat}$B<br>$^{12}$C(a,a)$^{12}$C<br>$^{14}$N(a,a)$^{14}$N<br>$^{16}$O(a,a)$^{16}$O<br>$^{1}$H(a,p)$^{4}$He | C.J.Wetteland et al. LA-UR-98-4867 |   |
| 65 | 27.01.07 | $^{27}$Al(d,p)$^{28}$Al<br>$^{27}$Al(d,a)$^{25}$Mg<br>$^{27}$Al(p,g)$^{28}$Si | to be published |   |
| 64 | 11.01.07 | $^{16}$O(a,a$_0$)$^{16}$O | J. Appl. Phys. 100 (2006) 124909 | File for 170$^{o}$ was corrected |
| 63 | 03.01.07 | $^{16}$O(a,a$_0$)$^{16}$O | J. Appl. Phys. 100 (2006) 124909 | Thanks to Julien Demarche |
| 62 | 06.10.06 | $^{16}$O(a,a$_0$)$^{16}$O | Feng at al., 1994 | Comment concerning the data source was added |
| 61 | 27.07.06 | $^{7}$Li(p,p'g)$^{7}$Li<br>$^{7}$Li(p,ng)$^{7}$Be<br>$^{19}$F(p,p'g)$^{19}$F<br>$^{19}$F(p,ag)$^{16}$O | Nucl. Instr. and Meth. B 249 (2006) 98 | Thanks to M.Chiari |
| 60 | 27.07.06 | $^{7}$Li(p,p$_0$)$^{7}$Li<br>$^{12}$C(p,p$_0$)$^{12}$C<br>$^{19}$F(p,p$_0$)$^{19}$F | Nucl. Instr. and Meth. B 249 (2006) 95 | Thanks to M.Chiari |
| 59 | 20.07.06 | $^{14}$N(a,p$_0$)$^{17}$O | Nucl. Instr. and Meth. B 249 (2006) 85 | Thanks to Peng Wei |
| 58 | 29.05.06 | $^{12}$C(d,p$_0$)$^{13}$C | Nucl. Instr. and Meth. B 227 (2005) 450 |   |
| 57 | 28.05.06 | $^{12}$C(d,p$_0$)$^{13}$C | Phys.Rev. 104 (1956) 1008 | Data for 160$^{o}$ scattering angle were added (EXFOR C0993006) |
| 56 | 27.05.06 | $^{12}$C(d,p$_0$)$^{13}$C | Phys.Rev. 117 (1960) 1289 | Data for 158.4$^{o}$ scattering angle were added (EXFOR C1007003) |
| 55 | 22.05.06 | $^{12}$C(d,p$_{1-3}$)$^{13}$C | to be published | Q-values were corrected in Kokkoris' files (see #44 in this table). |
| 54 | 16.05.06 | $^{16}$O(d,p$_0$)$^{17}$O<br>$^{16}$O(d,p$_1$)$^{17}$O<br>$^{16}$O(d,a$_0$)$^{14}$N | J. Rad. Chem. 38 (1977) 193 |   |
| 53 | 16.05.06 | $^{16}$O(d,p$_0$)$^{17}$O<br>$^{16}$O(d,p$_1$)$^{17}$O<br>$^{16}$O(d,a$_0$)$^{14}$N | Nucl.Phys. 45 (1963) 647 |   |
| 52 | 16.05.06 | $^{14}$N(p,p$_0$)$^{14}$N | Phys. Rev. 114 (1959) 1552 | The file with mistakes was substituted by a new one. Thanks to Rita Ramos. |
| 51 | 14.05.06 | $^{16}$O(d,p$_{0,1}$)$^{17}$O | Nucl. Instr. and Meth. B 226 (2004) 637 | Mistakes were corrected in the data in the energy range of 1330-1620 keV. |
| 50 | 12.05.06 | $^{16}$O(d,p$_0$)$^{17}$O<br>$^{16}$O(d,p$_1$)$^{17}$O<br>$^{16}$O(d,a$_0$)$^{14}$N | Nucl. Phys. 57 (1964) 526 |   |
| 49 | 10.05.06 | $^{14}$N(a,a$_0$)$^{14}$N | Phys. Rev. 112 (1958) 1210.<br>Nucl. Instr.& Meth. B66 (1992) 237<br>Nucl.Instr.& Meth. B71 (1992) 324 | 4 data files added. Thanks to Rita Ramos. |
| 48 | 09.05.06 | $^{16}$O(d,p$_1$)$^{17}$O | Nucl. Instr. and Meth. B43 (1989) 187 |   |
| 47 | 09.05.06 | $^{16}$O(d,p$_1$)$^{17}$O<br>$^{12}$C(d,p$_0$)$^{13}$C | Nucl. Instr. and Meth. 168 (1980) 611 |   |
| 46 | 08.05.06 | $^{12}$C(d,p$_0$)$^{13}$C<br>$^{16}$O(d,p$_1$)$^{17}$O | Nucl. Instr. and Meth. B61 (1991) 1 |   |
| 45 | 06.05.06 | $^{14}$N(p,p$_0$)$^{14}$N | Phys. Let. 24B (1967) 287<br>Phys. Rev. 115 (1959) 1655<br>Phys. Rev. 105 (1957) 210<br>Phys. Rev. 105 (1957) 219<br>Phys. Rev. 112 (1958) 475 | 13 data files added. Thanks to Rita Ramos. |

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| 44 | 03.05.06 | $^{12}C(d,p_{1-3})^{13}C$ | to be published | Thanks to M.Kokkoris |
| 43 | 30.03.06 | $^{6}Li(^{3}He,p_{0,1})^{8}Be$ | Phys.Rev. 104 (1956) 1064 | Correction of mistakes. Thanks to M.Mayer for pointing out the mistakes. |
| 42 | 07.03.06 |   | Modification of the files was perfomed by Ian Vickridge because of the update of the R33 format. | |
| 41 | 25.11.05 | $^{2}H(^{3}He,p_{0})^{4}He$ | Nucl. Instr. Meth. B234 (2005) 169 | Thanks to M.Mayer |
| 40 | 24.10.05 | $^{12}C(p,p_{0})^{12}C$ | Nucl. Instr. and Meth. B 77 (1993) 110 | Data for 150$^{o}$ and 110$^{o}$ scattering angles were added |
| 39 | 24.10.05 | $^{12}C(p,p_{0})^{12}C$ | Nucl. Instr. and Meth. B 227 (2005) 450 | See comment in NIM B 229 (2005) 157 |
| 38 | 14.10.05 | $^{9}Be(p,a_{0})^{6}Li$<br>$^{9}Be(p,d_{0})^{8}Be$ | Phys. Rev. 75 (1949) 1612 |   |
| 37 | 14.10.05 | $^{13}C(^{3}He,p_{0})^{15}N$ | Phys. Rev. 107 (1957) 538 |   |
| 36 | 14.10.05 | $^{13}C(p,p_{0})^{13}C$ | Phys. Rev. 122(3) (1961) 884 | Special thanks to M.Mayer |
| 35 | 14.10.05 | $^{12}C(p,p_{0})^{12}C$ | Nucl.Instr.& Meth. B 136-138 (1999) 86 |   |
| 34 | 14.10.05 | $^{11}B(p,p_{0})^{11}B$ | Nucl.Instr.& Meth. B 184 (2001) 309 |   |
| 33 | 14.10.05 | $^{10}B(p,p_{0})^{10}B$ | Nucl.Instr.& Meth. B 184 (2001) 309 |   |
| 32 | 10.10.05 | $^{24}Mg(p,p_{0})^{24}Mg$ | Chin. J. of Phys. 10 (1972) 1 | EXFOR O1030001 |
| 31 | 06.10.05 | $^{24}Mg(p,p_{0})^{24}Mg$ | JETP 44 (1963) 57 |   |
| 30 | 05.10.05 | $Mg(p,p_{0})Mg$ | Nucl.Instr.& Meth. B 201 (2003) 551 |   |
| 29 | 25.07.05 | $^{13}C(d,p_{0})^{14}C$ | to be published | Thanks to Tristan Thome |
| 28 | 05.07.05 | $^{12}C(d,d_{0})^{12}C$ | to be published | Thanks to M.Kokkoris |
| 27 | 24.05.05 | $^{12}C(d,p_{0})^{13}C$ | to be published | Thanks to M.Kokkoris |
| 26 | 12.04.05 | PIGE and DIGE data | A lot of different papers |   |
| 25 | 05.04.05 | $^{2}H(^{3}He,p_{0})^{4}He$ | Nucl.Instr.& Meth. B in press |   |
| 24 | 18.03.05 | $^{32}S(a,p_{0})^{35}Cl$ | Nucl.Instr.& Meth. B 113 (1966) 399 | EXFOR O0870001 |
| 23 | 15.03.05 | $^{23}Na(d,p_{0})^{24}Na$ | Nucl.Phys. v.74 (1965) 225 |   |
| 22 | 29.01.05 | $^{1}H(^{7}Li,^{1}H)^{7}Li$ | Nucl. Instr. Meth. B 229 (2005) 180 | Thanks to Iva Bogdanovic Radovic |
| 21 | 14.12.04 | $^{27}Al(p,p_{0})^{27}Al$ | J. of Radioanal. and Nucl. Chem. 250 (2001) 555 | EXFOR O1076002 |
| 20 | 10.12.04 | $^{14}N(a,p_{0})^{17}O$ | Nucl. Instr. Meth. B149 (1999) 390 |   |
| 19 | 09.12.04 | $^{3}H(a,t)^{4}He$ | Nucl. Instr. Meth. B219-220 (2004) 317 |   |
| 18 | 09.12.04 | $^{2}H(a,d)^{4}He$ | Nucl. Instr. Meth. B219-220 (2004) 317 |   |
| 17 | 09.12.04 | $^{1}H(a,p)^{4}He$ | Nucl. Instr. Meth. B219-220 (2004) 317 |   |
| 16 | 09.12.04 | $^{10}B(a,p_{0})^{13}C$ | Nucl. Instr. Meth. B211 (2003) 1 |   |
| 15 | 09.12.04 | $^{16}O(a,a_{0})^{16}O$ | Nucl. Instr. Meth. B207 (2003) 453 |   |
| 14 | 09.12.04 | $^{16}O(d,a_{0})^{14}N$ | Nucl. Instr. Meth. B207 (2003) 453 |   |
| 13 | 09.12.04 | $^{7}Li(a,a_{0})^{7}Li$ | Nucl. Phys. A179 (1972) 504 | Corrupted file was corrected. Thanks to C.Jeynes |
| 12 | 09.12.04 | $^{16}O(d,p_{1})^{17}O$ | Nucl. Instr. Meth. B207 (2003) 453 |   |
| 11 | 09.12.04 | $^{14}N(a,a_{0})^{14}N$ | Surf. Interface Anal. 36 (2004) ? |   |
| 10 | 09.12.04 | $^{14}N(p,p_{0})^{14}N$ | Surf. Interface Anal. 36 (2004) ? |   |
| 9 | 09.12.04 | $^{nat}Ti(p,p_{0})^{nat}Ti$ | Nucl. Instr. and Meth. B 217 (2004) 551 |   |
| 8 | 08.12.04 | $^{14}N(a,a_{0})^{14}N$ | Nucl. Instr. and Meth. B 192 (2002) 249 |   |
| 7 | 07.12.04 | $^{16}O(d,p_{0,1})^{17}O$ | Nucl. Instr. and Meth. B 226 (2004) 637 |   |
| 6 | 03.06.04 | $^{27}Al(d,p)^{28}Al$<br>$^{27}Al(d,a)^{25}Mg$ | S. Pellegrino, to be published | Thanks to S. Pellegrino |
| 5 | 03.06.04 | $^{14}N(d,p_{3})^{15}N$ | G.Amsel and D.David, Revue de Physique Appliquee, 4 (1969) 383 | Missed sigma scale factor was inserted. Thanks to S. Pellegrino |
| 4 | 01.06.04 | | Nucl. Instr. and Meth. B 219-220 (2004) 140 | Thanks to S. Pellegrino |

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|---|---|---|---|---|
| | | $^{14}N(d,p_{0-5})^{15}N$<br>$^{14}N(d,a_{0-1})^{12}C$ | | |
| 3 | 26.05.04 | $^{28}Si(d,p_{0,1})^{29}Si$ | Nucl. Instr. and Meth. B 226 (2004) 637 |   |
| 2 | 20.04.04 | $^{nat}Cr(p,p_0)^{nat}Cr$ | Helvetica Physics Acta 44 (1971) 757 | Thanks to Svetlana Dunaeva. See EXFOR |
| 1 | 31.03.04 | Some ~100 data files | SIMNRA library | Thanks to Matej Mayer |

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*by V.Zerkin, IAEA*